

Waveserver Ai

Waveserver Ai and 6500/5170 Interworking

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CHAPTER 1

About this document

This document provides detailed descriptions for Waveserver Ai and 6500/5170 interworking.

Most of the procedures in this document are CLI-based for 6500 Release 12.1 and higher and for 5170 Release 8.6.1 and higher. For a description of each CLI command see *6500 CLI Reference*, 323-1851-193 or *5170 Command Reference*, 380-1876-810.

The Waveserver Ai, 6500, 5170, and MCP documentation can be used with this guide for supplemental information.

For information on Waveserver Ai, refer to the *Waveserver Ai User Guide*, 323-4002-100.

For information on 6500, refer to the 6500 Packet-Optical Platform (6500) documentation library.

For information on 5170, refer to the 5170 Service Aggregation Switch (5170) documentation library.

For information on MCP, refer to *Blue Planet MCP User Guide*, 450-3709-301.

This section covers:

- [Publication history](#)

Publication history

Issue 3

Issue 3 has been updated to include the following information:

- [Managing Waveserver Ai through 6500](#)
- [Supported 6500 photonic configurations](#)

Managing Waveserver Ai through 6500

The Waveserver Ai can be managed through the 6500, when there is limited or no DCN connectivity at the sites. As 6500 supports GNE and Layer-3 routing functionality, the Waveserver Ai subtends the 6500 in this setup. See [Managing Waveserver Ais through 6500](#).

Supported 6500 photonic configurations

The [Photonics configurations](#) section provides a few examples of the supported photonic configurations. Additional supported configurations can be found in section [Supported 6500 photonic configurations](#).

The list of [Supported 6500 photonic configurations](#) is updated.

Issue 2

Issue 2 has been updated to include the following information:

- [Supported 6500 photonics configurations](#)
- [Waveserver Ai Release 1.4 and 6500 Release 12.4 interworking](#)
- [Waveserver Ai and 6500 WLAI MOTR interworking considerations](#)
- [Waveserver Ai Release 1.3 \(and higher\) and 6500 Release 12.3 interworking](#)
- [Waveserver Ai Release 1.2 \(and higher\) and 5170 Release 8.6.1 interworking](#)

Supported 6500 photonics configurations

The list of 6500 photonics configurations supported in 56GBaud mode is updated to include the following:

- Passive photonic line system with CMD48 and MLA3 (Waveserver Ai Release 1.4 and higher, 6500 Release 12.4 and higher)
- Passive photonic line system with CMD48 and MLA3/ESAM (Waveserver Ai Release and higher, 6500 Release 12.4 and higher)

See [Supported 6500 photonic configurations](#).

Waveserver Ai Release 1.4 and 6500 Release 12.4 interworking

Waveserver Ai Release 1.4 introduces support for fast tuning mode via SPLI. When a new channel is created on the 6500 (Release 12.4 and higher) with the channel capacity modes “Mode-2” or “Mode-3”, the ADJ-Tx Tuning mode is set to fast tuning and this mode is pushed to the Waveserver Ai via SPLI.

Waveserver Ai Release 1.4 and 6500 Release 12.4 supports Waveserver Ai 1xWLAI + 10xQSFP+/QSFP28 modules interworking with the 6500 WLAI MOTR circuit packs. The 10GE traffic coming into the Waveserver Ai 1xWLAI + 10xQSFP+/QSFP28 module hands off as an OTL4.4 signal to the 6500 WLAI circuit pack.

This section also provides details on the supported 6500 photonic line systems which interwork with the Waveserver Ai 2x400G 35/56GBaud EDFA C-Band w/OPS 8xQSFP28 module.

See [Waveserver Ai Release 1.4 and 6500 Release 12.4 interworking](#).

Waveserver Ai and 6500 WLAI MOTR interworking considerations

Information about the supported settings used on the 6500 for interworking with Waveserver Ai in x00G and x50G modes is updated. The 6500 provisioning methods to be used for x00G and x50G transmission modes in a regen configuration for Waveserver Ai Release 1.3 and higher are also updated. See [Waveserver Ai and 6500 WLAI MOTR interworking considerations](#).

Waveserver Ai Release 1.3 (and higher) and 6500 Release 12.3 interworking

For Waveserver Ai Release 1.4 (and higher), there is added support for Waveserver Ai 2x400G 35/56GBaud EDFA C-Band Premium 8xQSFP28 (186-1601-901) and Waveserver Ai 2x400G 35/56GBaud EDFA C-Band w/OPS 8xQSFP28 (186-1607-900) modules for certain interworking configurations.

Adds information about MCP 18.06 support for some of the interworking configurations and applicable considerations. See [Waveserver Ai Release 1.3 \(and higher\) and 6500 Release 12.3 interworking](#).

Waveserver Ai Release 1.2 (and higher) and 5170 Release 8.6.1 interworking

Adds information about MCP 18.06 support for the interworking configurations and applicable considerations. See [Waveserver Ai Release 1.2 \(and higher\) and 5170 Release 8.6.1 interworking](#).

Issue 1

Issue 1 is the first release of this document for Waveserver Ai and 6500/5170 Interworking.

The following topics are covered in this document:

- [Overview](#)
 - [Waveserver Ai overview](#)
 - [6500 Overview](#)
 - [5170 Overview](#)
- [6500 installation for Waveserver Ai](#)
- [6500 commissioning for Waveserver Ai](#)
- [Waveserver Ai and 6500 interworking](#)
- [Waveserver Ai and 5170 interworking](#)

CHAPTER 2

Overview

This section provides an overview on the Waveserver Ai, 6500, 5170 and MCP. This section also provides an overview of supported Waveserver Ai and 6500/5170 interworking configurations. See the following:

- [Waveserver Ai overview](#)
- [6500 Overview](#)
- [5170 Overview](#)
- [MCP Overview](#)

Waveserver Ai overview

Waveserver Ai supports 100G to 400G per wavelength to maximize capacity at any distance. Waveserver Ai features two line-side interfaces per pluggable module. Each is a duplex LC optical connector. Waveserver Ai line ports support 56GBaud and 35GBaud transmission modes. At 56GBaud, Waveserver Ai supports 100G, 150G, 200G, 250G, 300G, and 400G rates. At 35GBaud, Waveserver Ai supports 100G, 150G, 200G and 250G rates. Each Waveserver Ai module has either four, eight or ten client interfaces. The client ports on one 2x400G module can support up to 800 Gb/s of Ethernet traffic depending on the provisioned modem transceiver mode. Waveserver Ai supports up to three modules.

Aside from the 2x400G modules (including the 2x400G module with OPS) being supported for Waveserver Ai and 6500 interworking; Waveserver Ai 1x400G 35/56GBaud EDFA C-Band 4xQSFP28 (186-1603-900), Waveserver Ai 1x400G Encryption 35/56GBaud EDFA C-Band 4xQSFP28 (186-1606-900) and Waveserver Ai 1x400G 35/56GBaud EDFA C-Band 10xQSFP+/QSFP28 (186-1608-900) modules are also supported for interworking.

For information on Waveserver Ai, refer to the *Waveserver Ai User Guide*, 323-4002-100.

6500 Overview

The 6500 leverages a programmable infrastructure that can scale and respond on demand, to meet changing customer expectations and unpredictable traffic requirements. 6500 converges, packet, OTN and photonics in a single platform.

6500 Photonics can be used to interwork with the Waveserver Ai through the line side interfaces. The 6500 also supports OTU4 (OTL4.4) client interworking with the Waveserver Ai on some 6500 traffic cards.

5170 Overview

The 5170 Service Aggregation Switch enables cost-effective delivery of carrier class 100GE services direct to the customer premise in a compact, 1RU form factor. The Waveserver Ai connects to the 5170 through the 100GE client-side interfaces.

MCP Overview

Blue Planet MCP is Ciena's next-generation multi-layer Software Defined Networking (SDN) and Network Management System (NMS) platform with integrated network planning functionality that combines a web-scale platform, industry-leading SDN functionality and open interfaces.

CHAPTER 3

6500 installation for Waveserver Ai

For information on how to install the Ciena 6500 Packet-Optical Platform (6500) shelves and related equipment, as applicable to your configuration, refer to the following technical publications, as applicable to your shelf type:

- *6500 Installation - 2-slot Shelves*, 323-1851-201.1
- *6500 Installation - 7-slot & 6500-7 packet-optical Shelves*, 323-1851-201.2

For safety and regulatory information and for fiber cleaning information, refer to *6500 Installation-General Information*, 323-1851-201.0.

For more information that can be useful in preparing to install the shelf assembly, refer to the following sections in this technical publication and to the following 6500 technical publications:

- the sections on the following topics in *6500 Planning* and *6500 Ordering Information*, 323-1851-151, as applicable:
 - bay/rack configurations— *6500 Planning*, Part 1, feature overview chapter
 - bay equipping rules, including 6500 equipment rack space requirements and rack unit spacing—*6500 Ordering Information*, 323-1851-151, configuration rules chapter
 - shelf assemblies— *6500 Planning*, Part 1, shelf and equipment descriptions chapter
 - 6500 shelf and equipment—*6500 Ordering Information*, 323-1851-151, chapter on bays, shelf assemblies, and hardware
 - additional front exhaust considerations—*6500 Planning*, Part 1, feature overview chapter
 - engineering rules regarding the venting options—*6500 Ordering Information*, 323-1851-151, configuration rules chapter and chapter on circuit packs, modules, pluggable modules, and interface hardware
 - supported venting options on shelf variants—*6500 Planning*, Part 1, Shelf and equipment descriptions chapter

- power specifications— *6500 Planning*, Part 3
- supported shelf processors—*6500 Ordering Information*, 323-1851-151, chapter on circuit packs, modules, pluggable modules, and interface hardware

For more information on the 6500 2-slot shelves and 7-slot shelves, refer to the following 6500 technical publications:

- Part 1 of *6500 Planning*
- *6500 Ordering Information*, 323-1851-151

Before you begin the installation procedures, make sure that

- you have determined your site prerequisites
- you have determined your equipment prerequisites
- you have available all required tools and materials

Refer to the chapter on preparing for installation in *6500 Installation-General Information*, 323-1851-201.0 for information on the above.

- you have read the chapter on observing product and personnel safety guidelines in *6500 Installation-General Information*, 323-1851-201.0

For equipment specific to each of the supported configurations, refer to [Photonics configurations](#).

CHAPTER 4

6500 commissioning for Waveserver Ai

This section describes how to commission a new 6500 shelf for interworking with Waveserver Ai. Topics include:

- [6500 ROADM configurations—commissioning overview](#)
- [6500 non-ROADM configurations—commissioning overview](#)
 - [Standard reach and Extended reach configurations](#)
 - [Minimum channel count requirement](#)
 - [Assumptions for span reach](#)
 - [Link budget tables](#)
- [6500 commissioning prerequisites](#)
 - [Technical publication requirements](#)
 - [Equipment requirements](#)
 - [Supporting equipment requirements](#)
 - [6500 commissioning — provisioning information](#)
 - [Other customer provisioning information](#)

The following table provides a guide for locating information related to 6500 commissioning and setting up 6500 configurations.

Table 1 Where to find information

Topic	Location
Configuration support	Sub-section on Photonic configurations of this technical publication.
Detailed commissioning information for 6500 ROADM configurations	<i>6500 Commissioning and Testing</i> , 323-1851-221
Detailed commissioning information for 6500 configurations other than ROADM	This section

Topic	Location
Adding Waveserver Ai node name(s) to the SPLI list	Using CLI: This section Using Site Manager: <i>6500 Administration and Security</i> , 323-1851-301
Commissioning the Waveserver Ai	<i>Waveserver Ai User Guide</i> , 323-4002-100
• Interconnecting 6500 and Waveserver Ai nodes • Setting up the 6500 configurations other than ROADM (additional provisioning that requires the Waveserver Ai wavelength being connected to the 6500 photonic link) • Configuring the 6500 and Waveserver Ai nodes to ensure interworking • Verifying traffic carried by the Waveserver Ai-6500 solution	Waveserver Ai and 6500 interworking section of this technical publication
6500 safety and regulatory information and fiber cleaning information	<i>6500 Installation - General Information</i> , 323-1851-201.0
Detailed information on 6500 Photonics equipment	<i>6500 Photonics Equipment</i> , 323-1851-102.6
Detailed information on 6500 Coherent Select configurations	<i>6500 WaveLogic Photonics Coherent Select</i> , 323-1851-980
6500 CLI command syntax	<i>6500 CLI Reference</i> , 323-1851-193

6500 ROADM configurations—commissioning overview

For the list of supported ROADM configurations, refer to the sub-section on Photonics configurations of this technical publication.

For the roadmap to set up 6500 ROADM configurations, refer to [Roadmap for 6500 ROADM configurations](#).

6500 non-ROADM configurations—commissioning overview

Figure [6500 photonic line systems](#) illustrates the supported types of photonic line systems. Figure 2 to Figure 10 in procedure [Connecting intra-site fibers for the 6500 and performing 6500 site testing](#) illustrate examples of supported 6500 node configurations.

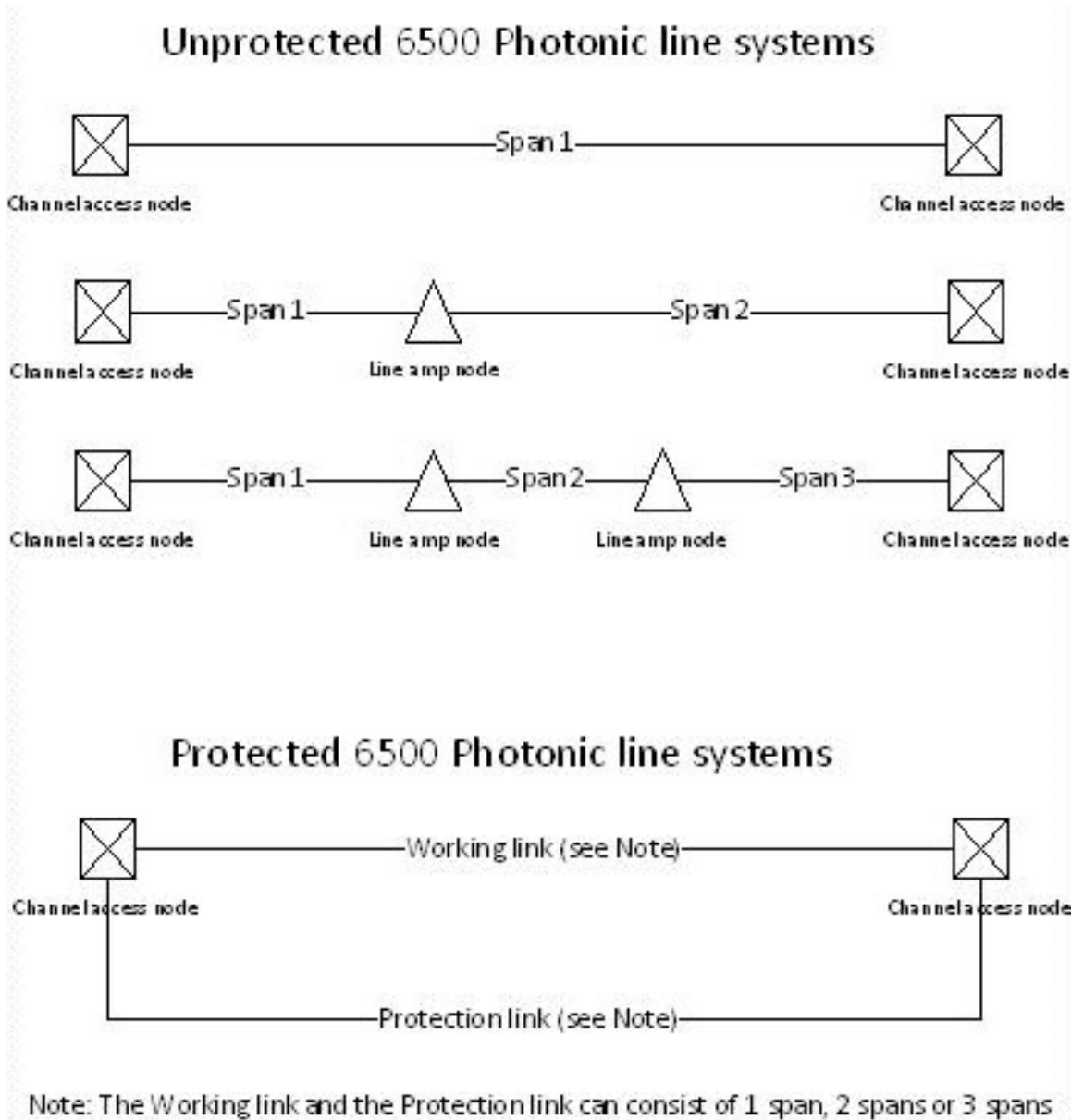
The “install create” commands are provided for each of the supported types of configurations—except Line Amp configurations—that perform all (for Flex-12 [50GHZ] (see [Figure 2](#) and [Figure 3](#)) or most of the provisioning required for commissioning in just one step. Tables “[Install create” commands for unprotected configurations](#) and “[Install create” commands for protected configurations](#) provide the “install create” command for unprotected and protected configurations. The Line Amp configurations must be provisioned manually.

The “install create” commands cannot be used for supported configurations that include XLAs or SRAs. For these configurations, you must commission the 6500 nodes as per standard 6500 procedures and manually provision the equipment/slot sequence. Refer to *6500 Commissioning and Testing*, 323-1851-221, and Part 2 of *6500 Configuration - Provisioning and Operating*, 323-1851-310.2.

Note 1: For the SRA, the “FORCEDSHUTOFF” parameter value must be set to “FALSE” to bring up the RAMAN pump lasers to ON.

Note 2: For CMD64 configurations with MLA3/ESAM, XLA/ESAM, or XLA/SRA, the “Tx Power Reduction Control” parameter value for the LOC facility must be set to “Always Off” so the transponders/Waveserver Ai ports associated with the CMD64 do not remain in reduced power state when SPLI is established. Refer to Part 1 of *6500 Configuration - Provisioning and Operating*, 323-1851-310.1, for details.

After you complete the commissioning of the 6500 shelf and connect the intra-site fibers for the 6500, proceed to the Waveserver Ai and 6500 interworking section.

Figure 1 6500 photonic line systems

Standard reach and Extended reach configurations

The configurations can have Standard reach or Extended reach. Standard reach configurations require that all spans be lower than the maximum span loss indicated in the link budget tables. If one or more spans exceed the indicated maximum loss, it must be considered as Extended reach. This is also the case for protected configurations. For example, if the working or the protection span exceeds the indicated maximum loss, then both the working and the protection span are considered Extended reach.

Multi-span Line Amp nodes consist of:

- For Standard reach: MLA3 and LIM
- For Extended reach: MLA3 pair with 10 dB pad

Minimum channel count requirement

A minimum channel count is required in order for automatic switching to occur in response to a fiber cut, see the following table.

If the network has one or more spans whose loss exceed(s) the limits described in the following table, generic rules do not apply. The minimum channel count is customized based on actual network specifications. For more information, contact Ciena.

Table 2 Minimum channel count

Configurations using	Span count and maximum span loss	Minimum channel count
		Fixed-64 band plan
SLA	1 span <10.2 dB span loss	N/A
MLA3	1 span <24.7 dB span loss	3
MLA3	2 spans <22.0 dB span loss	4
MLA2/MLA3	3 spans <22.0 dB span loss	5

Assumptions for span reach

The following elements are assumed for span reach:

- Fiber type is NDSF
- The fiber attenuation coefficient is 0.25 dB/km
- Short spans are padded so that net span loss exceeds the minimum span loss quoted in the link budget tables
- Head-end and tail-end patch panels have 0.5 dB loss

Link budget tables

This section provides the link budgets for the following:

- [Two wavelengths unamplified link budgets - Waveserver Ai](#)
- [Eight wavelengths unamplified link budgets - Waveserver Ai](#)
- [Twelve wavelengths unamplified link budgets - Waveserver Ai](#)

- [Forty-four wavelengths link budgets - Waveserver Ai](#)
- [Fifty-eight wavelengths link budgets - Waveserver Ai](#)
- [Sixty-four wavelengths link budgets - Waveserver Ai](#)

Table 3 Two wavelengths unamplified link budgets - Waveserver Ai

Data rate (GB/s)	Channels	Mux/Demux	Amp	Protected	Span Count	Min Span Loss (dB)	Max Span Loss (dB)	Max Reach (km)	Rx Input Power (dBm)
400	2	BMD2	None	No	1	0	16.9	64.0	-20.5
400	2	BMD2	None	Yes	1	0	10.8	39.0	-20.5
300	2	BMD2	None	No	1	0	21.9	84.0	-25.5
300	2	BMD2	None	Yes	1	0	15.8	59.0	-25.5
200	2	BMD2	None	No	1	0	23.1	88.0	-26.7
200	2	BMD2	None	Yes	1	0	17.0	64.0	-26.7

Note 1: Span loss consists of fiber plus head and tail patch panel loss of 0.5 dB each.

Note 2: Maximum reach calculated based on Tx maximum power of +4dB, and 0.25 dB attenuation per kilometer.

Table 4 Eight wavelengths unamplified link budgets - Waveserver Ai

Data rate (GB/s)	Channels	Mux/Demux	Amp	Protected	Span Count	Min Span Loss (dB)	Max Span Loss (dB)	Max Reach (km)	Rx Input Power (dBm)
400	8	OMDF8	None	No	1	0	18.8	71.0	-20.5
400	8	OMDF8	None	Yes	1	0	12.7	47.0	-20.5

Note 1: Span loss consists of fiber plus head and tail patch panel loss of 0.5 dB each.

Note 2: Maximum reach calculated based on Tx maximum power of +4dB, and 0.25 dB attenuation per kilometer.

Table 5 Twelve wavelengths unamplified link budgets - Waveserver Ai

Data rate (GB/s)	Channels	Mux/Demux	Amp	Protected	Span Count	Min Span Loss (dB)	Max Span Loss (dB)	Max Reach (km)	Rx Input Power (dBm)
400	12	CCMD12	None	No	1	20.0	26.0	100.0	-12.7
300	12	CCMD12	None	No	1	20.0	33.8	131.0	-20.0

Data rate (GB/s)	Channels	Mux/Demux	Amp	Protected	Span Count	Min Span Loss (dB)	Max Span Loss (dB)	Max Reach (km)	Rx Input Power (dBm)
200	12	CCMD12	None	No	1	20.0	39.5	154.0	-25.7
Note 1: Span loss consists of fiber plus head and tail patch panel loss of 0.5 dB each. Note 2: Maximum reach calculated based on Tx maximum power of +4dB, and 0.25 dB attenuation per kilometer. Note 3: The average output power from the CCMD12 Tx amp is 4.2 dBm/ch (TOP limit for 12 ch), which corresponds to a Tx output power of approximately -2 dBm; Tx amp in CCMD12 is provisioned to 20 dB gain, Rx amp in CCMD12 is provisioned to 23 dB gain to maximize Rx power; Reddest 12 channels on 75 GHz grid									

Table 6 Forty-four wavelengths link budgets - Waveserver Ai

Data rate (GB/s)	Channels	Mux/Demux	Amp	Protected	Span Count	Min Span Loss (dB)	Max Span Loss (dB)	Max Reach (km)	Rx Input Power (dBm)
400	44	eCMD44	None	No	1	0	12.5	46.0	-20.5
400	44	eCMD44	MLA3	No	1	10	29.5	114.0	-4.0
400	44	eCMD44	MLA3	No	2	10	25.0	192.0	-1.0
400	44	eCMD44	MLA3	No	3	10	21.5	246.0	0
400	44	eCMD44	MLA2	No	1	10	28.3	109.0	-5.0
400	44	eCMD44	MLA2	No	2	10	24.5	188.0	-1.0
400	44	eCMD44	MLA2	No	3	10	21.5	246.0	0
400	44	eCMD44	None	Yes	1	0	6.4	22.0	-20.5
400	44	eCMD44	MLA3	Yes	1	10	29.5	114.0	-6.0
400	44	eCMD44	MLA3	Yes	2	10	25.0	192.0	-3.0
400	44	eCMD44	MLA3	Yes	3	10	21.5	246.0	-2.0
400	44	eCMD44	MLA2	Yes	1	10	28.3	109.0	-8.0
400	44	eCMD44	MLA2	Yes	2	10	24.5	188.0	-3.0
400	44	eCMD44	MLA2	Yes	3	10	21.5	246.0	-2.0
300	44	eCMD44	None	No	1	0	17.1	64.0	-25.1
300	44	eCMD44	MLA3	No	1	10	36.8	143.0	-10.0
300	44	eCMD44	MLA3	No	2	10	32.8	254.0	-6.0

Data rate (GB/s)	Channels	Mux/Demux	Amp	Protected	Span Count	Min Span Loss (dB)	Max Span Loss (dB)	Max Reach (km)	Rx Input Power (dBm)
300	44	eCMD44	MLA3	No	3	10	29.2	338.0	-4.0
300	44	eCMD44	MLA2	No	1	10	34.2	133.0	-10.0
300	44	eCMD44	MLA2	No	2	10	30.5	236.0	-6.0
300	44	eCMD44	MLA2	No	3	10	28.0	324.0	-4.0
300	44	eCMD44	None	Yes	1	0	11.0	40.0	-25.1
300	44	eCMD44	MLA3	Yes	1	10	36.8	143.0	-12.0
300	44	eCMD44	MLA3	Yes	2	10	32.8	254.0	-8.0
300	44	eCMD44	MLA3	Yes	3	10	29.2	338.0	-6.0
300	44	eCMD44	MLA2	Yes	1	10	34.2	133.0	-12.0
300	44	eCMD44	MLA2	Yes	2	10	30.5	236.0	-8.0
300	44	eCMD44	MLA2	Yes	3	10	28.0	324.0	-6.0
200	44	eCMD44	None	No	1	0	18.7	71.0	-26.7
200	44	eCMD44	MLA3	No	1	10	43.1	168.0	-16.0
200	44	eCMD44	MLA3	No	2	10	39.9	311.0	-13.0
200	44	eCMD44	MLA3	No	3	10	37.7	440.0	-11.0
200	44	eCMD44	MLA2	No	1	10	40.0	156.0	-16.0
200	44	eCMD44	MLA2	No	2	10	36.5	284.0	-13.0
200	44	eCMD44	MLA2	No	3	10	34.3	400.0	-11.0
200	44	eCMD44	None	Yes	1	0	12.6	46.0	-26.7
200	44	eCMD44	MLA3	Yes	1	10	42.8	167.0	-18.0
200	44	eCMD44	MLA3	Yes	2	10	39.8	310.0	-15.0
200	44	eCMD44	MLA3	Yes	3	10	37.7	440.0	-13.0
200	44	eCMD44	MLA2	Yes	1	10	40.0	156.0	-18.0
200	44	eCMD44	MLA2	Yes	2	10	36.5	284.0	-15.0
200	44	eCMD44	MLA2	Yes	3	10	34.3	400.0	-13.0

Data rate (GB/s)	Channels	Mux/Demux	Amp	Protected	Span Count	Min Span Loss (dB)	Max Span Loss (dB)	Max Reach (km)	Rx Input Power (dBm)
Note 1: Span loss consists of fiber plus head and tail patch panel loss of 0.5 dB each. Note 2: Maximum reach calculated based on Tx maximum power of +4dB, and 0.25 dB attenuation per kilometer. Note 3: If OSC is required max span loss = 42 dB using NTK592NVE5 OSC. Note 4: The link budgets are valid for CMD44 100GHz and eCMD44 100GHz.									

Table 7 Fifty-eight wavelengths link budgets - Waveserver Ai

Data rate (GB/s)	Channels	Mux/Demux	Amp	Protected	Span Count	Min Span Loss (dB)	Max Span Loss (dB)	Max Reach (km)	Rx Input Power (dBm)
400	58	CMD64	MLA2	No	1	10	27.0	104.0	-5.0
400	58	CMD64	MLA2	No	2	10	22.6	173.0	-2.0
400	58	CMD64	MLA2	No	3	10	20.1	229.0	-1.0
400	58	CMD64	MLA2	Yes	1	10	27.0	104.0	-7.0
400	58	CMD64	MLA2	Yes	2	10	22.6	173.0	-4.0
400	58	CMD64	MLA2	Yes	3	10	20.1	229.0	-3.0
300	58	CMD64	MLA2	No	1	10	32.5	126.0	-11.0
300	58	CMD64	MLA2	No	2	10	28.9	223.0	-8.0
300	58	CMD64	MLA2	No	3	10	26.4	305.0	-7.5
300	58	CMD64	MLA2	Yes	1	10	32.5	126.0	-13.0
300	58	CMD64	MLA2	Yes	2	10	28.9	223.0	-10.0
300	58	CMD64	MLA2	Yes	3	10	26.4	305.0	-9.0
Note 1: Span loss consists of fiber plus head and tail patch panel loss of 0.5 dB each. Note 2: Maximum reach calculated based on Tx maximum power of +4dB, and 0.25 dB attenuation per kilometer. Note 3: MLA2 bandwidth supports channels 4 to 61 on CMD64.									

Table 8 Sixty-four wavelengths link budgets - Waveserver Ai

Data rate (GB/s)	Channels	Mux/Demux	Amp	Protected	Span Count	Min Span Loss (dB)	Max Span Loss (dB)	Max Reach (km)	Rx Input Power (dBm)
400	64	CMD64	none	No	1	0	10.9	40.0	-20.5
400	64	CMD64	MLA3	No	1	10	28.9	112.0	-5.0
400	64	CMD64	MLA3	No	2	10	23.7	182.0	-2.0
400	64	CMD64	MLA3	No	3	10	20.3	232.0	-1.0
400	64	CMD64	none	Yes	1	0	4.8	15.0	-20.5
400	64	CMD64	MLA3	Yes	1	10	28.9	112.0	-7.0
400	64	CMD64	MLA3	Yes	2	10	23.7	182.0	-4.0
400	64	CMD64	MLA3	Yes	3	10	20.3	232.0	-3.0
300	64	CMD64	none	No	1	0	15.5	58.0	-25.1
300	64	CMD64	MLA3	No	1	10	35.0	136.0	-11.0
300	64	CMD64	MLA3	No	2	10	31.1	241.0	-8.0
300	64	CMD64	MLA3	No	3	10	28.4	329.0	-7.5
300	64	CMD64	none	Yes	1	0	9.4	34.0	-25.1
300	64	CMD64	MLA3	Yes	1	10	35.0	136.0	-13.0
300	64	CMD64	MLA3	Yes	2	10	31.1	241.0	-10.0
300	64	CMD64	MLA3	Yes	3	10	28.4	329.0	-9.0
Note 1: Span loss consists of fiber plus head and tail patch panel loss of 0.5 dB each.									
Note 2: Maximum reach calculated based on Tx maximum power of +4dB, and 0.25 dB attenuation per kilometer.									

6500 commissioning prerequisites

Note the following considerations before commissioning a 6500 shelf:

- Engineering data specific to the deployment must be generated in the planning stages in the form of an engineering data sheet for each site. This engineering data contains items such as provisioning information, fiber connections, circuit pack placements, data communications strategy, and link budget design information.
- Make sure that the installation of the 6500 shelf has been completed. For information, refer to the Installation section in this technical publication.

- Make sure that the shelf is equipped with a shelf processor (6500 Rel 12.1 or higher NTK555NA/NB Shelf Processor [SPAP-2w/2xOSC for 6500-2/7 Type 2]. Otherwise, contact Ciena.
- You must have the following information and equipment and meet the following prerequisites.

Technical publication requirements

Ensure you have access to the following 6500 technical publications:

- *6500 Installation-General Information*, 323-1851-201.0 — for safety and regulatory information and for fiber cleaning information
- *6500 Photonics Equipment*, 323-1851-102.6 — for information on 6500 Photonics equipment
- *6500 Commissioning and Testing*, 323-1851-221 — for more information on setting up 6500 equipment
- *6500 WaveLogic Photonics Coherent Select*, 323-1851-980 — for more information on Coherent Select configurations for the 6500

Equipment requirements

You must have the equipment specific to the shelf as specified in the engineering data sheet. The circuit packs and their associated slot numbers must be identified.

Supporting equipment requirements

You must have the following supporting equipment as listed in the following table.

Table 9 6500 commissioning—supporting equipment list

Item	Check
Craft PC	
Optical fiber numbers or labels for each port to be connected	
Ethernet 10/100Base-T RJ-45 straight-through or cross-over cable (to connect the craft PC to the network element)	
Fiberscope with LC adapter (and SC adapter for optical layer equipment as required)	
Spare patch cords and adapters	

6500 commissioning — provisioning information

Note the following considerations before provisioning a 6500 shelf:

- For supported configurations that include XLAs or SRAs, the “install create” commands cannot be used. For these configurations, you must commission the 6500 nodes as per standard 6500 procedures and

manually provision the equipment/slot sequence. Refer to *6500 Commissioning and Testing*, 323-1851-221, and Part 2 of *6500 Configuration - Provisioning and Operating*, 323-1851-310.2.

- For most configurations, the XLA mode should be left at the default (High Gain). For XLA/SRA configurations, refer to [Calibrating the SRA circuit packs](#) for recommendations dependent on span range. For XLA/ESAM configurations, contact Ciena.
- For CMD64 configurations with MLA3/ESAM, XLA/ESAM, or XLA/SRA, the “Tx Power Reduction Control” parameter value for the LOC facility must be set to “Always Off” so the transponders/Waveserver Ai ports associated with the CMD64 do not remain in reduced power state when SPLI is established. Refer to Part 1 of *6500 Configuration - Provisioning and Operating*, 323-1851-310.1, for details.

6500 commissioning—provisioning information for Terminal nodes and Line Amp nodes

The tables in this section provide information on:

- “Install create” commands used to commission the Terminal node configurations for unprotected and protected configurations.
- descriptions for the parameters of the “Install create” commands used to commission the Terminal nodes
- system, shelf and OTS parameters for CLI commands used to commission manually the Line Amp configurations

Consult the engineering data sheet for your site and ensure you have the information listed in the following tables as applicable to your Terminal or Line Amp configuration.

- [6500 “install create” commissioning — provisioning information](#) (Terminal nodes: Fixed-64, Flex-12)
- [6500 Line Amp commissioning — for System parameters](#) (CLI example: system config set), [6500 Line Amp commissioning — for Shelf parameters](#) (CLI example: shelves shelf create), [6500 Line Amp commissioning — for OTS parameters](#) (CLI example: topology ots create) (Line Amps)

The following tables listed below provide information on Install create commands used to commission the Terminal node configurations.

- [“Install create” commands for unprotected configurations](#)
- [“Install create” commands for protected configurations](#)

Table 10 “Install create” commands for unprotected configurations

Configuration (Examples)	“Install create” command
Fixed-64 [75GHz] unprotected (See Figure 4 and Figure 5)	install create band-plan fixed64 group-id <Group Identifier> site-id <Site ID> remote-site-id <Remote Site ID> site-name <Site name> ip <IP Address of the COLAN port> netmask <Netmask of the COLAN port> [def-gateway <Default gateway>] [otdr yes] [location <Location of the site>] [system-name <System ID>] [display <Display commands flag>] Example (no OTDR): install create band-plan fixed64 group-id 3 site-id 5 remote-site-id 10 site-name "Site_1" ip 192.168.0.2 netmask 255.255.255.0 def-gateway 192.168.0.255 install create band-plan fixed64 group-id 5 site-id 10 remote-site-id 5 site-name "Site_2" ip 192.168.0.3 netmask 255.255.255.0 def-gateway 192.168.0.255 Example (OTDR): install create band-plan fixed64 group-id 3 site-id 5 remote-site-id 10 site-name "Site_1" ip 192.168.0.2 netmask 255.255.255.0 def-gateway 192.168.0.255 otdr yes install create band-plan fixed64 group-id 5 site-id 10 remote-site-id 5 site-name "Site_2" ip 192.168.0.3 netmask 255.255.255.0 def-gateway 192.168.0.255 otdr yes
Flex-12 [50GHz] (See Figure 2 and Figure 3)	install create band-plan flex12 group-id <Group Identifier> site-id <Site ID> remote-site-id <Remote Site ID> site-name <Site name> ip <IP Address of the COLAN port> netmask <Netmask of the COLAN port> [def-gateway <Default gateway>] [otdr yes] [location <Location of the site>] [system-name <System ID>] [display <Display commands flag>] Example (no OTDR): install create band-plan flex12 group-id 5 site-id 5 remote-site-id 10 site-name "Site_1" ip 192.168.0.2 netmask 255.255.255.0 def-gateway 192.168.0.255 install create band-plan flex12 group-id 5 site-id 10 remote-site-id 5 site-name "Site_2" ip 192.168.0.3 netmask 255.255.255.0 def-gateway 192.168.0.255 Example (OTDR): install create band-plan flex12 group-id 5 site-id 5 remote-site-id 10 site-name "Site_1" ip 192.168.0.2 netmask 255.255.255.0 def-gateway 192.168.0.255 otdr yes install create band-plan flex12 group-id 5 site-id 10 remote-site-id 5 site-name "Site_2" ip 192.168.0.3 netmask 255.255.255.0 def-gateway 192.168.0.255 otdr yes

Table 11 “Install create” commands for protected configurations

Configuration (Examples)	“Install create” command
Fixed-64 [75GHz] protected (see Figure 8)	install create band-plan fixed64 group-id <Group Identifier> protected yes site-id <Site ID> remote-site-id <Remote Site ID> site-name <Site name> ip <IP Address of the COLAN port> netmask <Netmask of the COLAN port> [def-gateway <Default gateway>] [location <Location of the site>] [system-name <System ID>] [display <Display commands flag>] Example: install create band-plan fixed64 group-id 3 protected yes site-id 5 remote-site-id 10 site-name "Site_1" ip 192.168.0.2 netmask 255.255.255.0 def-gateway 192.168.0.255 install create band-plan fixed64 group-id 5 protected yes site-id 10 remote-site-id 5 site-name "Site_2" ip 192.168.0.3 netmask 255.255.255.0 def-gateway 192.168.0.255

The table below includes the descriptions for the parameters of the Install create commands used to commission the Terminal nodes ([Commissioning 6500 Terminal nodes](#)).

Table 12 6500 “install create” commissioning — provisioning information

Parameter and description	Value	“install create” command parameter	Mandatory to enter at commissioning stage
Band plan type Determines the provisioning of the equipment in the configuration and the channel-to-wavelength mappings. The band-plan on 6500 and Waveserver Ai must match. Not stored on the shelf. Note: For more information on band plans, refer to the wavelengths plans in the Photonic equipment description section of <i>6500 Photonic Equipment</i>, 323-1851-102.6.	• fixed96 • fixed88 • fixed64 • fixed44 • flex12	band-plan	yes

Parameter and description	Value	"install create" command parameter	Mandatory to enter at commissioning stage
Default gateway Used to provision a static route to the default gateway. Must be in the same subnet as the IP address.	Standard dot notation: n.n.n.n	def-gateway	no
Display commands flag If this flag is set, the commands will be printed to the screen, but not executed.	• yes • no (default)	display	no
Group Identifier This number must match the group-id of the Waveserver Ai modules connected to the line system. (For more information on the Waveserver Ai group-id, refer to the <i>Waveserver Ai User Guide</i> , 323-4002-100.) This group identifier is combined with the site name to generate a node name. This is stored in the shelf/baynum attribute and is also included in the TID of the system if the site-name is also provided.	Integer value in the range 1 to 99. For information on Waveserver Ai, refer to the <i>Waveserver Ai User Guide</i> , 323-4002-100.	group-id	yes
IP Address of the COLAN port Also used as the SHELF IP address. Note: COLAN (central office local area network) is a facility on the network element that allows the 6500 to communicate with the customer data comms network (DCN).	Standard dot notation: n.n.n.n 0.0.0.0 indicates "unnumbered".	ip	yes
Location of the site Stored in the shelf/location field.	GPS coordinates of the site (comma-separated longitude,latitude. For example, "45.34940,-75.85461".)	location	no

Parameter and description	Value	"install create" command parameter	Mandatory to enter at commissioning stage
Netmask of the COLAN port	standard dot notation: n.n.n.n.	netmask	yes
OPS Protection flag If set to yes, it indicates that an OPS will be provisioned with two Mux/Demux groups. Not stored on the shelf.	• no (default) • yes	protected	no
OTDR flag If set to yes, it indicates that an ESAM will be provisioned for OTDR functionality. Not stored on the shelf.	• no (default) • yes	otdr	no
Remote Site ID This remote site ID is used to generate a unique optical system ID (OSID), if a system-id is not provided. It is also used to properly provision the Path ID for the OTS equipment created. This is not stored on the shelf, but it is used to construct the following attributes ots/osid If a system-name is not provided, the OSID is constructed of the hex forms of the site-id and remote-site-id. If it is unprotected or a working link, the smaller of the two site IDs will come first. If it is a protection link, the smaller of the two site IDs will come last.	Integer value in the range 1 to 65535	remote-site-id	yes

Parameter and description	Value	"install create" command parameter	Mandatory to enter at commissioning stage
Site ID This number must be a unique ID within the comms connected network. It is used to associate all photonic equipment at the same site together into logical groupings for management interfaces. Stored in the shelf/siteid attribute. Also used in the ots/osid field if the system-name is not provided. If a system-name is not provided, the OSID is constructed of the hex forms of the site-id and remote-site-id. If it is unprotected or a working link, the smaller of the two site IDs will come first. If it is a protection link, the smaller of the two site IDs will come last.	Range is 1 to 65535	site-id	yes
Site name The site name is a human readable name common to the nodes at a site. If it is provided, it is combined with the provided group ID to provision the node name of the device also. Stored in the shelf/siteName field. If it is supplied, it is combined with the group-id to construct a TID. If it is omitted, the default TID will be used.	Alphanumeric string Maximum length of 17 characters. The name can include any combination of uppercase and lowercase letters, numbers, or special characters. The name cannot include the following characters: backslash(\), space, double-quote ("), colon (:), semicolon (;), ampersand (&), greater than (>), less than (<), or comma (,).	site-name	yes

Parameter and description	Value	“install create” command parameter	Mandatory to enter at commissioning stage
System ID The system-name is used to uniquely identify the links between sites in a network. If the system-name is not provided, it is generated based on the local site ID and the remote site ID. If the system is a protected system, the system ID is limited to seven characters (alphanumeric characters and underscore “_”), so that a 'W' or 'P' can be appended to indicate the working and protection lines. If supplied, this is used as the basis of the OSID. If the configuration is protected, a "W" or "P" is added to the end of the OSID. In this case, the system-name is limited to seven characters.	Alphanumeric string Up to seven characters for protected configurations or eight characters for unprotected configurations	system-name	no

The tables listed below include parameters for CLI commands used to manually commissioning the Line Amp configurations.

- [6500 Line Amp commissioning — for System parameters \(CLI example: system config set\)](#)
- [6500 Line Amp commissioning — for Shelf parameters \(CLI example: shelves shelf create\)](#)
- [6500 Line Amp commissioning — for OTS parameters \(CLI example: topology ots create\)](#)

Table 13 6500 Line Amp commissioning — for System parameters (CLI example: system config set)

Parameter description	Value	Command parameter	Mandatory to enter at commissioning stage
Alarm info Parameter is listable: a set of parameters can be defined (for example, CRDTYP, WVLNGTH). This parameter does not work in a cumulative manner, that is, to add/remove flag from an existing set, the desired new set must be fully specified as a list.	• DSBL (No alarm information is reported in Autonomous Outputs [AO].) • CRDTYP (Card Type is reported in AOs.) • WVLNGTH (Wavelength is reported in AOs.) • FIC (FIC is reported in AOs.) • SBNTNM (Subnet Name is reported in AOs.) • PHYSSHLF (Physical Shelf ID is reported in AOs.) • BYNMBR (Bay Number is reported in AOs.) • LABEL (Label is reported in AOs. This field is supported only on SP2 and SPTX cards.) • CLFI (CLFI is reported in AOs. This field is supported only on SP2 and SPTX cards.) • NM (NAME is reported in AOs. This field is supported only on SP2 card.)	alarm-info	no

Parameter description	Value	Command parameter	Mandatory to enter at commissioning stage
AutoRoute Sets the AutoRoute value in new channel access OTSs created in this node, unless the user provides an explicit value in the create command. When AutoRoute is disabled, cross-connects must be provisioned at all channel access sites. When AutoRoute is enabled, cross-connects are automatically derived as follows: passthrough cross-connects are created at intra-domain channel access sites and local add/drop cross-connects are created for in-service CMD Tx/Rx adjacencies.	• DISABLE Set to DISABLE for Line Amp site. • ENABLE (default)	autoroute-def	yes
Coherent Select (CS) control	Set to OFF	cs-ctrl	yes
Group Identifier For Line Amp, this refers to a physical bay number. It is stored in the shelf/baynum attribute and is also included in the TID of the system when the site-name is provided.	Integer value in the range 1 to 99	group-id	yes
Location of the site Stored in the shelf/location field.	GPS coordinate of the site (comma-separated longitude,latitude. For example, "45.34940,-75.85461".)	location	no
Minimum provisioned cooling requirements of the shelf	• LOWFLOW • HIGHFLOW (default)	min-cooling	no
NE mode typically set to SONET	• SONET • SDH • SDHJ • UNKNOWN (default)	ne-mode	yes

Parameter description	Value	Command parameter	Mandatory to enter at commissioning stage
OSC Mode State value used to control auto-provisioning of comms circuit when an OSC facility is provisioned.	• OSPF (default) OSPF circuits will be auto-created using default area 0.0.0.0. • DISABLED (To manually provision OSPF circuits using non-default areas, set to DISABLED.)	osc-mode	no
Site ID This number must be a unique ID within the comms connected network. It is used to associate all photonic equipment at the same site together into logical groupings for management interfaces.	Range is 1 to 65535	site-id	yes
Site name The site name is a human readable name common to the nodes at a site. If it is provided, it is combined with the provided group ID to provision the node name of the device also. Stored in the shelf/siteName field. If it is supplied, it is combined with the group-id to construct a TID. If it omitted, the default TID will be used.	Alphanumeric string Maximum length of 17 characters. The name can include any combination of uppercase and lowercase letters, numbers, or special characters. The name cannot include the following characters: backslash(\), space, double-quote ("), colon (:), semicolon (;), ampersand (&), greater than (>), less than (<), or comma (,).	site-name	yes

Table 14 6500 Line Amp commissioning — for Shelf parameters (CLI example: shelves shelf create)

Parameter description	Value	Command parameter	Mandatory to enter at commissioning stage
Bay number The bay number defines the bay ID where the shelf is located. It is optionally reported in AO messages.	Integer value. Range is 0 to 99.	bay-num	no
Primary Member Activation — TID Consolidation The parameter enables/disables the shelf as primary.	• enable • disable (default)	primary	no
Site ID This number must be a unique ID within the comms connected network. It is used to associate all photonic equipment at the same site together into logical groupings for management interfaces. Stored in the shelf/siteid attribute. Also used in the ots/osid field if the system-name is not provided. If a system-name is not provided, the OSID is constructed of the hex forms of the site-id and remote-site-id. If it is unprotected or a working link, the smaller of the two site IDs will come first. If it is a protection link, the smaller of the two site IDs will come last.	Range is 0 to 65535. 0 indicates not defined.	site-id	yes

Parameter description	Value	Command parameter	Mandatory to enter at commissioning stage
Name The site name is a human readable name common to the nodes at a site. If it is provided, it is combined with the provided group ID to provision the node name of the device also. Stored in the shelf/siteName field. If it is supplied, it is combined with the group-id to construct a TID. If omitted, the default TID will be used.	Alphanumeric string Maximum length of 20 characters. The name can include any combination of uppercase and lowercase letters, numbers, or special characters. The name cannot include the following characters: backslash(\), space, double-quote ("), colon (:), semicolon (;), ampersand (&), greater than (>), less than (<), or comma (,).	site-name	yes
TID Consolidation	• enable • disable (default)	tidc	no

Table 15 6500 Line Amp commissioning — for OTS parameters (CLI example: topology ots create)

Parameter description	Value	Command parameter	Mandatory to enter at commissioning stage
Configuration type	Set to AMP for Line Amp commissioning.	cfgtype	yes Set to AMP for Line AMP site.
Optical System ID	Alphanumeric string up to 7 characters for protected configurations or 8 characters for unprotected configurations	osid	yes Must match the OSIDs provisioned on the Terminal nodes.

Parameter description	Value	Command parameter	Mandatory to enter at commissioning stage
tx-path Parity must be the opposite of the tx-path for the far-end neighbor. That is, if remote tx-path is an odd number, then the local tx-path must be an even number. Typically, only 1 and 2 are used alternating across the network.	Integer value from 1 to 254.	tx-path	yes

Other customer provisioning information

If there are additional customer provisioning requirements, make sure that you have all the necessary information.

List of procedures

- [Procedure 1, "Logging into the 6500 CLI and preparing to commission the 6500 shelf"](#)
- [Procedure 2, "Commissioning 6500 Terminal nodes"](#)
- [Procedure 3, "Commissioning 6500 Line Amp nodes"](#)
- [Procedure 4, "Connecting intra-site fibers for the 6500 and performing 6500 site testing"](#)
- [Procedure 5, "Adding Waveserver Ai information for SPLI"](#)
- [Procedure 6, "Connecting the inter-site fibers for the 6500, performing additional provisioning, and testing the 6500 photonic link"](#)
- [Procedure 7, "Roadmap for 6500 ROADM configurations"](#)

Procedure 1 Logging into the 6500 CLI and preparing to commission the 6500 shelf

Use this procedure to connect the craft PC to the 6500 shelf and log into the shelf command line interface (CLI). For information on CLI command syntax, refer to *6500 CLI Reference*, 323-1851-193.

Requirements

- Review [6500 commissioning prerequisites](#).
- Have a craft PC.
- Have a terminal emulator (for example, PuTTY) installed on the craft PC.
- Have the Ethernet cable required to connect the craft PC to the 6500 shelf.
- Make sure that the shelf is equipped with a shelf processor running Release 12.1 or higher.
- Have the engineering data sheet with the information for your configuration.
- Use an account with a UPC level of 4 or higher.

Steps

- 1 Connect the craft PC to the Craft port on the 6500 shelf.
- 2 Configure your PC to obtain an IP address automatically (the 6500 shelf processor is a DHCP server and can assign an IP address).
- 3 Connect to the SP via an SSH connection (port 20002) using a terminal emulator. The default IP address of the network element is 10.0.0.1.

Note: The SSH client software may display a security warning related to the SSH client performing a validity check of the SSH server prior to the client sending authentication data to the server. This is expected when the 6500 node has been decommissioned after an SSH connection was established to the node. (The information for the SSH server on the node is reset upon node restart and no longer matches the information the client had stored during the initial SSH connection.) To clear the warning message and enable the client to connect, refer to the documentation for the SSH client.

- 4 Select the type of configuration you are commissioning.

If your configuration is a

Then log into the CLI with default credentials (ADMIN/ADMIN) and

Terminal node (listed in tables
“Install create” commands for
unprotected configurations and
“Install create” commands for
protected configurations)

perform procedure [Commissioning
6500 Terminal nodes](#).

Line Amp node

perform procedure [Commissioning
6500 Line Amp nodes](#).

—end—

Procedure 2 Commissioning 6500 Terminal nodes

Use this procedure to commission 6500 Terminal nodes (also referred to as Channel access nodes). This procedure refers to CLI commands. For more information on CLI command syntax, refer to *6500 CLI Reference*, 323-1851-193.

Requirements

Have the engineering data sheet for each site with the commissioning information specific to your configuration. See [6500 “install create” commissioning — provisioning information](#) for commissioning parameters.

The configurations can have Standard reach or Extended reach. For more information, refer to [Standard reach and Extended reach configurations](#).

Note 1: The “install create” commands cannot be used for supported configurations that include XLAs or SRAs. For these configurations, you must commission the 6500 nodes as per standard 6500 procedures and manually provision the equipment/slot sequence. Refer to *6500 Commissioning and Testing*, 323-1851-221, and Part 2 of *Configuration - Provisioning and Operating*, 323-1851-310.2.

Note 2: For the SRA, the “Forced Shut Off” parameter value for the RAMAN facility must be set to “False” in order to bring up the RAMAN pump lasers to ON. Refer to Part 1 of *6500 Configuration - Provisioning and Operating*, 323-1851-310.1, for details.

Note 3: For CMD64 configurations with MLA3/ESAM, XLA/ESAM, or XLA/SRA, the “Tx Power Reduction Control” parameter value for the LOC facility must be set to “Always Off” so the transponders/Waveserver Ai ports associated with CMD64 do not remain in reduced power state when SPLI is established. Refer to Part 1 of *6500 Configuration - Provisioning and Operating*, 323-1851-310.1, for details.

Steps

- 1 Select your configuration (see the Configuration column in [“Install create” commands for unprotected configurations](#) and [“Install create” commands for protected configurations](#)).

If your configuration is	Then
Flex-12 (See Figure 2 and Figure 3)	step 2
Flex-64 [75GHz] unprotected (See Figure 4 and Figure 5)	step 3
Flex-64 [75GHz] protected (See Figure 8)	

Commissioning a Flex-12 [50GHz] configuration

- 2 Enter the “install create” command for Flex-12 [50GHz], see [“Install create” commands for unprotected configurations](#) or [“Install create” commands for protected configurations](#).

The system sets the following parameters automatically:

- CS Control is set to ON.

Note: To check, use the following CLI command:

```
system config show
```

and verify the value of the cs-ctrl parameter.

- For both amplifiers embedded within the CCMD12:

-Target Gain is set to 23dB

-LOS thresholds are left at default values as follows:

Input LOS threshold = -29.0 dBm

Output LOS threshold = -12 dBm

Shut Off threshold = -35.0 dBm

Note: To check, use the following CLI command:

```
amps amp show amp-1-1-<25|26>
```

You have completed the commissioning of the Flex-12 [50GHz] configuration, Next perform procedure [Connecting intra-site fibers for the 6500 and performing 6500 site testing](#).

Commissioning a Fixed-64 [75GHz] configuration (protected or unprotected)

- 3 Enter the “install create” command as applicable to your configuration, see [“Install create” commands for unprotected configurations](#) and [“Install create” commands for protected configurations](#).

The system sets the following parameters automatically:

- CS Control is set to ON.

Note: To check, use the following CLI command:

```
system config show
```

and verify the value of the cs-ctrl parameter.

- If your configuration is equipped with MLA2 or MLA3, the “install create” command sets parameter values as indicated below. For protected configurations, the parameter values are set for both MLA2/MLA3 modules in the shelf.

-Line A (AMP port 8) Input LOS Threshold is left at default value of -36 dBm.

Note 1: To check, use the following CLI command:

```
amps amp show amp-1-<1|2>-8
```

-Line B (AMP port 6) Gain Offset is set to 0.0 dB (for Fixed-64 configurations) and Input LOS Threshold is set to -28 dBm.

Note 2: To check, use the following CLI command:

```
amps amp show amp-1-<1|2>-6
```

- If your configuration is equipped with SLA, the “install create” command sets parameter values as indicated below. For protected configurations, the parameter values are set for both SLA modules in the shelf.

-Line A (AMP port 8) Input LOS Threshold is set to -35 dBm.

Note 1: To check, use the following CLI command:

```
amps amp show amp-1-<1|2>-8
```

-Line B (OPTMON port 6) LOS Threshold is set to -28 dBm.

Note 2: To check, use the following CLI command:

```
optmons optmon show optmon-1-<1|2>-6
```

- SLA/MLA2/MLA3 Line B (ADJ-LINE port 5) parameters are set as follows. For protected configurations, the parameter values are set for both SLA/MLA2/MLA3 modules in the shelf.

- fiber-type: NDSF. If your deployed fiber type is not NDSF, you can edit the fiber-type parameter in [step 4](#).

-min-span-loss for SLA: 0 dB

-min-span-loss for MLA2/MLA3: 16 dB

-span-loss-margin: 3 dB

-target-span-loss: 21.5 dB (see [Note 2](#))

Note 1: To check, use the following CLI command:

```
adjacencies line show adj-1-<1|2>-5
```

Note 2: In procedure [Connecting the inter-site fibers for the 6500, performing additional provisioning, and testing the 6500 photonic link](#), after you connect the far-end node, you will set the target span loss of SLA/MLA2/MLA3 Line B (ADJ-LINE port 5) to the expected span loss value.

- 4 If your deployed fiber type is not NDSF, edit the fiber-type parameter (of SLA/MLA2/MLA3 Line B [ADJ-LINE port 5]) to match the actual fiber type deployed, using the following CLI command:

```
adjacencies line set adj-1-<1|2>-5 fiber-type <fiber-type Value>
```

- For protected configurations, edit the fiber-type parameter for both line interface modules (that is, both SLA/MLA2/MLA3 modules in the shelf).

- 5 Select your next step. For more information on Standard and Extended reach, see [Standard reach and Extended reach configurations](#).

If your configuration is being provisioned	Then
for Standard reach	you have completed this procedure. Next perform procedure Connecting intra-site fibers for the 6500 and performing 6500 site testing. Note: If your configuration is protected, additional manual provisioning is required to complete setting up the protected Terminal nodes with Standard reach. After the Waveserver Ai wavelengths are connected to the 6500 Photonic system (in the Waveserver Ai-6500 interworking section of this technical publication), the procedures will direct you to the required steps.
for Extended reach	go to step 6 to perform additional provisioning for Extended reach configurations

Performing additional provisioning at Terminal nodes for Extended reach configurations

- 6 Set CS control to OFF by entering:

```
system config set cs-ctrl off
```
- 7 Set the following parameters for Line A (port 8)
 - Target Gain Tilt to -3.0 dB
 and
 - Target Gain to 27.0 dB
 by entering

```
amps amp set amp-1-<1|2>-8 targ-gain-tilt -3 targ-gain 27
```

 - For protected configurations, set the Line A (port 8) Target Gain Tilt and Target Gain for both line interface modules (that is, both MLA2/MLA3 modules in the shelf).
- 8 Set Line B (port 6) Target Gain Tilt to -2.5 dB by entering:

```
amps amp set amp-1-<1|2>-6 targ-gain-tilt -2.5
```

 - For protected configurations, set the Line B (port 6) Target Gain Tilt for both line interface modules (that is, both MLA2/MLA3 modules in the shelf).

You have completed this procedure.

Next perform procedure [Connecting intra-site fibers for the 6500 and performing 6500 site testing](#).

Note: Additional manual provisioning is required to complete setting up the Terminal nodes with Extended reach (protected or unprotected). After the Waveserver Ai wavelengths are connected to the 6500 Photonic system (in the Waveserver Ai and 6500 interworking section of this technical publication), the procedures will direct you to the required steps.

—end—

Procedure 3 Commissioning 6500 Line Amp nodes

Use this procedure to commission 6500 Line Amp nodes. This procedure refers to CLI commands. For more information on CLI command syntax, refer to *6500 CLI Reference*, 323-1851-193.

Requirements

Have the engineering data sheet with the commissioning information specific to your configuration.

For Line Amp commissioning parameters see tables:

- [6500 Line Amp commissioning — for System parameters \(CLI example: system config set\)](#)
- [6500 Line Amp commissioning — for Shelf parameters \(CLI example: shelves shelf create\)](#)
- [6500 Line Amp commissioning — for OTS parameters \(CLI example: topology ots create\)](#)

Examples of Line Amp configurations are illustrated in [Example of Line Amp node - Standard reach](#) and [Example of Line Amp node - Extended reach](#).

Steps

- 1 Enter the following CLI commands to commission the shelf:

```
system sid set name <Site name>-<group-id>
shelves shelf create shelf-1 site-id <Site ID> site-name
<Site name> bay-num <group-id> [location <Location of the
site>]
system config set ne-mode sonet autoroute-def disable cs-
ctrl <Value> [alarm-info <alarm-info Value>]
```

where cs-ctrl Value is

 - on, if your configuration includes MLAx+LIM
 - off, if your configuration includes MLAx+MLAx

where MLAx is MLA2 or MLA3

Provisioning comms

- 2 Enter the following CLI commands to provision comms:
- ```
comms ip create shelf-1 ip-addr <COLAN ipv4 address>
netmask <Value>

comms ospf router create shelf-1 router-id <COLAN ipv4
address>

comms ospf circuit create shelf-1 net-area <n.n.n.n>

comms ospf circuit create osc-1-<slot>-1 net-area
<n.n.n.n>

comms ospf circuit create osc-1-<slot>-2 net-area
<n.n.n.n>
```

**Note:** <n.n.n.n> is used on the Terminal nodes, where n is the first octet of the IP address.

**Confirming that the equipment is provisioned**

- 3 Confirm the equipment is provisioned.
- ```
equipment config show ALL
```

Configuring the line-facing modules

- 4 Select your next step.

Note: In procedure [Connecting the inter-site fibers for the 6500, performing additional provisioning, and testing the 6500 photonic link](#), after you connect the far-end node, you will set the target span loss of MLAx Line B (ADJ-LINE port 5) to the expected span loss value.

If your Line Amp node includes	Then go to
MLAx+MLAx (see Figure 10)	step 5
MLAx+LIM (see Figure 9)	step 9

Setting amplifies parameters for MLA3+MLA3 configurations

- 5 Enter the following CLI commands to configure the amplifies in the shelf:
- ```
amps amp set amp-1-1-6 input-los-thres -28
amps amp set amp-1-2-6 input-los-thres -28
```
- 6 Provision the pad loss for 10 dB pads at midstage:
- ```
adjacencies fiber set ADJ-1-<1|2>-7 pad-loss 10
```
- 7 Set Line A (AMP port 8) Target Gain Tilt to 0 dB and Target Gain to 23.5 dB on both MLAx:
- ```
amps amp set amp-1-<1|2>-8 targ-gain-tilt 0 targ-gain
23.5
```
- 8 Set Line B (AMP port 6) Target Gain Tilt to -2.5 dB on both MLAx:
- ```
amps amp set amp-1-<1|2>-6 targ-gain-tilt -2.5
```

Go to [step 10](#) for OTS provisioning.

Note: After the Waveserver Ai wavelengths are connected to the 6500 Photonic system (in the Waveserver Ai and 6500 interworking section of this technical publication), you will set the Line B (AMP port 6) Target Gain on both MLAx. The procedures will direct you to the required step.

Setting line-facing module parameters for MLA3+LIM configurations

9 Enter the following CLI commands:

```
amps amp set amp-1-<MLAx slot number>-6 input-los-thres -
8
```

```
optmons optmon set optmon-1-<LIM slot number>-6 los-thres
-20
```

OTS provisioning

10 Enter the following CLI commands:

Note that:

- At the Line Amp site, the line amp in slot 1 should be facing the remote site with the lowest site-id, and the line amp in slot 2 should be facing the remote site with the highest site-id.
- The Optical System ID (OSID) must match the OSIDs of the Terminal nodes. At Terminal nodes, OSID is provisioned/queried as system-name (see [Table 6500 “install create” commissioning — provisioning information](#). For more information on OSID at Terminal nodes, also see [site-id and remote-site-id](#).)

```
topology ots create ots-1-1 osid <Value> tx-path
<parity_value> cfgtype amp amp-mate ots-1-2 osc osc-1-15-
<1|2> lim lim-1-1
```

```
topology ots set ots-1-2 osc osc-1-15-<1|2>
```

```
topology ots set ots-1-2 lim lim-1-2
```

11 Verify the OTS has been created:

```
topology ots show ALL
```

You have completed the commissioning of the Line Amp node.

Next perform procedure [Connecting intra-site fibers for the 6500 and performing 6500 site testing](#).

—end—

Procedure 4 Connecting intra-site fibers for the 6500 and performing 6500 site testing

Use this procedure to perform intra-site fiber connections for 6500 equipment and to perform intra-site testing for the 6500. For information on the equipment supported in configurations, refer to *6500 Photonics Equipment*, 323-1851-102.6.

Requirements

- Ensure that you have the engineering data sheet for your node.
- Ensure that cables are connected between the access panel and subtending modules (CMD).
- Ensure that the fibers have been cleaned and verified to be in working condition, and the loss between fibers is not greater than 0.5 dB. It is very important that fibers are cleaned and scoped prior to being connected. Refer to the cleaning instructions in *6500 Installation - General Information*, 323-1851-201.0.
- Ensure you have access to *6500 Commissioning and Testing*, 323-1851-221.



CAUTION

Risk of damage to equipment

Electrostatic discharge (ESD) can damage electrostatic-sensitive devices. Always connect yourself to ground before you handle an electrostatic-sensitive device.



CAUTION

Risk of injury

Observe the safety guidelines in *6500 Installation-General Information*, 323-1851-201.0, and those recommended by your company at all times.

Steps

- 1 Review the following figures and engineering data sheet to identify the example for your node configuration type:
 - Flex-12 50 GHz unprotected - [Example of CCMD12 50GHz unprotected Channel access node \(also known as Flex-12 \[50GHz\] configuration—unprotected\)](#)
 - Flex-12 50 GHz unprotected with ESAM - [Example of CCMD12 50GHz unprotected \(with ESAM\) Channel access node \(also known as Flex-12 \[50GHz\] configuration—unprotected\)](#)
 - Fixed-64 75 GHz unprotected - [Example of CMD64 75GHz unprotected Channel access node \(also known as Fixed-64 \[75GHz\] configuration—unprotected\)](#)
 - Fixed-64 75 GHz unprotected with ESAM - [Example of CMD64 75GHz unprotected \(with ESAM\) Channel access node \(also known as Fixed-64 \[75GHz\] configuration—unprotected\)](#)
 - Fixed-64 75 GHz unprotected with XLA/SRA - [Example of CMD64 75GHz unprotected \(with XLA and SRA\) Channel access node \(also known as Fixed-64 \[75GHz\] configuration—unprotected\)](#)
 - Fixed-64 75 GHz unprotected with XLA/ESAM - [Example of CMD64 75GHz unprotected \(with XLA and ESAM\) Channel access node \(also known as Fixed-64 \[75GHz\] configuration—unprotected\)](#)
 - Fixed-64 75 GHz protected - [Example of CMD64 75GHz protected Channel access node \(also known as Fixed-64 \[75GHz\] configuration—protected\)](#)
- 2 Connect the 6500 equipment as illustrated in the respective example.
- 3 Verify that there are no unexpected alarms. To clear alarms, see 6500 *Fault Management - Alarm Clearing*, Parts 1 and 2. You have completed the connecting of the intra-site fibers for the 6500 at this site.
Next perform procedure [Adding Waveserver Ai information for SPLI](#).
—end—

Figure 2 Example of CCMD12 50GHz unprotected Channel access node (also known as Flex-12 [50GHz] configuration—unprotected)

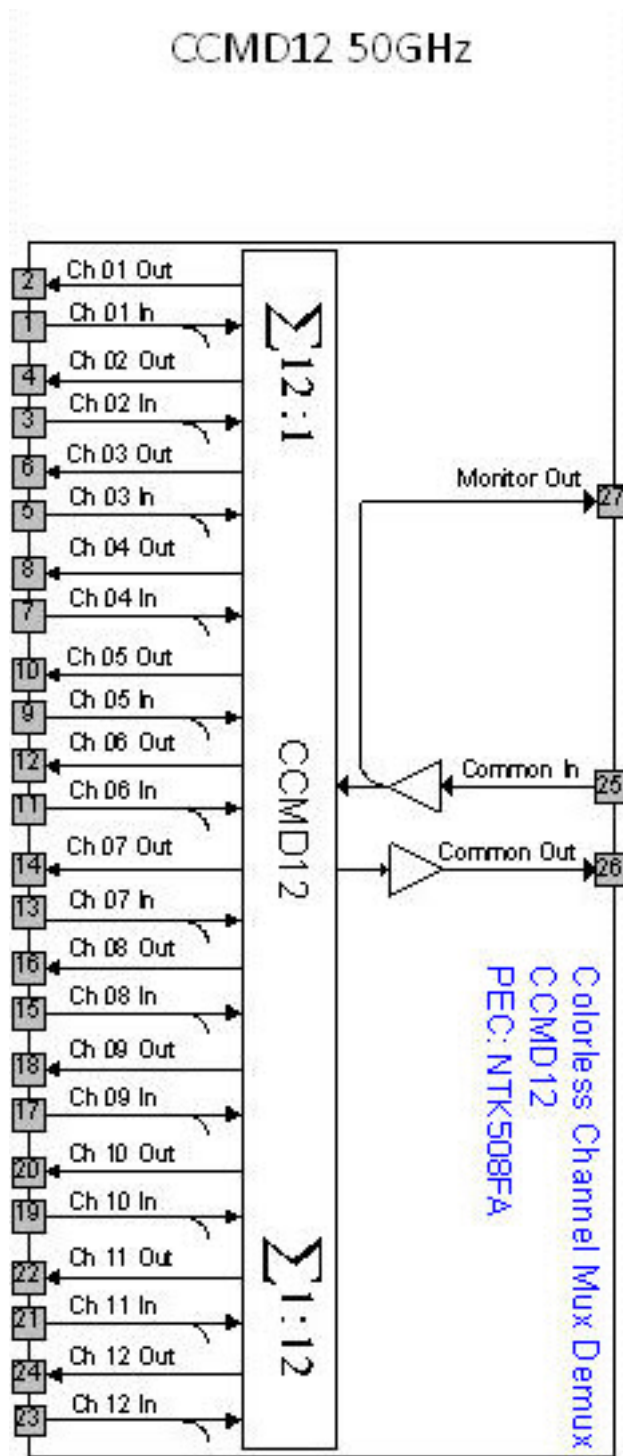


Figure 3 Example of CCMD12 50GHz unprotected (with ESAM) Channel access node (also known as Flex-12 [50GHz] configuration—unprotected)

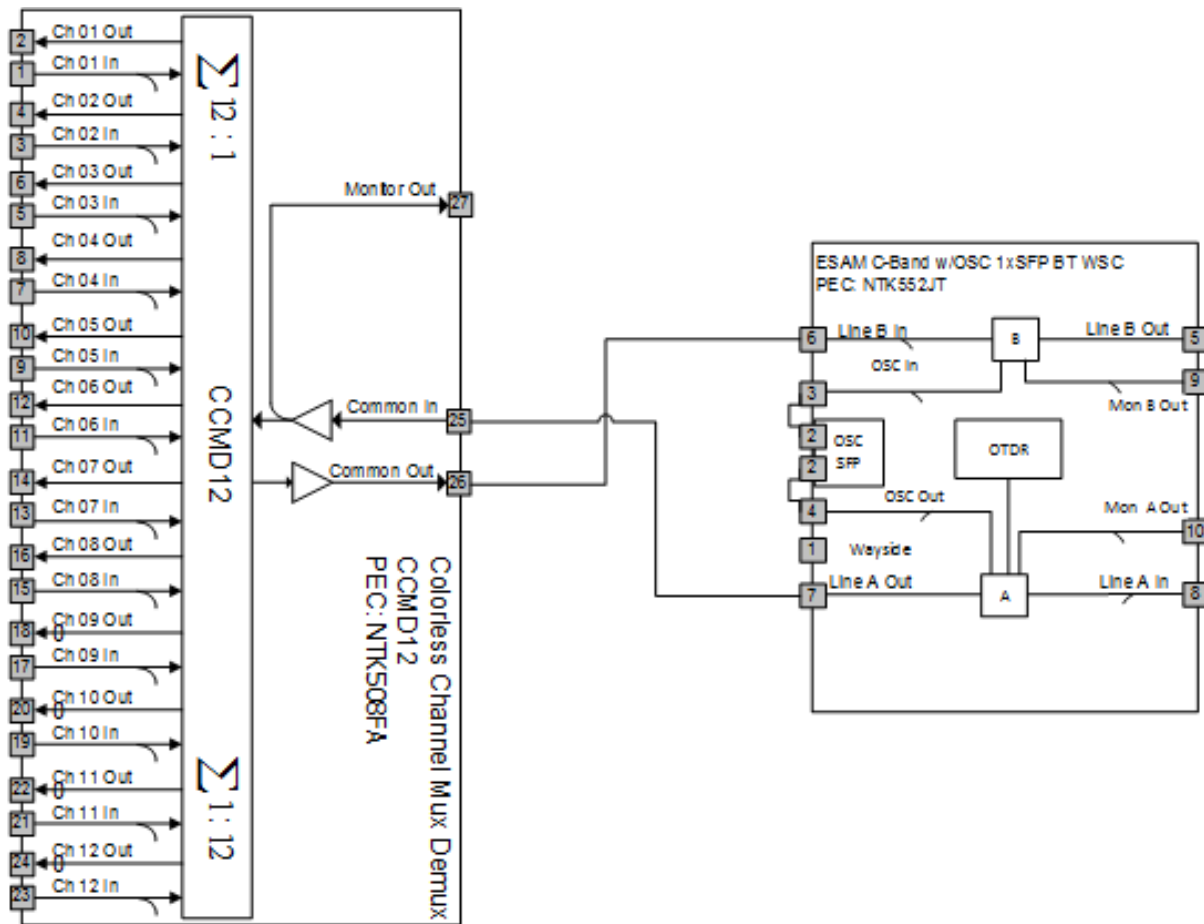


Figure 4 Example of CMD64 75GHz unprotected Channel access node (also known as Fixed-64 [75GHz] configuration—unprotected)

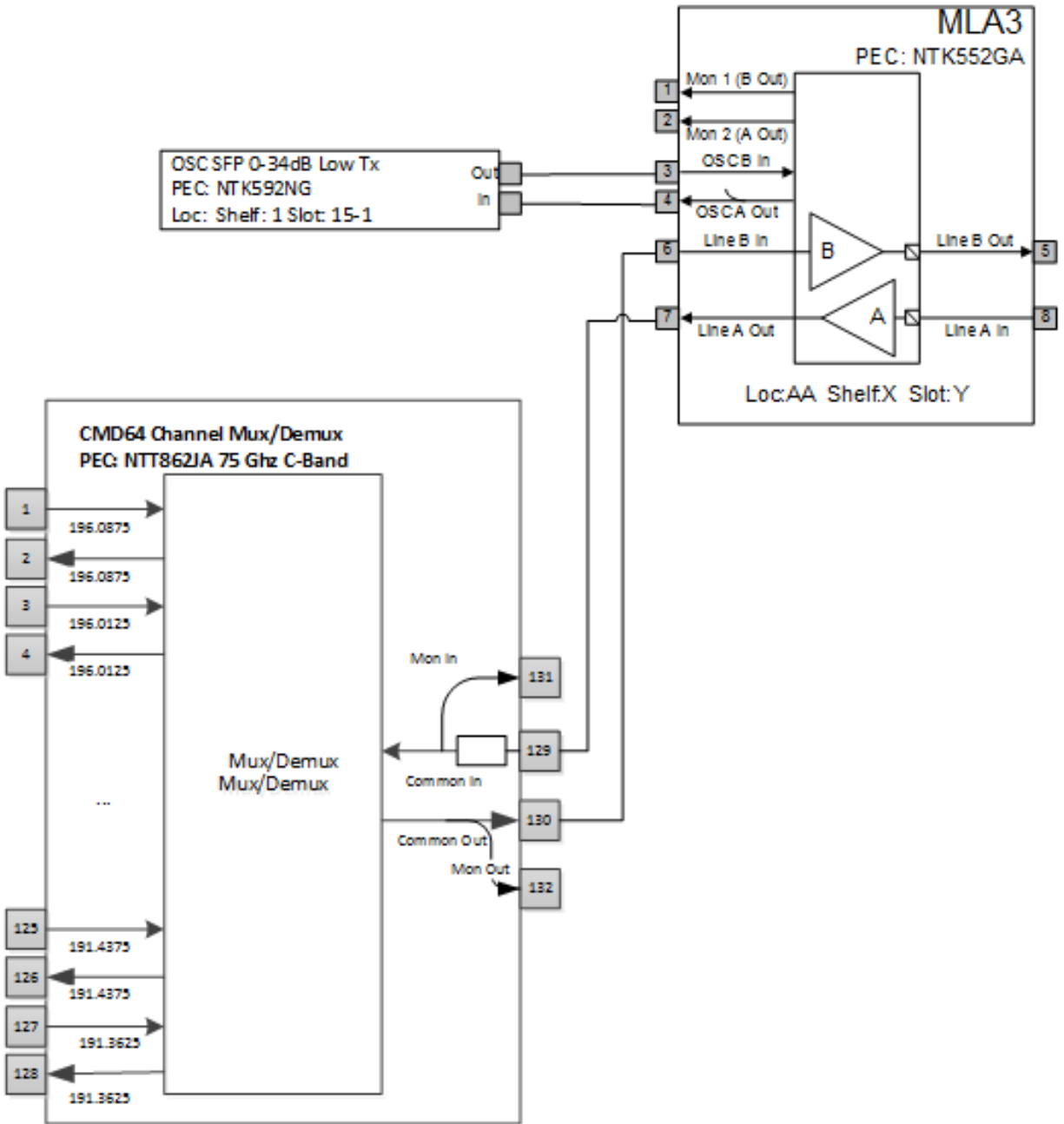


Figure 5 Example of CMD64 75GHz unprotected (with ESAM) Channel access node (also known as Fixed-64 [75GHz] configuration—unprotected)

Figure 6 Example of CMD64 75GHz unprotected (with XLA and SRA) Channel access node (also known as Fixed-64 [75GHz] configuration—unprotected)

Figure 7 Example of CMD64 75GHz unprotected (with XLA and ESAM) Channel access node (also known as Fixed-64 [75GHz] configuration—unprotected)

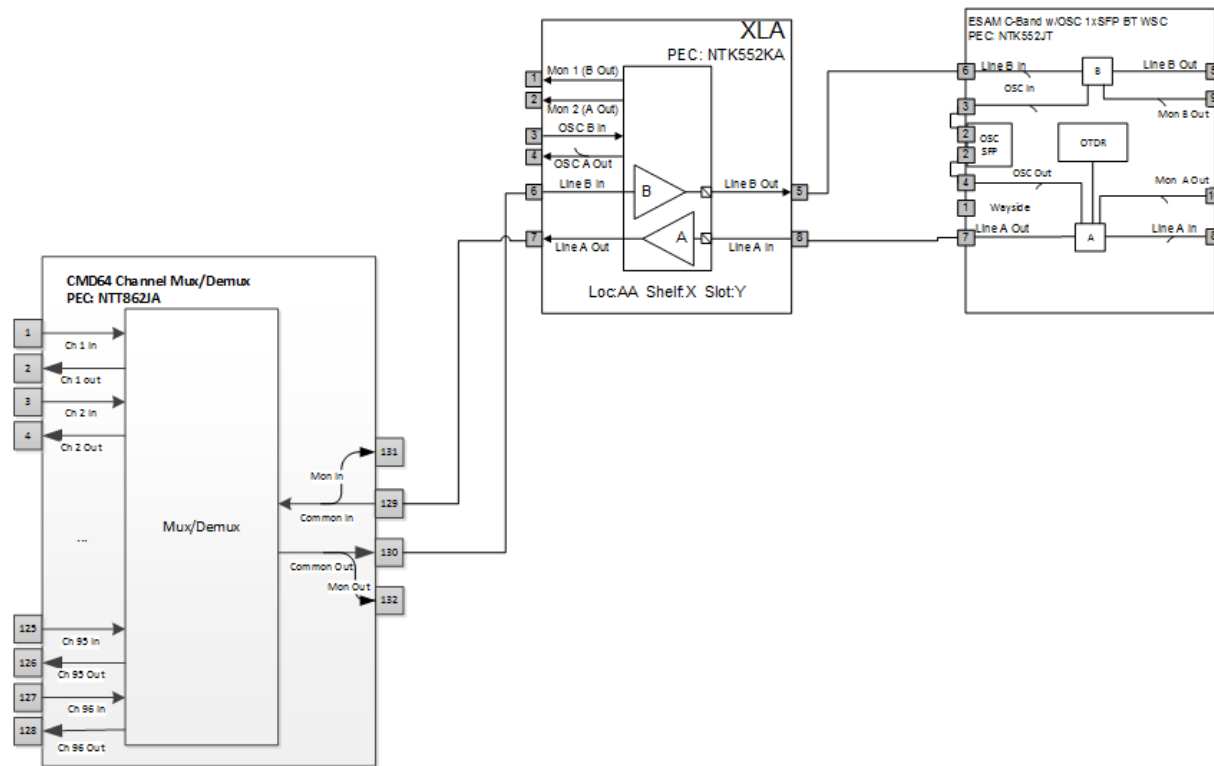


Figure 8 Example of CMD64 75GHz protected Channel access node (also known as Fixed-64 [75GHz] configuration—protected)

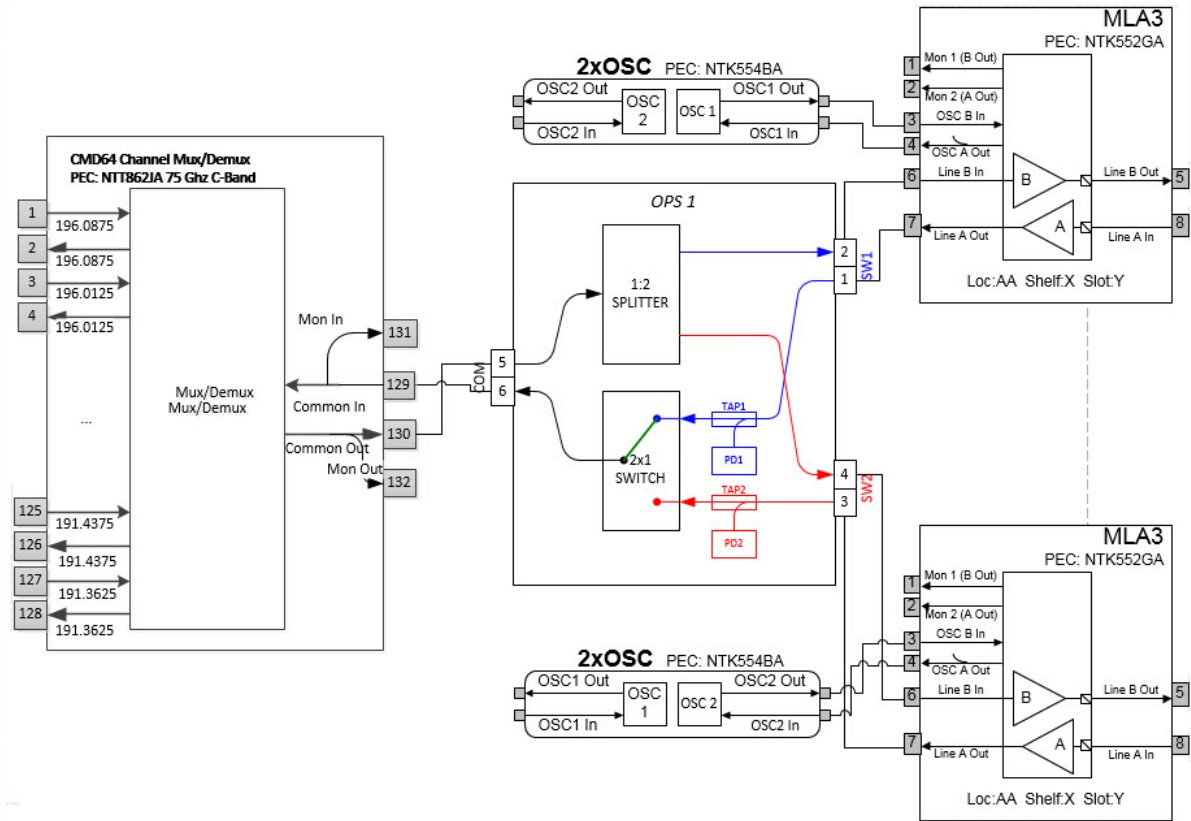
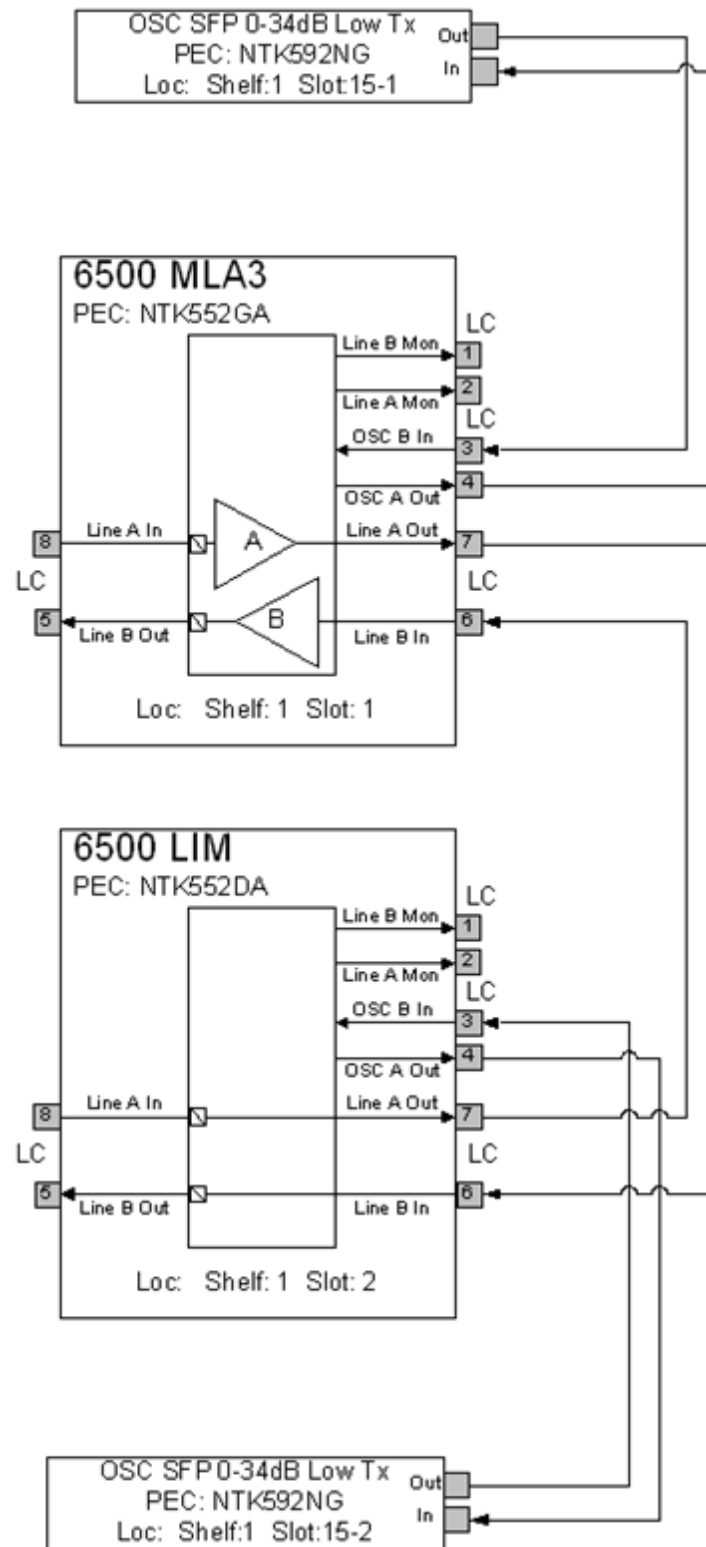
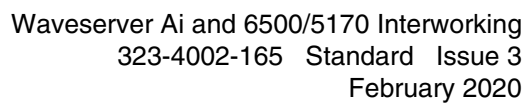


Figure 9 Example of Line Amp node - Standard reach



Procedure 5 Adding Waveserver Ai information for SPLI

Use this procedure to add a Waveserver Ai node name to the 6500 SPLI list and provision the Waveserver Ai line port as Far End Address in 6500's ADJ facility.

Requirements

- Have the IPv4 address(es) of the Waveserver Ai node(s).
- If you are using Site Manager, have Site Manager software installed on your craft computer for this 6500 software load release.
- Use an account with a UPC level of 4 or higher.
- On the Waveserver Ai, SPLI must be enabled globally and on the PTP connecting to the 6500. For more information, see SPLI with 6500 in *Waveserver Ai User Guide*, 323-4002-100.
- For more information on Waveserver Ai and SPLI information sharing, see [Waveserver Ai \(Release 1.2 and higher\) and 6500 \(Release 12.3\) SPLI information sharing](#).
- This procedure refers to CLI commands. For more information on CLI command syntax, refer to *6500 CLI Reference*, 323-1851-193.

Note: The Waveserver Ai node name will be added on the 6500 NE to which the Waveserver Ai line port is connected. That is, in a TIDc configuration, if the Waveserver Ai node is connected to the CMD on the member shelf, the Waveserver Ai node name will be added to the member shelf instead of the primary shelf.

Steps

- 1 Log into the 6500 NE using Site Manager or use the CLI, as applicable.
- 2 Add the IPv4 address(es) of the Waveserver Ai node name(s) to the SPLI list of the 6500 node as follows:
 - If you are using Site Manager, refer to the procedure on adding SPLI entries in *6500 Administration and Security*, 323-1851-301.
 - If you are using the CLI, enter:


```
adjacencies spli create SHELF-<shelf number> fend-ip-addr
<FEIPADD> spli-id <TID>
```

 where
 -FEIPADD is the IPv4 address of the Waveserver Ai node being added to the 6500 SPLI list. Surround the IP address with double-quotes.

-TID is a string defined by the user to represent the Waveserver Ai node. Surround the string with double-quotes. It is recommended that the Waveserver Ai's member-name string be used as TID.

Note: DHCP is not recommended to be used on Waveserver Ai when SPLI is used as the Waveserver Ai could get assigned a different IPv4 address through DHCP. When that occurs, the user would have to return to this step to manually update the IPv4 address provisioned in the 6500 SPLI table.

3 Provision the Waveserver Ai line port as Far End Address in 6500's ADJ facility as follows.

- If you are using Site Manager, refer to *6500 Commissioning and Testing*, 323-1851-221, and *Configuration - Provisioning and Operating Part 1*, 323-1851-310 for more information.

Note: Refer to the following for information on the supported Waveserver Ai far end addresses that can be entered in Site Manager far end address builder dialog.

- If you are using the CLI, enter:

```
adjacencies adjacency set <AID> prov-addr-form <prov-addr-form> prov-fe-addr <prov-fe-addr>
```

where

-<AID> is the CMD adjacency in the form ADJ-SH-SL-PORT for the CMD port connected to the Waveserver Ai.

-<prov-addr-form> defines the format of the far end address. For example, 'NODENAME-SL-PRT' for Waveserver Ai node name-Slot-Port.

-<prov-fe-addr> is a string defined by the user to represent the far end address of the Waveserver Ai in the form TID-Slot-Port.

For Waveserver Ai, the Slot can be '1', '2', or '3' and the Port is either '1' or '2' (that is, slot-port is '1-1', '1-2', '2-1', '2-2', '3-1', or '3-2'), with the exception if you are using the Waveserver Ai 2x400G 35/56GBaud EDFA C-Band w/OPS 8xQSFP28 module (186-1607-900) with SPLI.

For Waveserver Ai (Release 1.4 or higher) and 6500 (Release 12.4 or higher), when using the Waveserver Ai 2x400G 35/56GBaud EDFA C-Band w/OPS 8xQSFP28 module (186-1607-900) with SPLI, the Slot can be '1', '2', or '3' and the Ports are '1' and '11' or '2' and '22'. See the following table for the Waveserver Ai PTP port -> Waveserver Ai SPLI CLI port -> 6500 port far end address mapping.

Table 16 Waveserver Ai PTP port -> Waveserver Ai SPLI CLI port -> 6500 port far end address mapping

Waveserver Ai PTP port	Waveserver Ai SPLI CLI port	6500 port
1	1W	1
1	1P	11
2	2W	2
2	2P	22

If following recommendations, the TID is the Waveserver Ai's member-name string. Surround the string with double-quotes.

For example, if connecting slot 3 port 2 of a Waveserver Ai with a TID of 'WSTEST' to port 26 of the 6500 CMD in shelf 2 slot 83, you would enter:

```
adjacencies adjacency set ADJ-3-83-26 prov-addr-form  
NODENAME-SL-PRT prov-fe-addr "WSTEST-3-2"
```

- 4 If you have more sites to commission, proceed to the next 6500 site, where you will start with procedure [Logging into the 6500 CLI and preparing to commission the 6500 shelf](#).

If you have finished performing all the tasks up to this procedure on all the 6500 sites, go to procedure [Connecting the inter-site fibers for the 6500, performing additional provisioning, and testing the 6500 photonic link](#).

—end—

Procedure 6 Connecting the inter-site fibers for the 6500, performing additional provisioning, and testing the 6500 photonic link

Use this procedure to connect the inter-site fibers for the 6500 and test the 6500 photonic link.

Steps

- 1 Connect the inter-site fibers between the local 6500 site and the remote 6500 site.
- 2 Wait for five minutes and make sure there are no alarms indicating that the line ports were not connected correctly. DOC invalid photonic domain alarms should clear.

Provisioning the target span loss for Terminal nodes

- 3 For Terminal nodes, after you connect the remote node, you can set the target span loss of SLA/MLA2/MLA3 Line B (ADJ-LINE port 5) to the expected span loss value using the following CLI command:

```
adjacencies line set adj-1-<1|2>-5 targ-span-loss
<targ-span-loss Value>
```

-For protected configurations, set the target span loss value for both line interface modules (that is, both SLA/MLA2/MLA3 modules in the shelf).

-For spans using MLA2/MLA3, if the actual span loss is 16 dB, set the Target Span Loss to 16.1 dB (because it must exceed Minimum Span Loss = 16 dB).

View the reported span loss using the following CLI command:

```
adjacencies line show adj-1-<1|2>-5
```

and verify the value of the osc-span-loss parameter.

Although the value is reported against port 5, the reported span loss is in the direction from the remote node to this node.

Provisioning the target span loss of Line Amp nodes

- 4 For Line Amp nodes (equipped with MLAx-MLAx or MLAx-LIM), after you connect the remote nodes, you can set the target span loss of MLAx Line B (ADJ-LINE port 5) to the expected span loss value using the following CLI command:

```
adjacencies line set adj-1-<1|2>-5 targ-span-loss
<targ-span-loss Value>
```

If the actual span loss is 16 dB, set the Target Span Loss to 16.1 dB (because it must exceed Minimum Span Loss = 16 dB).

View the reported span loss using the following CLI command:

```
adjacencies line show adj-1-<1|2>-5
```

and verify the value of the osc-span-loss parameter.

Although the value is reported against port 5, the reported span loss is in the direction from the remote node to this node.

Provisioning the fiber type for Line Amp nodes

- 5 For Line Amp nodes, edit the fiber-type parameter of both line facing circuit packs to match the actual fiber type deployed, using the following CLI command:

```
adjacencies line set adj-1-<1|2>-5 fiber-type <fiber-type Value>
```

Testing the 6500 photonic link

- 6 Verify discovered far end addresses and span loss OSC using the following commands.

- For Fixed-64 configurations, enter:

```
adjacencies adjacency show adj-1-1-5
```

```
adjacencies line show adj-1-1-5
```

- In addition, for protected configurations, enter:

```
adjacencies adjacency show adj-1-2-5
```

```
adjacencies line show adj-1-2-5
```

Note: Additional testing (pm amp counts) will be performed after the Waveserver Ai wavelengths are connected to the 6500 Photonic system (in the Waveserver Ai and 6500 interworking section of this technical publication). The procedures will direct you to the required steps.

- 7 Verify that there are no unexpected alarms. To clear alarms, see *6500 Fault Management - Alarm Clearing, Parts 1 and 2*, 323-1851-543. Note that Shutoff Threshold Crossed alarms raised against all the amplifiers are expected (because the Waveserver Ai equipment is not connected yet).

Then go to the Waveserver Ai and 6500 interworking section of this technical publication.

—end—

Procedure 7 Roadmap for 6500 ROADM configurations

Use this roadmap to commission the 6500 shelf at the ROADM site and SLAT the 6500 ROADM site.

Requirements

- Ensure you have access to the following technical publications
 - *6500 Commissioning and Testing*, 323-1851-221
 - *6500 Administration and Security*, 323-1851-301
- Be familiar with the 6500 SLAT process and prerequisites, including provisioning parameters for your 6500 mode turning-up solution. Refer to SLAT process overview in *6500 Commissioning and Testing*, 323-1851-221.

Steps

- 1 To commission the 6500 shelf at the ROADM site and SLAT the 6500 ROADM site:
 - Commission the 6500 shelf. Refer to Nodal SLAT procedures in *6500 Commissioning and Testing*, 323-1851-221.
 - Perform photonic layer provisioning procedures. Refer to Photonic layer provisioning procedures in *6500 Commissioning and Testing*, 323-1851-221.
 - If your configuration is Low channel count DIA ROADM, on the WSS connected to the CCMD12, set the unused OPM monitor port 2 to the automatic in-service (AINS) state.
 - Add Waveserver Ai node name(s) to the SPLI list. Refer to *6500 Administration and Security*, 323-1851-301
 - Connect the intra-site fibers for your 6500 ROADM site.

Refer to: illustrations for examples of the following supported 6500 ROADM configuration types:

 - CDA using WSS Flex C-Band w/OPM 20x1
 - For illustrations of Coherent Select configuration examples, refer to Photonic layer site testing procedures in *6500 Commissioning and Testing*, 323-1851-221.

-The procedure on connecting intra-site fibers for your 6500 ROADM configuration. Refer to Photonic layer site testing procedures in *6500 Commissioning and Testing*, 323-1851-221.

-
- Perform the photonic layer site testing procedure for your 6500 ROADM configuration. Refer to Photonic layer site testing procedures in *6500 Commissioning and Testing*, 323-1851-221.
- Perform the procedure on connecting inter-NE fibers and testing the photonic layer for 6500. Refer to *Photonic layer site testing procedures in 6500 Commissioning and Testing*, 323-1851-221.
- After you complete setting up your 6500 ROADM configuration, proceed to the Waveserver Ai and 6500 interworking section of this technical publication.

—end—

CHAPTER 5

Waveserver Ai and 6500 interworking

This section includes the common configurations between Waveserver Ai and 6500.

Topics include:

- [Waveserver Ai and 6500 interworking summary](#)
- [Photonics configurations](#)
- [Supported 6500 photonic configurations](#)
- [Comms channel for 6500-based SLTE configurations](#)
- [Managing Waveserver Ais through 6500](#)
- [Waveserver Ai \(Release 1.2 and higher\) and 6500 \(Release 12.3\) SPLI information sharing](#)
- [Waveserver Ai \(Release 1.3 and lower\) and 6500 \(Release 12.4\) SPLI interworking considerations](#)
- [Waveserver Ai Release 1.3 \(and higher\) and 6500 Release 12.3 interworking](#)
- [Waveserver Ai Release 1.4 and 6500 Release 12.4 interworking](#)
- [Waveserver Ai and 6500 WLAi MOTR interworking considerations](#)

Note 1: For information on Waveserver Ai regen applications, see *Waveserver Ai User Guide*, 323-4002-100.

Note 2: For information on configuring Waveserver Ai, refer to *Waveserver Ai User Guide*, 323-4002-100 and for information on 6500 refer to the 6500 Packet-Optical Platform (6500) documentation library.

Waveserver Ai and 6500 interworking summary

The Waveserver Ai connection to the 6500 CMD is through its two line-side interfaces 1 and 2. OTL4.4 client interworking is also supported on Waveserver Ai and some 6500 traffic cards.

Photonics configurations

The Waveserver Ai interworks with the 6500 Photonic line through the following configurations:

Note: For information on other supported Waveserver Ai and 6500 interworking configurations, see [Supported 6500 photonic configurations](#).

- [Fixed-64 \(75GHz\) configuration](#)
- [Flex-12 configuration](#)
- [CDA using WSS Flex C-Band w/OPM 20x1 and CCMD12](#)

For interworking, the line protection plan used on the Waveserver Ai and the 6500 Photonic Layer must match.

Waveserver Ai adjacency types are supported by the 6500 Service and Photonic Layer Interoperability (SPLI) feature. Provisioning the far-end address on the adjacency facility allows the Photonic line to discover the proper type of transmitter and autoprovision basic configuration information, such as the wavelength and nominal power.

For information on adding Waveserver Ai node names to the SPLI list, refer to the “6500 commissioning for Waveserver Ai” chapter in this technical publication.

Note 1: All nodes in a Fixed-64 or Flex-12 configuration must run the same software release. The system does not support a mix of software releases.

Note 2: For CCMD12 thin terminal configurations, frequency assignments are manual and no additional validation is done. Users should not reuse already used wavelengths on the ports while using the SPLI application for frequency assignments.

Fixed-64 (75GHz) configuration

The fixed-64 (75GHz) configuration consists of the following Photonics elements:

- CMD64 75 GHz C-Band module
- MLA3, MLA2, MLA, or SLA C-Band amplifier
- 1+1 Trunk/OPS protected configurations
- ESAM circuit pack, for OTDR configurations
- SRA and XLA circuit packs, for SRA-based configurations

The fixed-64 (75GHz) configuration supports protected and unprotected configurations:

- Unprotected configurations are supported in the 6500 2-slot optical Type 2 shelf (NTK503LA variant) and the 6500 7-slot optical Type 2 shelf (NTK503KA variant) equipped with the SPAP-2 w/2xOSC (NTK555NA/NB) shelf processor.
- Protected configurations are supported in the 7-slot optical Type 2 shelf (NTK503KA variant) equipped with the SPAP-2 w/2xOSC (NTK555NA/NB) shelf processor.

Note: SRA-based configurations are unprotected. In addition, the “install create” command used for commissioning cannot be used for SRA-based configurations. The network element must be provisioned manually.

The following tables summarize the hardware components used in the fixed-64 (75GHz) configuration and provide a reference for more information.

- [Fixed-64 \(75GHz\) hardware components - shelves, shelf processor, shelf processor pluggables, mux/demux](#)
- [Fixed-64 \(75GHz\) hardware components - for configurations without OTDR](#)
- [Fixed-64 \(75GHz\) hardware components - for configurations with OTDR](#)
- [Fixed-64 \(75GHz\) hardware components - for SRA-based configurations](#)

For a complete list of 6500 ordering codes, see *6500 Ordering Information*, 323-1851-151.

Table 17 Fixed-64 (75GHz) hardware components - shelves, shelf processor, shelf processor pluggables, mux/demux

Function	Name	PEC	Supported slots 2-slot shelf	Supported slots 7-slot shelf	Reference document
Shelves	2-slot optical Type 2	NTK503LA	N/A	N/A	<i>6500 Common Equipment</i> , 323-1851-102.1
	7-slot optical Type 2 (required for protected configurations)	NTK503KA	N/A	N/A	<i>6500 Common Equipment</i> , 323-1851-102.1
Shelf processor	SPAP-2 w/2xOSC	NTK555NA NTK555NB	15	15	<i>6500 Common Equipment</i> , 323-1851-102.1

Function	Name	PEC	Supported slots 2-slot shelf	Supported slots 7-slot shelf	Reference document
Shelf processor pluggables	OC-3/STM-1 CWDM 1511nm (0-15 dB span)	NTK592NP E6	15-1	15-1 15-2	6500 Common Equipment, 323-1851-102.1
	OC-3/STM-1 CWDM 1511nm (10-30 dB span)	NTK592NB E6			
	OC-3/STM-1 CWDM 1511nm (20-34 dB span)	NTK592NH E6			
	OC-3/STM-1 CWDM 1511nm (0-34 dB span)	NTK592NG E5			
	OC-3/STM-1 CWDM 1511nm (12-42 dB span)	NTK592NV E5			
Mux/Demux	CMD64 100 GHz C-Band module	NTT862JA	83 (external slot)	83 (external slot)	6500 Photonics Equipment, 323-1851-102.6

Table 18 Fixed-64 (75GHz) hardware components - for configurations without OTDR

Function	Name	PEC	Supported slots 2-slot shelf	Supported slots 7-slot shelf	Reference document
Amplifier	Midstage Line Amplifier 3 (MLA3 C-Band)	NTK552GAE5	1	1 (unprotected) 1 and 2 (protected)	6500 Photonics Equipment, 323-1851-102.6
	Midstage Line Amplifier 2 (MLA2 C-Band)	NTK552FAE5			
	Single Line Amplifier (SLA C-Band)	NTK552AAE5			
Protection	4xOptical Protection Switch (OPS)	NTK554TA	N/A	3	6500 Common Equipment, 323-1851-102.1

Table 19 Fixed-64 (75GHz) hardware components - for configurations with OTDR

Function	Name	PEC	Supported slots 2-slot shelf	Supported slots 7-slot shelf	Reference document
Amplifier	Midstage Line Amplifier 3 (MLA3 C-Band)	NTK552GAE5	2	2 (unprotected) 2 and 4 (protected)	6500 Photonics Equipment, 323-1851-102.6
Protection	4xOptical Protection Switch (OPS)	NTK554TA	N/A	5	6500 Common Equipment, 323-1851-102.1
Service access	Enhanced Service Access Module (ESAM)	NTK552JT	1	1 (unprotected) 1 and 3 (protected)	6500 Photonics Equipment, 323-1851-102.6
ESAM pluggables	OC-3/STM-1 CWDM 1511nm (0-34 dB span)	NTK592NGE5	slot 1, port 2	slot 1, port 2 slot 3, port 2	6500 Common Equipment, 323-1851-102.1
	OC-3/STM-1 CWDM 1511nm (12-42 dB span)	NTK592NVE5			

Table 20 Fixed-64 (75GHz) hardware components - for SRA-based configurations

Function	Name	PEC	Supported slots 2-slot shelf	Supported slots 7-slot shelf	Reference document
Amplifier	Single Line Raman Amplifier (SRA C-Band)	NTK552JA	1	1	6500 Photonics Equipment, 323-1851-102.6
	Switchable Line Amplifier (XLA C-Band)	NTK552KA	2	2	6500 Photonics Equipment, 323-1851-102.6

Flex-12 configuration

The flex-12 configuration consists of the CCMD12 C-Band circuit pack in unprotected configurations in the 2-slot optical Type 2 shelf (NTK503LA variant) or the 7-slot optical Type 2 shelf (NTK503KA variant) equipped with the SPAP-2 w/2xOSC (NTK555NA/NB) shelf processor.

The CCMD12 C-Band circuit pack provides an erbium-doped fiber amplifier (EDFA) at the common input port of the demux path and the common output port of the mux path.

The following tables summarize the hardware components used in the flex-12 configuration and provide a reference for more information.

- [Flex-12 hardware components - shelves, shelf processor and shelf processor pluggables](#)
- [Flex-12 hardware components - for configurations without OTDR](#)
- [Flex-12 hardware components - for configurations with OTDR](#)

For a complete list of 6500 ordering codes, see *6500 Ordering Information*, 323-1851-151.

Table 21 Flex-12 hardware components - shelves, shelf processor and shelf processor pluggables

Function	Name	PEC	Supported slots 2-slot shelf	Supported slots 7-slot shelf	Reference document
Shelves	7-slot optical Type 2	NTK503KA	N/A	N/A	<i>6500 Common Equipment</i> , 323-1851-102.1
	2-slot optical Type 2	NTK503LA	N/A	N/A	<i>6500 Common Equipment</i> , 323-1851-102.1
Shelf processor	SPAP-2 w/2xOSC	NTK555NA NTK555NB	15	15	<i>6500 Common Equipment</i> , 323-1851-102.1

Function	Name	PEC	Supported slots 2-slot shelf	Supported slots 7-slot shelf	Reference document
Shelf processor pluggables	OC-3/STM-1 CWDM 1511nm (0-15 dB span)	NTK592NPE 6	15-1	15-2	6500 Common Equipment, 323-1851-102.1
	OC-3/STM-1 CWDM 1511 nm (10-30 dB span)	NTK592NBE 6			
	OC-3/STM-1 CWDM 1511 nm (20-34 dB span)	NTK592NH E6			
	OC-3/STM-1 CWDM 1511 nm (0-34 dB span)	NTK592NG E5			
	OC-3/STM-1 CWDM 1511nm (12-42 dB span)	NTK592NVE 5			

Table 22 Flex-12 hardware components - for configurations without OTDR

Function	Name	PEC	Supported slots 2-slot shelf	Supported slots 7-slot shelf	Reference document
Mux/Demux	CCMD12 C-Band	NTK508FAE5	1	1	6500 Photonics Equipment, 323-1851-102.6

Table 23 Flex-12 hardware components - for configurations with OTDR

Function	Name	PEC	Supported slots 2-slot shelf	Supported slots 7-slot shelf	Reference document
Mux/Demux	CCMD12 C-Band	NTK508FAE5	2	2	6500 Photonics Equipment, 323-1851-102.6
Service access	Enhanced Service Access Module (ESAM)	NTK552JT	1	1	6500 Photonics Equipment, 323-1851-102.6

Function	Name	PEC	Supported slots 2-slot shelf	Supported slots 7-slot shelf	Reference document
ESAM pluggables	OC-3/STM-1 CWDM 1511nm (0-34 dB span)	NTK592NGE5	slot 1, port 2	slot 1, port 2	6500 Common Equipment, 323-1851-102.1
	OC-3/STM-1 CWDM 1511nm (12-42 dB span)	NTK592NVE5			

CDA using WSS Flex C-Band w/OPM 20x1 and CCMD12

The Colorless Direct Attach (CDA) with CCMD12 provides colorless directional functionality using the WSS Flex C-Band w/OPM 20x1 and CCMD12 circuit packs along with Fiber Interconnect Modules (FIM).

This configuration uses the following hardware:

- WSS Flex C-Band w/OPM 20x1 (NTK553MA)
The WSS Flex C-Band w/OPM 20x1 is a wavelength-selective switch (WSS) equipped with 20 switch ports. Compared to other WSS modules with fewer switch ports, this version has a wavelength-selective switch in the demux direction instead of a passive splitter. This feature adds selective functions to the demux direction, essential for this configuration.
- FIM Type 4 (NTK504CD), FIM Type 5 (NTK504CE), or FIM Type 6 (NTK504CF)
 - FIM Type 4 is used at add/drop nodes. MPO ports on the FIM Type 4 connect to the WSS w/OPM Flex C-Band 20x1. LC ports on the FIM Type 4 connect to CCMD12 equipment.
 - FIM Type 5 is used at branching nodes with five degrees or less. MPO ports on the FIM Type 5 connect to the WSS w/OPM Flex C-Band 20x1.
 - FIM Type 6 is used at branching nodes that require more than five degrees. MPO ports on the FIM Type 6 connect to the WSS w/OPM Flex C-Band 20x1.
- CCMD12 (NTK508FA)
The CCMD12 functions as a power combiner/splitter with an embedded EDFA in the ingress and egress, providing up to 12 channel mux/demux ports without a filter function.

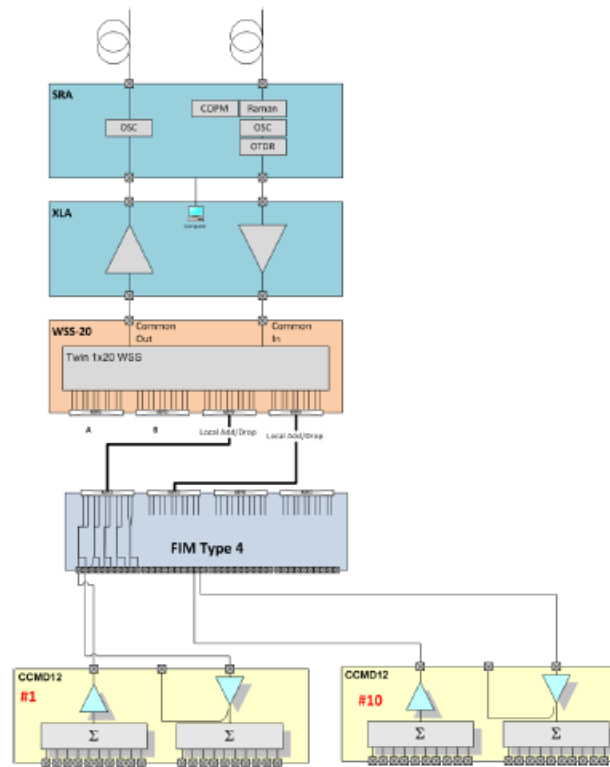
This configuration also uses an XLA with SRA combination.

The following figure shows the CDA with CCMD12 direct attach configuration.

For information on provisioning this configuration using Site Manager, refer to *6500 Configuration - Provisioning and Operating*, Parts 1 and 2, 323-1851-310.

For technical specifications on the modules used in this configuration, refer to *6500 Photonics Equipment*, 323-1851-102.6.

Figure 11 CDA using WSS Flex C-Band w/OPM 20x1



Supported 6500 photonic configurations

This section provides the following information:

- [Waveserver Ai interworking with 6500 photonic configurations](#)
- [6500 point-to-point configurations](#)

Waveserver Ai interworking with 6500 photonic configurations

This section provides information on Waveserver Ai interworking with 6500 photonic configurations. These topics are covered:

- [Waveserver Ai interworking with 6500 photonic configurations in 56GBaud mode](#)

- [Waveserver Ai interworking with 6500 photonic configurations in 35GBaud mode](#)
- [Waveserver Ai encryption module with 6500 photonic configurations in 56GBaud mode](#)
- [Waveserver Ai encryption module with 6500 photonic configurations in 35GBaud mode](#)
- [Waveserver Ai with 6500 submarine photonic line systems in 56GBaud mode](#)
- [Waveserver Ai with 6500 submarine photonic line systems in 35GBaud mode](#)

Waveserver Ai interworking with 6500 photonic configurations in 56GBaud mode

Waveserver Ai supports interworking with the following 6500 photonic configurations in 56GBaud mode:

- Passive photonic line system with CMD64 and MLA3 (Waveserver Ai 1.0 and higher, 6500 Release 12.1 and higher)
- Passive photonic line system with CMD64 and MLA3/ESAM (Waveserver Ai 1.0 and higher, 6500 Release 12.1 and higher)
- ROADM with CMD64 (Waveserver Ai Release 1.0 and higher, 6500 Release 12.1 and higher)
- 1x9 CDA ROADM (Waveserver Ai Release 1.0 and higher, 6500 Release 12.1 and higher)
- 1x20 CDA ROADM (Waveserver Ai Release 1.0 and higher, 6500 Release 12.1 and higher)
- 1x9 CD ROADM (Waveserver Ai Release 1.0 and higher, 6500 Release 12.1 and higher)
- 1x20 CD ROADM (Waveserver Ai Release 1.0 and higher, 6500 Release 12.1 and higher)
- 1x20 CDC ROADM (Waveserver Ai Release 1.0 and higher, 6500 Release 12.1 and higher)
- 1x5 RLA with CMD64 (Waveserver Ai Release 1.2 and higher, 6500 Release 12.1 and higher)
- Coherent Select (Waveserver Ai Release 1.0 and higher, 6500 Release 12.1 and higher)
- CMD64 (Waveserver Ai Release 1.0 and higher, 6500 Release 12.1 and higher)
- CCMD12 (Waveserver Ai Release 1.0 and higher, 6500 Release 12.1 and higher)

- CMDi4 (Waveserver Ai Release 1.0 and higher, 6500 Release 12.1 and higher)
- Passive photonic line system with CMD48 and MLA3 (Waveserver Ai Release 1.4 and higher, 6500 Release 12.4 and higher)
- Passive photonic line system with CMD48 and MLA3/ESAM (Waveserver Ai Release 1.4 and higher, 6500 Release 12.4 and higher)
- Passive photonic line system with CMD48/CMD64 and 6500 RLS TPS and DLM (Waveserver Ai Release 1.5 and higher, 6500 Release 12.6 and higher, 6500 RLS 1.0 and higher)

Waveserver Ai interworking with 6500 photonic configurations in 35GBaud mode

Waveserver Ai supports interworking with the following 6500 photonic configurations in 35GBaud mode:

- Passive photonic line system with CMD96 and MLA3 (Waveserver Ai Release 1.2 and higher, 6500 Release 12.1 and higher)
- Passive photonic line system with CMD96 and MLA3/ESAM (Waveserver Ai Release 1.2 and higher, 6500 Release 12.1 and higher)
- Passive photonic line system with CMD44 and SLA/MLA/MLA2/MLA3 (Waveserver Ai Release 1.2 and higher, 6500 Release 12.1 and higher)
- Passive photonic line system with BMD2, CMD44 and SLA/MLA/MLA2/MLA3 (Waveserver Ai Release 1.2 and higher, 6500 Release 12.1 and higher)
- Colored ROADM (Waveserver Ai Release 1.2 and higher, 6500 Release 12.1 and higher)
- 1x9 CDA ROADM (Waveserver Ai Release 1.2 and higher, 6500 Release 12.1 and higher)
- 1x20 CDA ROADM (Waveserver Ai Release 1.2 and higher, 6500 Release 12.1 and higher)
- 1x9 CD ROADM (Waveserver Ai Release 1.2 and higher, 6500 Release 12.1 and higher)
- 1x20 CD ROADM (Waveserver Ai Release 1.2 and higher, 6500 Release 12.1 and higher)
- 1x20 CDC ROADM (Waveserver Ai Release 1.2 and higher, 6500 Release 12.1 and higher)
- Coherent Select (Waveserver Ai Release 1.2 and higher, 6500 Release 12.1 and higher)

Waveserver Ai encryption module with 6500 photonic configurations in 56GBaud mode

Waveserver Ai encryption module in 56GBaud mode supports interworking with the following 6500 photonic configurations:

- Passive photonic line system with CMD64 and MLA3 (Waveserver Ai 1.3 and higher, 6500 Release 12.3 and higher)
- Passive photonic line system with CMD64 and MLA3/ESAM (Waveserver Ai 1.3 and higher, 6500 Release 12.3 and higher)
- Passive photonic line system with CMD44 (100GHz) and SLA/MLA/MLA2/MLA3 (Waveserver Ai Release 1.3 and higher, 6500 Release 12.3 and higher)
- ROADM with CMD64 (Waveserver Ai Release 1.3 and higher, 6500 Release 12.3 and higher)
- ROADM with CMD44 (Waveserver Ai Release 1.3 and higher, 6500 Release 12.3 and higher)
- 1x9 CDA ROADM (Waveserver Ai Release 1.3 and higher, 6500 Release 12.3 and higher)
- 1x20 CDA ROADM (Waveserver Ai Release 1.3 and higher, 6500 Release 12.3 and higher)
- 1x9 CD ROADM (Waveserver Ai Release 1.3 and higher, 6500 Release 12.3 and higher)
- 1x20 CD ROADM (Waveserver Ai Release 1.3 and higher, 6500 Release 12.3 and higher)
- 1x20 CDC ROADM (Waveserver Ai Release 1.3 and higher, 6500 Release 12.3 and higher)
- 1x5 RLA with CMD64 (Waveserver Ai Release 1.3 and higher, 6500 Release 12.3 and higher)
- Coherent Select (Waveserver Ai Release 1.3 and higher, 6500 Release 12.3 and higher)
- CCMD12 (Waveserver Ai Release 1.3 and higher, 6500 Release 12.3 and higher)

Waveserver Ai encryption module with 6500 photonic configurations in 35GBaud mode

Waveserver Ai encryption module in 35GBaud mode supports interworking with the following 6500 photonic configurations:

- Passive photonic line system with CMD96 and MLA3 (Waveserver Ai Release 1.3 and higher, 6500 Release 12.3 and higher)
- Passive photonic line system with CMD96 and MLA3/ESAM (Waveserver Ai Release 1.3 and higher, 6500 Release 12.3 and higher)

- Passive photonic line system with CMD44 and SLA/MLA/MLA2/MLA3 (Waveserver Ai Release 1.3 and higher, 6500 Release 12.3 and higher)
- Passive photonic line system with BMD2, CMD44 and SLA/MLA/MLA2/MLA3 (Waveserver Ai Release 1.3 and higher, 6500 Release 12.3 and higher)
- ROADM with CMD44 (Waveserver Ai Release 1.3 and higher, 6500 Release 12.3 and higher)
- 1x9 CDA ROADM (Waveserver Ai Release 1.3 and higher, 6500 Release 12.3 and higher)
- 1x20 CDA ROADM (Waveserver Ai Release 1.3 and higher, 6500 Release 12.3 and higher)
- 1x9 CD ROADM (Waveserver Ai Release 1.3 and higher, 6500 Release 12.3 and higher)
- 1x20 CD ROADM (Waveserver Ai Release 1.3 and higher, 6500 Release 12.3 and higher)
- 1x20 CDC ROADM (Waveserver Ai Release 1.3 and higher, 6500 Release 12.3 and higher)
- Coherent Select (Waveserver Ai Release 1.3 and higher, 6500 Release 12.3 and higher)

Waveserver Ai with 6500 submarine photonic line systems in 56GBaud mode

Waveserver Ai in 56GBaud mode supports the following submarine photonic line systems:

- Flexible Grid 1x9 GMD10 SLTE (Waveserver Ai Release 1.1 and higher, 6500 Release 12.2 and higher)
- Flexible Grid 1x9 CDA SLTE (Waveserver Ai Release 1.1 and higher, 6500 Release 12.2 and higher)

Waveserver Ai with 6500 submarine photonic line systems in 35GBaud mode

Waveserver Ai in 35GBaud mode supports the following submarine photonic line systems:

- Fixed Grid 1x9 SLTE (Waveserver Ai Release 1.2 and higher, 6500 Release 12.2 and higher)
- Flexible Grid 1x9 GMD10 SLTE (Waveserver Ai Release 1.2 and higher, 6500 Release 12.1 and higher)
- Flexible Grid 1x9 CDA SLTE (Waveserver Ai Release 1.2 and higher, 6500 Release 12.1 and higher)

The ROADM configurations are set up using Site Manager. To set up Waveserver Ai-6500 interoperability, you must provision the Waveserver Ai far-end address and also add the Waveserver Ai to the defined platform types in Site Manager. For more information about configuring the 6500 SPLI

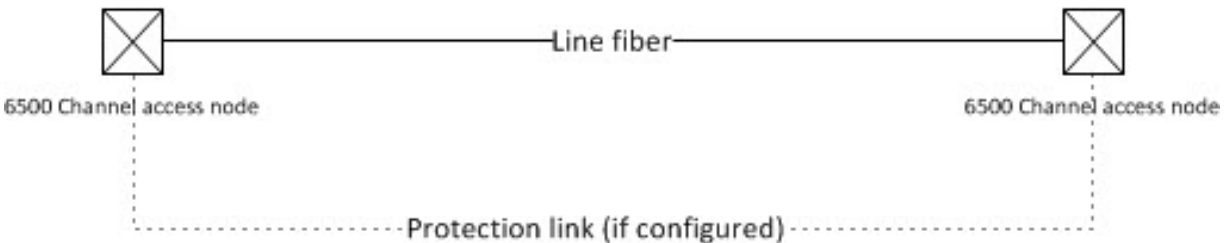
parameters, see technical publications *6500 Administration and Security*, 323-1851-301, *6500 Commissioning and Testing*, 323-1851-221, and *Equipment and Facility Provisioning*, Part 1 of 2, 323-1851-310. For information about configuring Waveserver Ai SPLI parameters, see *Waveserver Ai User Guide*, 323-4002-100.

6500 point-to-point configurations

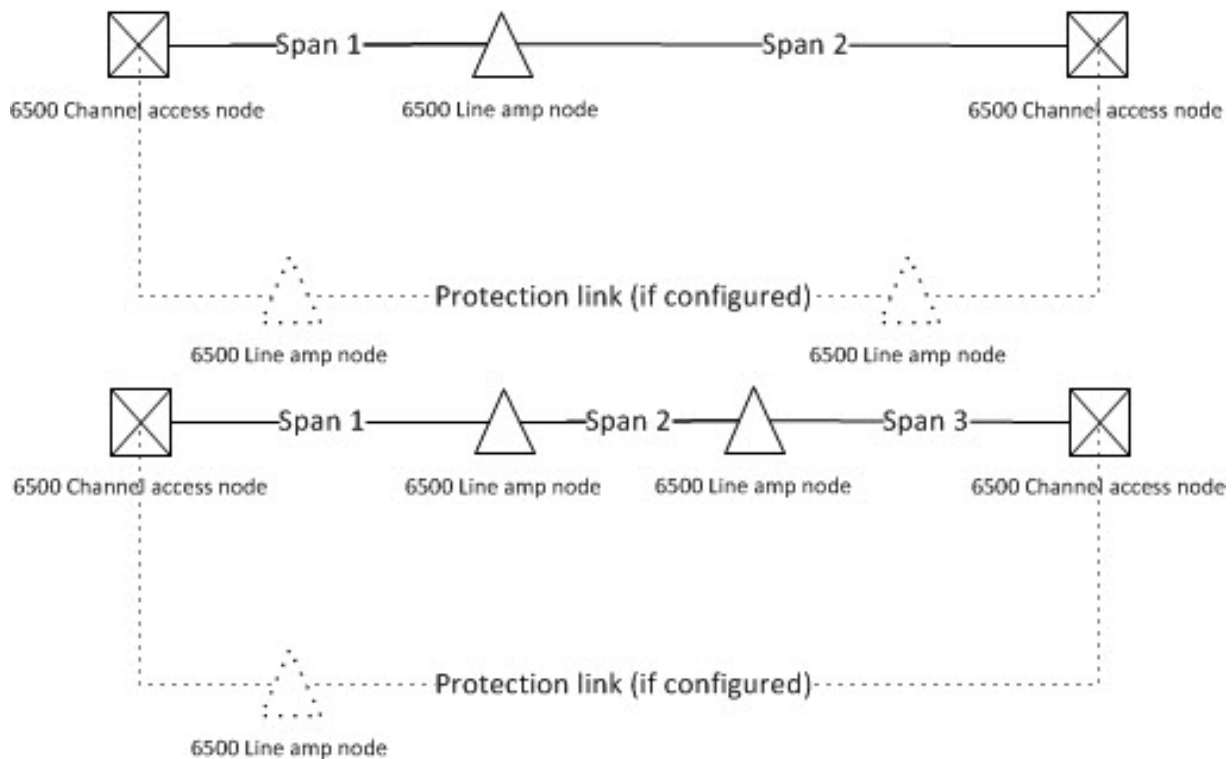
Waveserver Ai is designed to be configured as a point-to-point network. There are a number of options for the configuration of the 6500 Photonic section, but all traffic on a Waveserver Ai must terminate on its neighboring Waveserver Ai node. In addition to the ROADM configuration, for the 6500 photonic configurations, there are two possible point-to-point configurations:

- two 6500 terminal nodes (configured as unprotected or protected), directly connected with line fiber. See the following figure.

Figure 12 6500 point-to-point configuration



- two 6500 terminal nodes (configured as unprotected or protected), with line amps configured between them. For 6500, the photonic line supports 1 or 2 line amps on the working and/or protection path depending on the type of amplifiers used. This will result in 1, 2, or 3 line spans between the 6500 terminal nodes. See following figure.

Figure 13 6500 line amp configuration

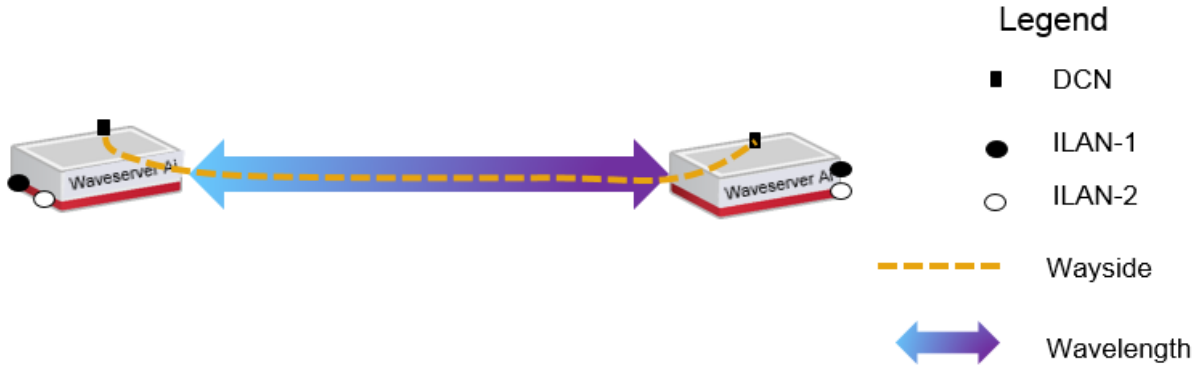
Note: Waveserver Ai is supported on 6500 (Release 12.1 and above) fixed and Flexible grid ROADM based photonics line system as Ciena native wavelengths. In 6500 configurations running pre-12.1 releases, Waveserver Ai is treated as foreign wavelengths.

Comms channel for 6500-based SLTE configurations

Waveserver Ai supports a wayside channel for use in submarine applications when interworking with a 6500 as the photonic line system. In submarine applications, due to distance, no OSC is available for photonic control messaging, and since Waveserver Ai provides the wavelengths, 6500 GCC is also not available. In this situation, the Waveserver Ai wayside channel can be used to provide a channel for 6500-to-6500 communications across the submarine link that are required by the photonic applications.

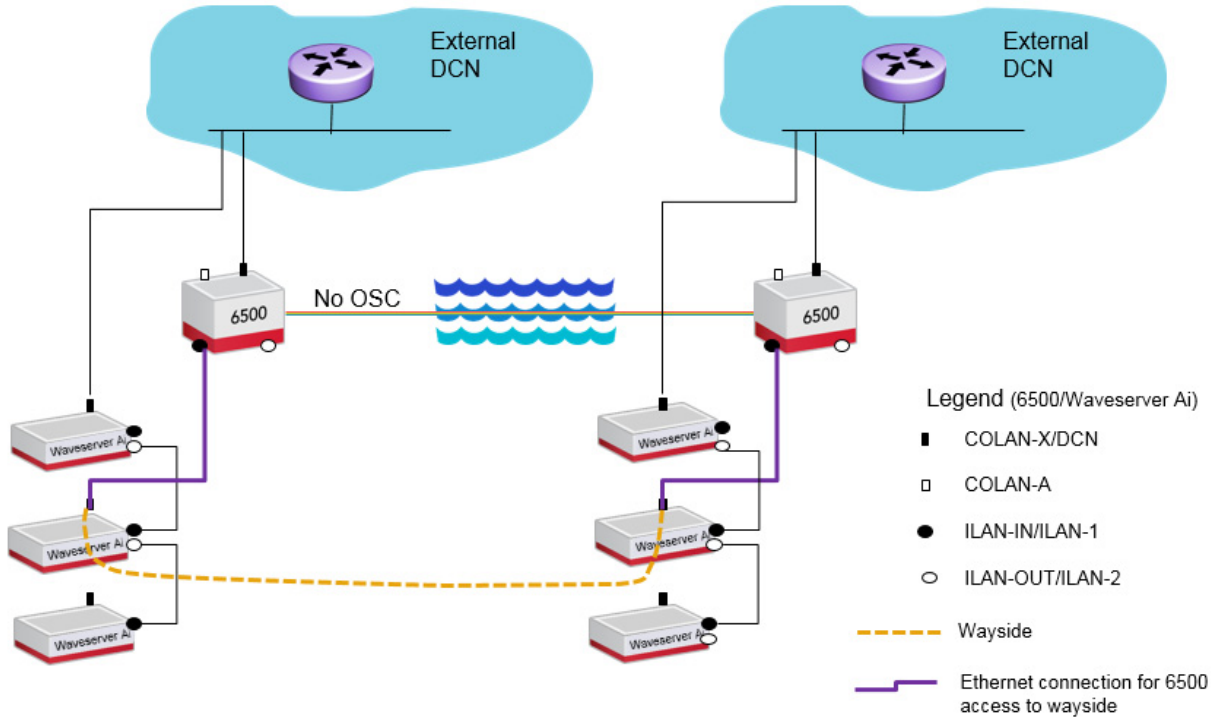
To create an end-to-end wayside channel, the DCN port on a pair of Waveserver Ai nodes (one at each end of the submarine link) is re-purposed as a wayside access port using CLI command `chassis port set`, as shown in the following figure.

Figure 14 Waveserver Ai wayside access via DCN port



When the DCN port on a Waveserver Ai chassis is used for a wayside channel it cannot be used for normal external DCN connectivity. This means that the chassis must be connected via ILAN port to another Waveserver Ai or Waveserver Ai cluster that has connectivity to the external DCN for management access. This requirement means that at least two Waveserver Ai chassis are required at each site. See the following figure for an example of a typical application.

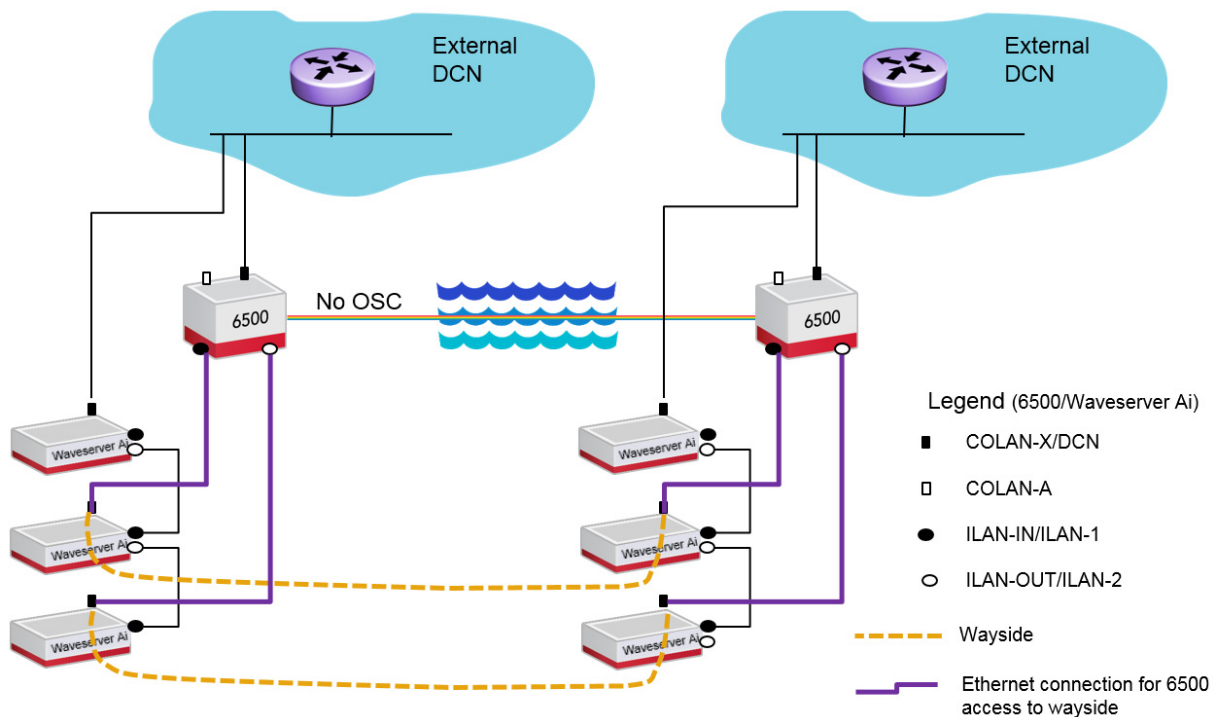
Figure 15 Waveserver Ai wayside for 6500 SLTE communications



As shown in the figure above, logically the 6500 nodes are using an ILAN-to-ILAN connection, and with OSPF configured on that connection, the 6500 photonic control software components are able to communicate and therefore manage the optical link.

You can create more than one wayside channel by using additional Waveserver Ai pairs. This provides improved resiliency for certain failures; see the following figure for an example.

Figure 16 Multiple Waveserver Ai wayside channels used for 6500 SLTE communications



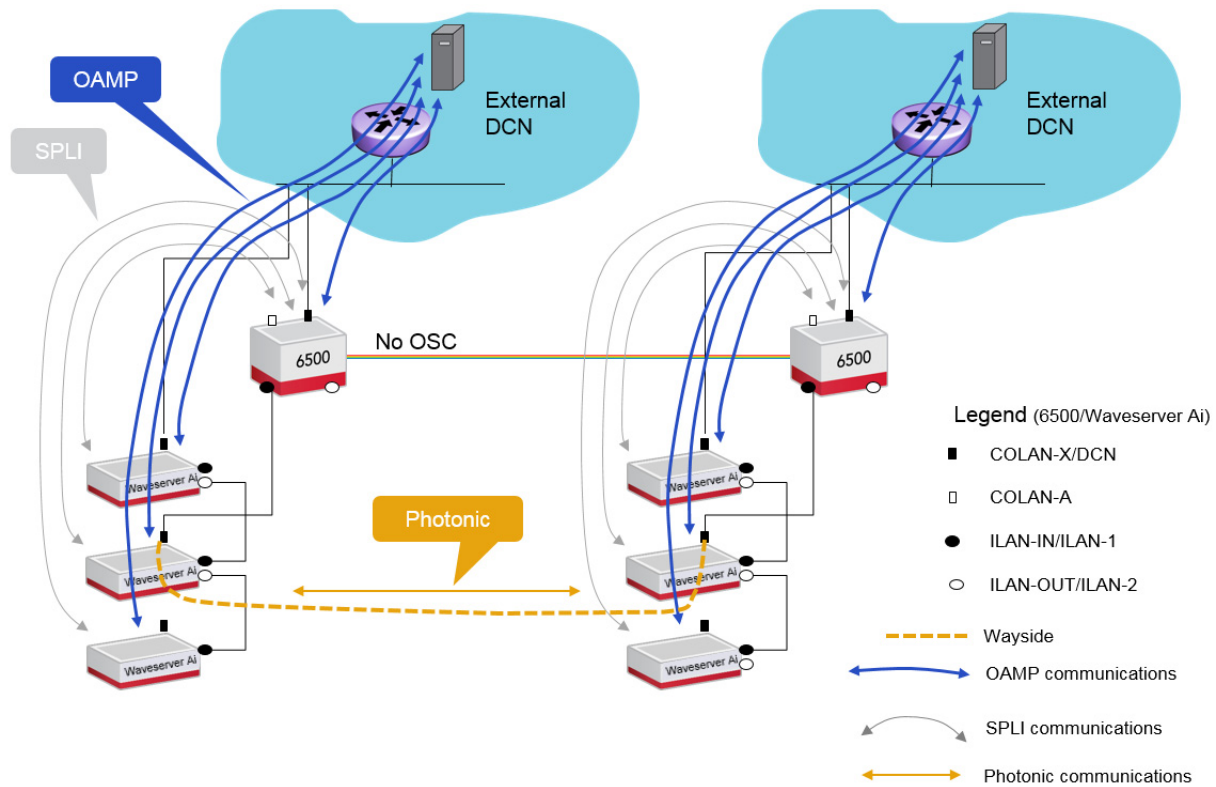
Wayside channel guidelines

The following section provides both general and more detailed guidelines for implementing a wayside channel.

- The wayside channel is supported on Waveserver Ai only.
- The wayside channel is supported for IPv4 only.
- The wayside channel operates at Layer 2. When connecting to the wayside channel, take care to prevent loops. For example, do not connect the same two wayside channels into an L2 switch at each end of the submarine link.
- Ensure that the Waveserver Ai chassis to be used for the wayside channel has the MOTR module installed in slot 1. Only port 1/1 carries the wayside channel.

- To configure a comms channel for 6500-based SLTE configurations, set the DCN port on the local and remote Waveserver Ai cluster as “wayside-channel” using CLI command `chassis port set port dcn mode wayside-channel`.
- Ciena recommends using a 6500 ILAN port to connect to the Waveserver Ai DCN port for wayside access. A 6500 COLAN port can also be used, however COLAN-A port is not supported on 6500 14-slot shelves for this application.
- The 6500 ILAN or COLAN port connected to the Waveserver Ai DCN port must be:
 - Set to AUTO mode (auto-negotiate). The ports will auto-negotiate to 100BT full-duplex.
 - Set as a numbered interface. In routable IP configurations, the IP addresses must be routable. A /30 subnet is recommended.
 - Configured with OSPF with opaque LSAs enabled. Note that if a COLAN port is used, opaque LSAs are disabled by default.
- Waveserver Ai wayside throughput varies with frame size and achieves approximately 10Mbps at an average frame size of 512 bytes.
- The configuration should be engineered so that north/south OAMP traffic does not normally traverse the wayside channel, since the channel bandwidth is intended for 6500 photonics use.
- Waveserver Ai supports regen configurations with 6500-based SLTE comms. 6500-based SLTE comms cannot be added to a regen node, the 6500 must be connected to a terminal node. For information on Waveserver Ai regen applications, see *Waveserver Ai User Guide*, 323-4002-100.
- Use of the Waveserver Ai wayside channel does not restrict or affect the use of a 6500 GNE configuration, however the type of configuration can affect local SPLI communications. For SPLI communication between 6500 and Waveserver Ai, the 6500 shelf-IP address must be visible to the Waveserver Ai; therefore some GNE configurations, such as Private-IP, cannot be used. The following figure shows the various communications flows, including Waveserver Ai to 6500 SPLI.

Figure 17 Communication flows



For information on how to configure 6500-based SLTE configurations using Waveserver Ai wayside channel, see [Configuring 6500-based SLTE configurations using Waveserver Ai wayside channel](#).

Managing Waveserver Ais through 6500

This section provides information on managing Waveserver Ais through the 6500. You can use this configuration when there is limited or no DCN connectivity at the sites containing 6500s and Waveserver Ais. As 6500 supports GNE and Layer-3 routing functionality, the Waveserver Ai subtends the 6500 in this setup.

Note: Where there is sufficient DCN connectivity available at a site, Waveserver Ai and 6500 can be connected to separate DCNs. For more information on connecting Waveserver Ai directly to a DCN, see the “Direct DCN connection” in the *Waveserver Ai User Guide*, 323-4002-100 and for more information on connecting 6500 directly to a DCN, see *6500 Data Communications Planning and User Guide*, 323-1851-101.

The general approaches for management access involving the 6500 and Waveserver Ai are as follows:

- Where DCN connectivity to a site is limited to a single LAN drop, connect the 6500 directly to the DCN LAN drop, and connect the Waveserver Ai to the 6500 using the 6500 port preferences, see [6500 port preferences](#). For examples of this configuration, see [Example 1: DCN at all sites, statically routed GNE, no line amps](#) and [Example 2: DCN at all sites with multiple 6500s and OSPF GNEs](#).
- Where there is no direct DCN connectivity to a site, but there is, for example, OSC connectivity to the 6500 site, connect the Waveserver Ai to the 6500 using the 6500 port preferences, see [6500 port preferences](#). For an example of this configuration, see [Example 3: DCN at head-end only with 6500 OSC used to access site without DCN connectivity](#).

These topics are covered:

- [Waveserver Ai and 6500 interconnect model and 6500 port preferences](#)
- [DCN connectivity at all sites](#)
- [DCN connectivity at head-end site only](#)

Note: For information on Managing Waveserver through 6500, see *Waveserver and 6500 Interworking*, 323-4001-165.

Waveserver Ai and 6500 interconnect model and 6500 port preferences

The Waveserver Ai and 6500 interconnect model illustrates how to connect the Waveserver Ais to the 6500, when you want to manage Waveserver Ais through the 6500.

For the Waveserver Ai and 6500 interconnect model:

- 6500 Gateway Network Element (GNE) configurations with IP static route (non-redundant) and OSPF (redundant) configurations are supported, private IP (NAT) is not supported. For additional information on 6500 GNE setups, see *6500 Data Communications Planning and User Guide*, 323-1851-101.
- Auto-negotiation must be enabled and an OSPF passive circuit must be provisioned on the 6500 port that connects to the Waveserver Ai.
- The 6500 management port must be numbered. IP subnets must be large enough to accommodate the 6500 management port and all the subtending Waveserver Ais.
- When there are multiple Waveserver Ais subtending a 6500, a single Waveserver Ai DCN port must be connected to the 6500 COLAN/ILAN, with the remaining Waveserver Ais connected in an ILAN daisy-chain (either linear or in a ring configuration).

Note: Waveserver Ai ILAN loop closure is only supported on Waveserver Ai Release 1.4 and higher.

- If you are not using ZTP, Waveserver Ai must be provisioned with the appropriate IP address and netmask. The gateway must be provisioned with the respective 6500 COLAN/ILAN IP address.
- Supports IPv4 and IPv6
 - IPv4 support: Supported on all Waveserver Ai and 6500 releases.
 - IPV6 support: Supported on all Waveserver Ai releases. Supported on 6500 Release 11.2 and higher.

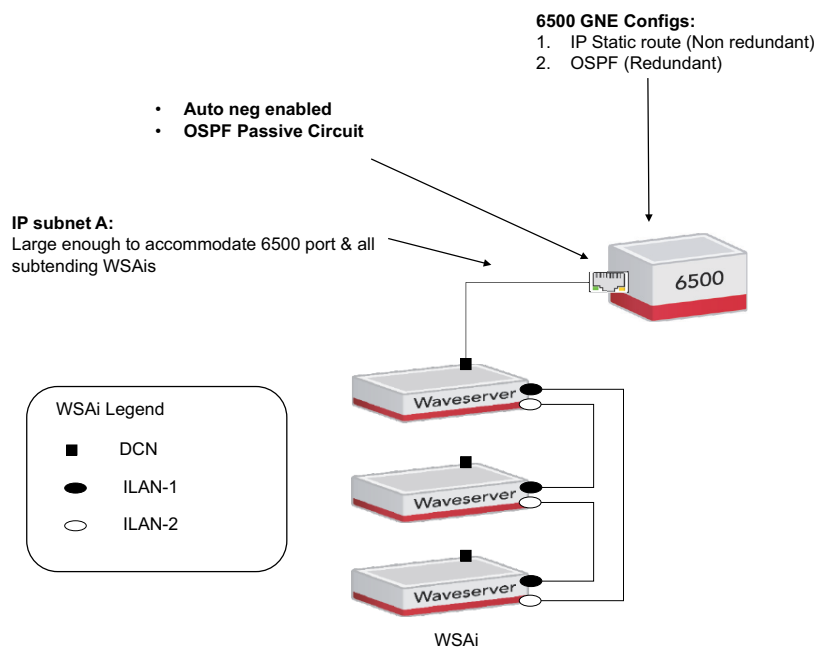
Note: IPv6 is not available on 6500 T-series.

- It is generally recommended to use COLAN-A on the 6500 for connecting the subtending Waveserver Ais, as COLAN-X is typically used for DCN connectivity and the ILANs are typically used for inter-6500 connectivity. However, depending on what ports are available, COLAN-X or ILAN can be used. For more information on 6500 port preferences, [6500 port preferences](#).

Note: COLAN-A is not recommended on the 6500 14-slot S-series chassis when it is in a protected dual-SP2 configuration. For more information, see *6500 Data Communications Planning and User Guide*, 323-1851-101.

- Waveserver Ai must not be connected between 6500 shelves. For example, Waveserver Ai must not be inserted into a 6500 ILAN ring. The Waveserver Ai must be connected to a 6500 port dedicated to Waveserver Ai.
- For this configuration, as 6500 and Waveserver Ai are directly connected, SPLI traffic between them flows over the local connection between the two devices, and not through the external DCN.

See the following figures for the Waveserver Ai and 6500 interconnect model and the 6500 interconnect port preferences used for this model. For the 6500 interconnect port preferences, preferences are listed with the most preferred first, though ILAN-IN and ILAN-OUT have equal preference.

Figure 18 Waveserver Ai and 6500 interconnect model**Figure 19** 6500 port preferences

6500 – 7 slot	WSAi / 6500 interconnect port preference:	Auto-neg must be enabled on 6500 ports that connect into WSAi
	COLAN-A, COLAN-X, ILAN-IN , ILAN-OUT	
6500 – 8 slot	COLAN-A, COLAN-X, ILAN-IN , ILAN-OUT	
6500 – 14 slot	COLAN-X, ILAN-IN , ILAN-OUT, COLAN-A	Use of COLAN-A not recommended on S-series 14 slot running SP2 in slots 15 & 16. For more information, see 6500 Data Communications Planning and User Guide, 323-1851-101.
6500 – 32 slot	COLAN-A, COLAN-X, ILAN-IN , ILAN-OUT	
6500 – T series	COLAN-A, COLAN-X, ILAN-IN-1, ILAN-OUT-1, ILAN-IN-2, ILAN-OUT-2, ILAN-IN-3, ILAN-OUT-3	

DCN connectivity at all sites

This section provides examples where each site has DCN access, as a result Waveserver Ai management traffic generally flows through the COLAN port of the local 6500 GNE.

Example 1: DCN at all sites, statically routed GNE, no line amps

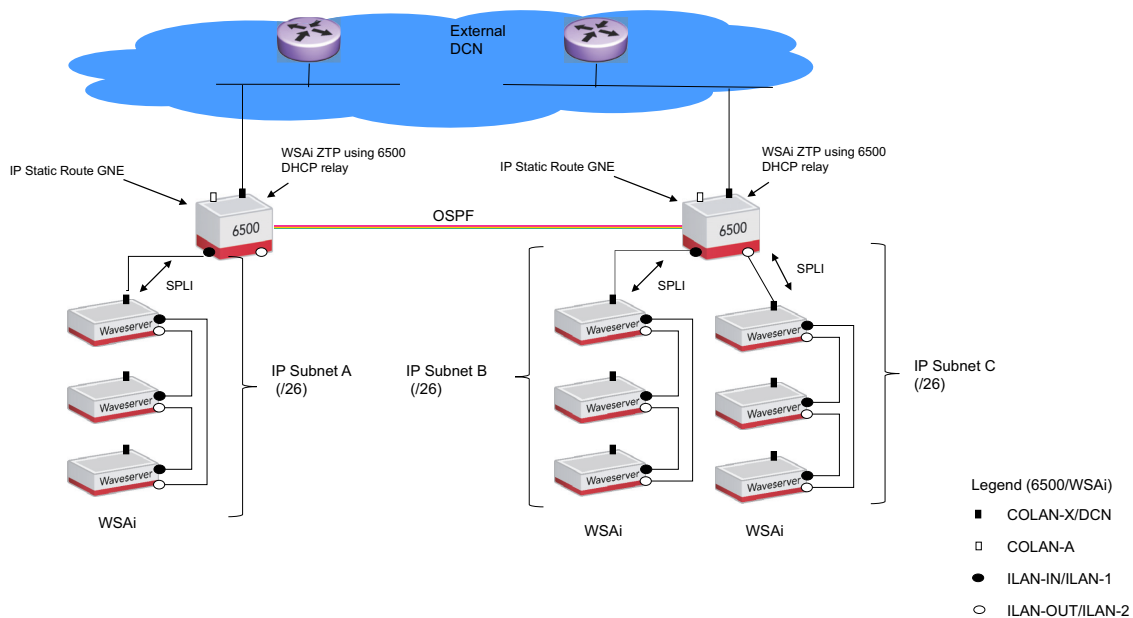
For this example, there is DCN access and a single 6500 chassis at each site. It is assumed that there is only a single connection to the external DCN available at each site, therefore requires the Waveserver Ai(s) to subtend the 6500. The 6500s are assumed to be 14-slot shelves, where COLAN-A is not recommended, thus the Waveserver Ais are connected to the 6500 ILAN ports.

These notes apply:

- An IP static route is provisioned for the non-redundant 6500 GNE.
- OSPF is used within the 6500 network only.
- A /26 subnet is allocated for the Waveserver Ais to allow for future growth.
- OSPF passive circuits are required on the 6500 ports to the Waveserver Ai.
- Waveserver Ai ZTP is supported using 6500 DHCP relay (plain mode).

See the following figure.

Figure 20 DCN at all locations



Example 2: DCN at all sites with multiple 6500s and OSPF GNEs

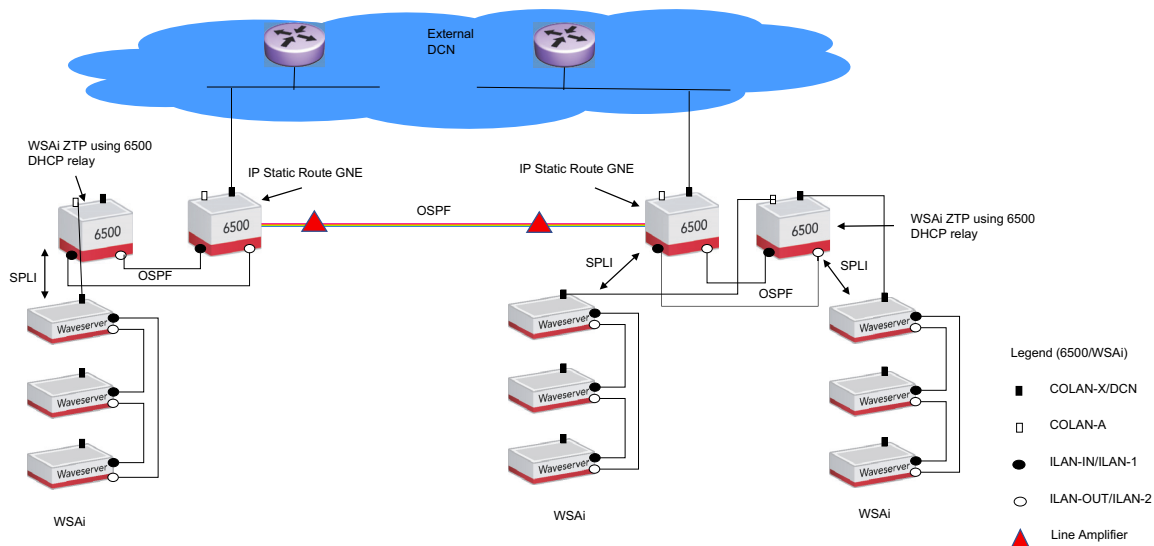
For this example, there is an external DCN access at each site, and because there are multiple 6500's at each site, the ILAN ports are all used for intra-site connectivity. Since the COLAN-X ports on the 6500 GNEs are used to connect to the external DCN, the example shows the Waveserver Ai chassis connected to the COLAN ports on the non-GNE 6500 shelves. OSPF GNEs are used to provide resilient access to the network, including the line amplifiers.

These notes apply:

- OSPF is configured on the COLAN-X of each site, providing resilient access to all 6500s and Waveserver Ais.
- OSPF is used within the 6500 network and also over COLAN-X to the external DCN to provide resilient access.
- OSPF passive circuits are required on the 6500 ports to the Waveserver Ai.
- Waveserver Ai ZTP is supported using 6500 DHCP relay (plain mode).

See the following figure.

Figure 21 DCN at all locations with multiple 6500s



DCN connectivity at head-end site only

When the Waveserver Ais are located at a site where the 6500 has indirect DCN connectivity, the Waveserver Ais management traffic can be routed from the External DCN through the 6500 management network to the site without DCN access using the available 6500 OSC comms channel.

The OSC channel is available and recommended when interworking with the 6500 chassis, as it provides the highest bandwidth channel available.

The 6500 provides several routing capabilities and it supports several management configurations.

See the following example.

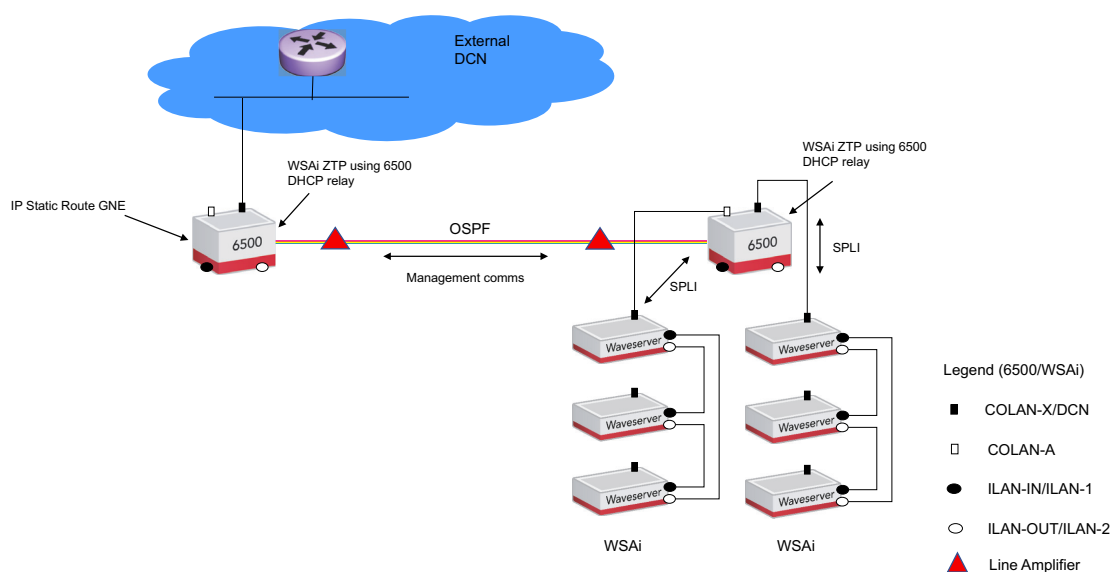
Example 3: DCN at head-end only with 6500 OSC used to access site without DCN connectivity

For this example, there is a DCN at the head-end only and the 6500 OSC is used to access the site without DCN connectivity.

These notes apply:

- An IP static route is provisioned for the non-redundant 6500 GNE.
- OSPF is used within the 6500 network only.
- OSPF passive circuits are required on the 6500 ports to the Waveserver Ai.
- Waveserver Ai ZTP is supported using 6500 DHCP relay (plain mode).

See the following figure.

Figure 22 DCN at head-end only with 6500 OSC used to access remote site

Waveserver Ai (Release 1.2 and higher) and 6500 (Release 12.3) SPLI information sharing

In a SPLI-enabled system, the Waveserver Ai (Release 1.2 and higher) and 6500 (Release) will share certain provisionable attributes. For information shared from 6500 to Waveserver Ai, see table [SPLI information shared from 6500 -> Waveserver Ai](#). For information shared from Waveserver Ai to 6500, see [SPLI information shared from Waveserver Ai -> 6500](#).

Table 24 SPLI information shared from 6500 -> Waveserver Ai

Parameter	6500 photonics (CMD/OPS) facility parameter name	Waveserver Ai transponder/ OPS facility parameter name
Tx Power	Actual Tx Power (dBm)	Tx Power (dBm)
Tx Wavelength	ITU Center Wavelength (nm)	Wavelength (nm)
Line System Type	OCH Line System Type	Line System Configuration - Line System Type
Tx Power Reduced State (TxBlanking)	Tx Power Reduced State	Tx Power Reduction State
Laser Centering Mode	Laser Centering	Laser Control - Centering Mode
Laser Centering Range	NA	Laser Control - Centering Range (GHz)

Parameter	6500 photonics (CMD/OPS) facility parameter name	Waveserver Ai transponder/ OPS facility parameter name
Tx Frequency	Center Frequency (THz)	Transmitter Frequency (GHz)
Transmission mode	Provisioned Transmission Mode	Modulation Scheme
NMC Allocated Spectrum Width	Allocated NMC Spectral Width (GHz)	Allocated Spectral Width (GHz)
Path dispersion	Discovered Path Dispersion (ps/nm)	Tx Dispersion Compensation Management - Path Dispersion value (ps/nm)
Line Optimization Mode	Discovered Line Optimization Mode	Link Optimization Mode
Transmitter dispersion compensation mode	Discovered Transmitter Dispersion Mode	Tx Dispersion Compensation Management - Mode
Tx dispersion precompensation value	Discovered Transmitter Dispersion Value (ps/nm)	Tx Dispersion Compensation Management - Pre-compensation value (ps/nm)
Expected Restoration Time	Expected Restoration Time (seconds)	Laser Control - Expected Restoration Time (seconds)

Table 25 SPLI information shared from Waveserver Ai -> 6500

Paramter	Waveserver Ai transponder/ OPS facility parameter name	6500 photonics (CMD/OPS) facility parameter name
Transmitter Type	Modem Type	Transmitter Type (not on OPS)
Platform Type	NA	Platform Type
Modulation Scheme/ Transmission Mode	Modulation Scheme	Transmission Mode
Discovered Far-end PEC	Equipment Code	Discovered Far-end PEC
Modem Output Type	Modem Output	NA
Discovered SPLI Management	Line System Configuration - SPLI Management	Discovered SPLI Management
Mate Info/Config Info	NA	Mate Info
Discovered Wavelength	Transmitter Wavelength (nm)	Discovered Wavelength (nm)
Tx Tuned	cleared Tx Tuning In Progress alarm	Tx Tuned
Discovered Provisioned Tx Power	Tx Power (dBm)	Discovered Tx Provisioned Power (dBm)

Paramter	Waveserver Ai transponder/ OPS facility parameter name	6500 photonics (CMD/OPS) facility parameter name
Discovered Provisioned Tx Blanking	Tx Power Reduction State	Discovered Tx Power Reduced State
Discovered Frequency	Frequency (GHz)	Discovered Center Frequency (THz)
Tx Frequency Tuning Resolution	Frequency Resolution (GHz)	Tx Frequency Tuning Resolution (GHz)
Discovered Laser Centering Mode	Laser Control - Centering Mode	Discovered Laser Centering
Discovered Laser Centering Range	Laser Control - Centering Range (GHz)	Discovered Laser Centering Range (GHz)
Discovered Transmission Mode	Modulation Scheme	Discovered Transmission Mode
Tx Minimum Spectral Width	Tx Minimum Spectral Width (GHz)	Tx Minimum Spectral Width (GHz)
Tx Signal Bandwidth 10dB	Tx Signal Bandwidth 10dB (GHz)	Tx Signal Bandwidth 10dB (GHz)
Tx Signal Bandwidth 3dB	Tx Signal Bandwidth 3dB (GHz)	Tx Signal Bandwidth 3dB (GHz)
Discovered Line Optimization Mode	Link Optimization Mode	Discovered Line Optimization Mode
Discovered Transmitter Dispersion Mode	Tx Pre-compensation Mode	Discovered Tx Dispersion Mode
Discovered Transmitter Dispersion Value	Tx Dispersion Compensation Management - Tx Path Dispersion (ps/nm)	Discovered Tx Dispersion Value (ps/nm)
Discovered Transmitter Pre-compensation	Tx Dispersion Compensation Management - Tx Pre-compensation (ps/nm)	Discovered Tx Pre-Compensation (ps/nm)

Waveserver Ai (Release 1.3 and lower) and 6500 (Release 12.4) SPLI interworking considerations

6500 Release 12.4 introduces L0 Control Plane Mesh Restoration Speed Improvements to enable faster traffic capacity changes and quicker L0 Control Plane Mesh Restoration times in comparison to previous releases. For this feature, channel capacity modes “Mode-1”, “Mode-2” and “Mode-3” are introduced.

When a new channel is created on the 6500 (Release 12.4) with channel capacity mode “Mode-1”, the ADJ-TX Tuning mode will be set to “Accelerated Tuning” and pushed to the modem (Waveserver Ai) via SPLI. Similarly, when

a new channel is created with channel capacity modes “Mode-2” or “Mode-3”, the ADJ-TX Tuning mode will be set to 'Fast Tuning' and pushed to the modem (Waveserver Ai) via SPLI.

If 6500 Release 12.4 is interworking with Waveserver Ai (Release 1.3 and lower), the "Fast Tuning" mode is not supported by Waveserver Ai SPLI. In this scenario, the 6500 (Release 12.4) must have its channel capacity mode set to "Mode-1". If the channel capacity mode is set to "Mode-2" or "Mode-3", the "Adjacency Mismatch" alarm will be raised on the ADJ-TX facility on the 6500 and the tuning mode will be either “Accelerated Tuning” or “Performance Optimized Tuning”. To clear this alarm, set the 6500 (Release 12.4) channel capacity mode to "Mode-1".

Note: “Fast Tuning” mode is supported on Waveserver Ai Release 1.4 and higher, for additional information see [Waveserver Ai Release 1.4 fast tuning mode support](#).

Waveserver Ai Release 1.3 (and higher) and 6500 Release 12.3 interworking

The following Waveserver Ai Release 1.3 and 6500 Release 12.3 interworking configurations are supported:

- [Waveserver Ai 2xWLAi + 8xQSFP28 module interworking with 6500 WLAi MOTR circuit pack as a transponder](#)
- [Waveserver Ai 2xWLAi + 8xQSFP28 module interworking with 6500 WLAi MOTR circuit pack in a regen configuration](#)
- [Waveserver Ai 1xWLAi + 4xQSFP28 Encryption module interworking with 6500 WLAi MOTR circuit pack in a regen configuration](#)
- [6500 WLAi MOTR circuit pack interworking with Waveserver Ai 2xWLAi + 8xQSFP28 module in a regen configuration](#)
- [Waveserver Ai 2xWLAi + 8xQSFP28 module interworking with 6500 WLAi MOTR circuit pack in a hybrid regen configuration](#)
- [6500 WLAi MOTR interworking with Waveserver Ai 2xWLAi + 8xQSFP28 module circuit pack in a hybrid regen configuration](#)
- [Waveserver Ai 2xWLAi + 8xQSFP28 module interworking with 6500 WLAi MOTR circuit pack](#)
- [Waveserver Ai 2xWLAi + 8xQSFP28 module interworking with 6500 100G OCl circuit pack](#)
- [Waveserver Ai 2xWLAi + 8xQSFP28 module interworking with 6500 200G OCl circuit pack](#)
- [Waveserver Ai 2xWLAi + 8xQSFP28 module interworking with the 6500 100G OTR circuit pack](#)

Waveserver Ai 2xWLAi + 8xQSFP28 module interworking with 6500 WLAi MOTR circuit pack as a transponder

This section outlines the Waveserver Ai 2xWLAi + 8xQSFP28 module interworking with 6500 WLAi MOTR as a transponder. See the following figures.

Figure 23 Waveserver Ai 2xWLAi + 8xQSFP28 module interworking with 6500 WLAi MOTR circuit pack as a transponder

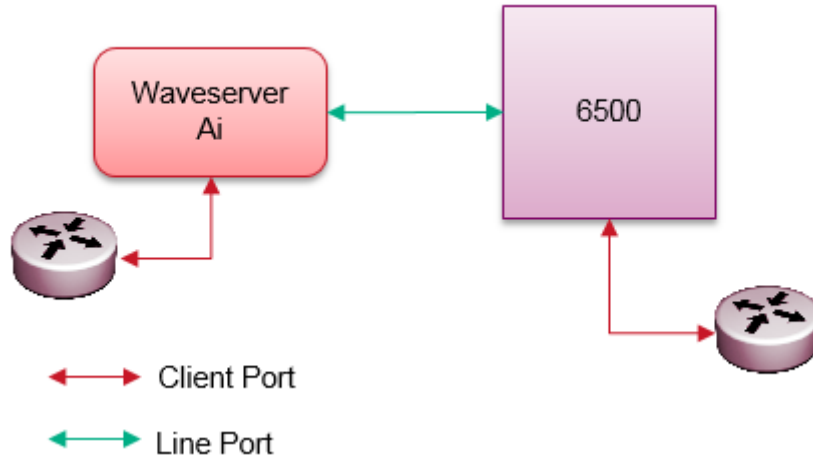
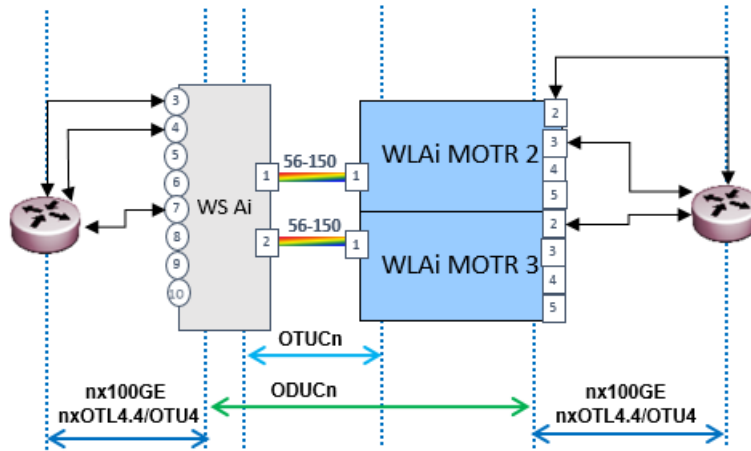


Figure 24 Waveserver Ai 2xWLAi + 8xQSFP28 module interworking with 6500 WLAi MOTR circuit pack as a transponder for 150G interworking



The following table provides the supported baud rates and line rates supported in [Figure 23](#).

Table 26 Supported baud rates and line rates

Baud rate (GBaud)	Line rate (Gbit/s)
56	100G and 100G SOP (Fast-Tracking) (Waveserver Ai Release 1.2 and higher)
	150G (Waveserver Ai Release 1.3 and higher)
	200G and 200G SOP (Fast-Tracking) (Waveserver Ai Release 1.2 and higher)
	250G (Waveserver Ai Release 1.3 and higher)
	300G (Waveserver Ai Release 1.2 and higher)
	400G (Waveserver Ai Release 1.2 and higher)
35	100G and 100G SOP (Fast-Tracking) (Waveserver Ai Release 1.2 and higher)
	150G (Waveserver Ai Release 1.3 and higher)
	200G (Waveserver Ai Release 1.2 and higher)

The following interworking considerations apply:

- Supports 56GBaud and 35/56GBaud interworking on line ports.
- Supports all client traffic types, alarms, conditioning, Performance Monitoring, Integrated Test Set, and L0CP (Layer 0 Control Plane).
- Regens cannot be used in conjunction with L0CP end to end.

For additional Waveserver Ai and 6500 WLAI interworking considerations, see [Waveserver Ai and 6500 WLAI MOTR interworking considerations](#).

Waveserver Ai 2xWLAI + 8xQSFP28 module interworking with 6500 WLAI MOTR circuit pack in a regen configuration

This section outlines the Waveserver Ai 2xWLAI + 8xQSFP28 module interworking with 6500 WLAI MOTR in a regen configuration. The 6500 WLAI MOTR is in a back-to-back client configuration and will act as a regen for Waveserver Ai. See the following figures.

Figure 25 Waveserver Ai 2xWLAi + 8xQSFP28 module interworking with 6500 WLAi MOTR circuit pack in a regen configuration

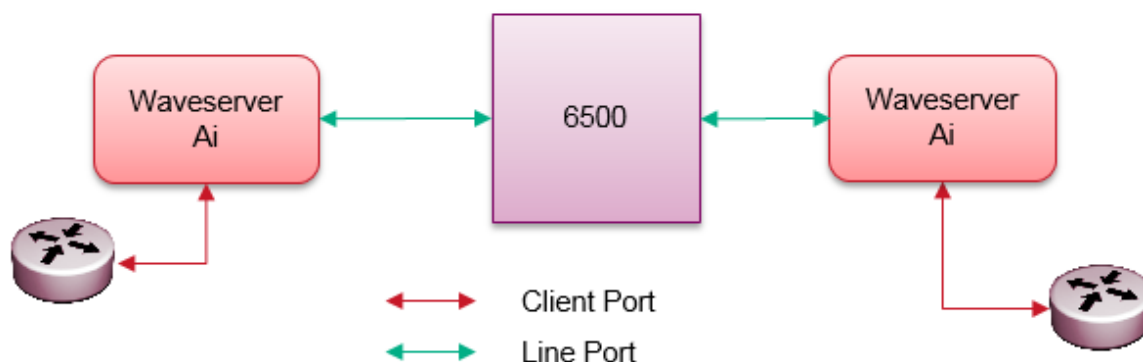
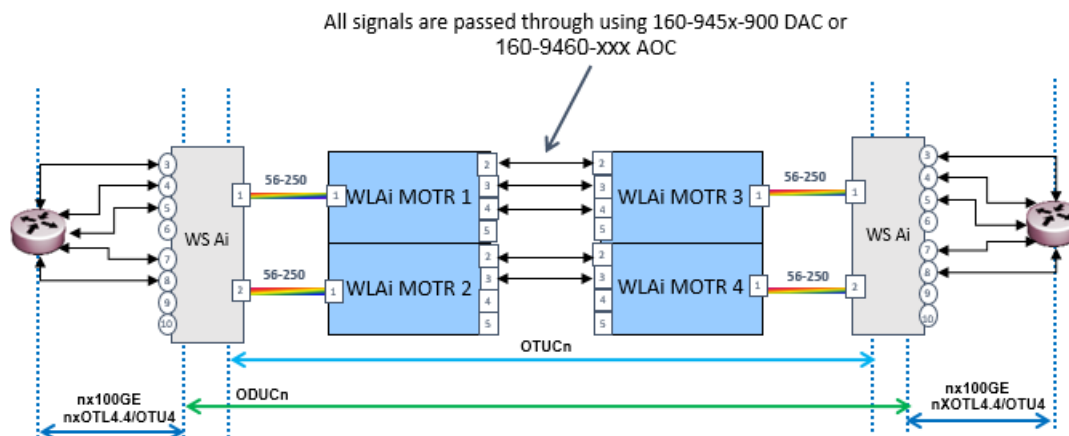


Figure 26 Waveserver Ai 2xWLAi + 8xQSFP28 module interworking with 6500 WLAi MOTR circuit pack in a regen configuration for 250G



The following table provides the supported baud rates and line rates supported in [Figure 25](#).

Table 27 Supported baud rates and line rates

Baud rate (GBaud)	Line rate (Gbit/s)
56	100G and 100G SOP (Fast-Tracking) (Waveserver Ai Release 1.2 and higher)
	150G (Waveserver Ai Release 1.3 and higher)
	200G and 200G SOP (Fast-Tracking) (Waveserver Ai Release 1.2 and higher)
	250G (Waveserver Ai Release 1.3 and higher)
	300G (Waveserver Ai Release 1.2 and higher)
	400G (Waveserver Ai Release 1.2 and higher)
35	100G and 100G SOP (Fast-Tracking) (Waveserver Ai Release 1.2 and higher)
	150G (Waveserver Ai Release 1.3 and higher)
	200G (Waveserver Ai Release 1.2 and higher)

The following interworking considerations apply:

- Supports 56GBaud and 35/56GBaud interworking on line ports.
- Supports all client traffic types, alarms, conditioning, Performance Monitoring, Integrated Test Set, and L0CP.
- DAC and AOC support at the regen site.
- x50G transmission modes are only supported on Waveserver Ai 2xWLAi + 8xQSFP28 modules (186-1601-900 and 186-1801-900) and 6500 WLAi MOTR circuit pack (NTK538DR/DS). See [Note](#).

Note: For Waveserver Ai Release 1.4 (and higher), Waveserver Ai 2x400G 35/56GBaud EDFA C-Band Premium 8xQSFP28 (186-1601-901) and Waveserver Ai 2x400G 35/56GBaud EDFA C-Band w/OPS 8xQSFP28 (186-1607-900) modules are also supported for this configuration.

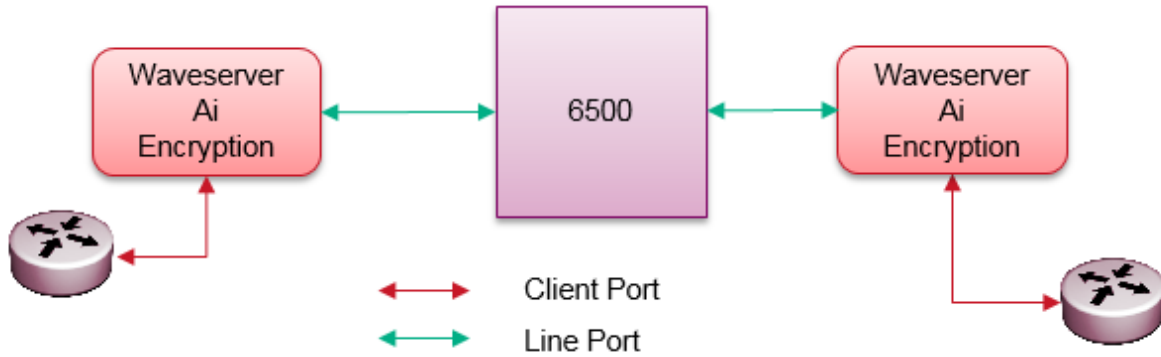
For additional Waveserver Ai and 6500 WLAi interworking considerations, see [Waveserver Ai and 6500 WLAi MOTR interworking considerations](#).

Waveserver Ai 1xWLAi + 4xQSFP28 Encryption module interworking with 6500 WLAi MOTR circuit pack in a regen configuration

This section outlines the Waveserver Ai 1xWLAi + 4xQSFP28 Encryption module interworking with 6500 WLAi MOTR circuit pack in a regen configuration. The 6500 WLAi MOTR clear text in a back-to-back client

configuration will act as a regen for Waveserver Ai. For more details on Waveserver Ai encryption, see *Waveserver Ai User Guide*, 323-4002-100. See the following figures.

Figure 27 Waveserver Ai 1xWLAi + 4xQSFP28 Encryption module interworking with 6500 WLAi MOTR circuit pack in a regen configuration



The following table provides the supported baud rates and line rates for this configuration.

Table 28 Supported baud rates and line rates

Baud rate (GBaud)	Line rate (Gbit/s)
56	100G and 100G SOP (Fast-Tracking) (Waveserver Ai Release 1.3 and higher)
	200G and 200G SOP (Fast-Tracking) (Waveserver Ai Release 1.3 and higher)
	300G (Waveserver Ai Release 1.3 and higher)
	400G (Waveserver Ai Release 1.3 and higher)
35	100G and 100G SOP (Fast-Tracking) (Waveserver Ai Release 1.3 and higher)
	200G (Waveserver Ai Release 1.3 and higher)

The following interworking considerations apply:

- The two end points of a secured data path must be equipped with encryption-enabled Waveserver Ai hardware for encryption/decryption to work. The nodes in the middle (for example, 6500 nodes with non-encryption circuit packs (regens) or a pure photonics system) do not participate in encryption/decryption operations.

- Waveserver Ai 1x400G Encryption 35/56GBaud EDFA C-Band 4xQSFP28 (186-1606-900) can interoperate with a 6500 regen constructed with the 6500 WLAi MOTR cards (NTK538DR/DS).
- Supports all client traffic types, alarms, conditioning, Performance Monitoring and LOCP.
- DAC and AOC support at the regen site.
- MCP 18.06 supports
 - Waveserver Ai and 6500 discovery
 - alarms reported on the Waveserver Ai and 6500

For information on Waveserver Ai Release 1.3, 6500 Release 12.3 and MCP 18.06 interworking considerations, see [Waveserver Ai Release 1.3](#), [6500 Release 12.3](#), and [MCP 18.06 interworking considerations](#).

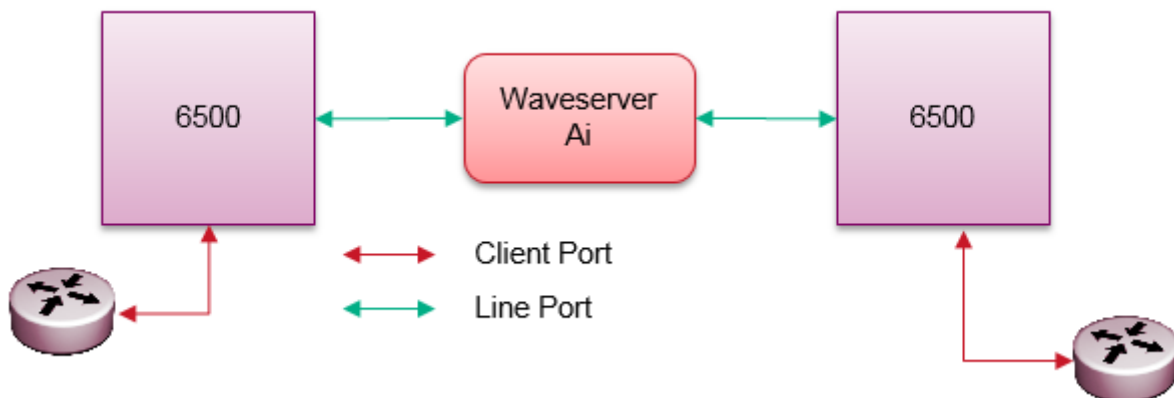
For additional Waveserver Ai and 6500 WLAi interworking considerations, see [Waveserver Ai and 6500 WLAi MOTR interworking considerations](#).

For more information on Waveserver Ai encryption, see *Waveserver Ai User Guide*, 323-4002-100

6500 WLAi MOTR circuit pack interworking with Waveserver Ai 2xWLAi + 8xQSFP28 module in a regen configuration

This section outlines the 6500 WLAi MOTR interworking with the Waveserver Ai 2xWLAi + 8xQSFP28 module in a regen configuration. The Waveserver Ai 2xWLAi + 8xQSFP28 module is in a back-to-back client configuration and will act as a regen for 6500. See the following figures.

Figure 28 6500 WLAi MOTR circuit pack interworking with Waveserver Ai 2xWLAi + 8xQSFP28 module in a regen configuration



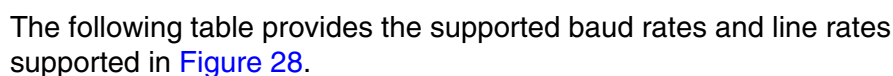


Table 29 Supported baud rates and line rates

Baud rate (GBaud)	Line rate (Gbit/s)
56	100G and 100G SOP (Fast-Tracking) (Waveserver Ai Release 1.2 and higher)
	150G (Waveserver Ai Release 1.3 and higher)
	200G and 200G SOP (Fast-Tracking) (Waveserver Ai Release 1.2 and higher)
	250G (Waveserver Ai Release 1.3 and higher)
	300G (Waveserver Ai Release 1.2 and higher)
	400G (Waveserver Ai Release 1.2 and higher)

Baud rate (GBaud)	Line rate (Gbit/s)
35	100G and 100G SOP (Fast-Tracking) (Waveserver Ai Release 1.2 and higher)
	150G (Waveserver Ai Release 1.3 and higher)
	200G (Waveserver Ai Release 1.2 and higher)

The following interworking considerations apply:

- Supports 56GBaud and 35/56GBaud interworking on line ports.
- Supports all client traffic types, alarms, conditioning, Performance Monitoring, and L0CP.
- AOC support at the regen site.
- x50G transmission modes are only supported on Waveserver Ai 2xWLAi + 8xQSFP28 modules (186-1601-900 and 186-1801-900) and 6500 WLAi MOTR circuit pack (NTK538DR/DS). See [Note](#).

Note: For Waveserver Ai Release 1.4 (and higher), Waveserver Ai 2x400G 35/56GBaud EDFA C-Band Premium 8xQSFP28 (186-1601-901) and Waveserver Ai 2x400G 35/56GBaud EDFA C-Band w/OPS 8xQSFP28 (186-1607-900) modules are also supported for this configuration.

For additional Waveserver Ai and 6500 WLAi interworking considerations, see [Waveserver Ai and 6500 WLAi MOTR interworking considerations](#).

Waveserver Ai 2xWLAi + 8xQSFP28 module interworking with 6500 WLAi MOTR circuit pack in a hybrid regen configuration

This section outlines the Waveserver Ai 2xWLAi + 8xQSFP28 module interworking with 6500 WLAi MOTR circuit pack in a hybrid regen configuration. The 6500 WLAi MOTR is in a back-to-back OTUCn client configuration and will act as a regen for Waveserver Ai. See the following figures.

Figure 31 Waveserver Ai 2xWLAi + 8xQSFP28 module interworking with 6500 WLAi MOTR circuit pack in a hybrid regen configuration

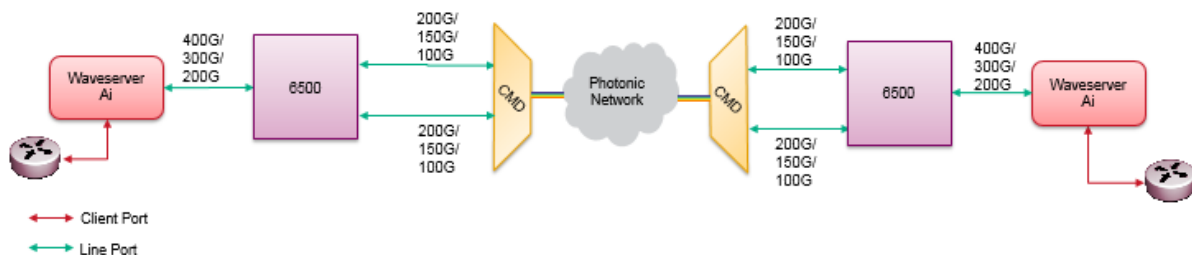


Figure 33 6500 WLAi MOTR interworking with Waveserver Ai 2xWLAi + 8xQSFP28 module circuit pack in a hybrid regen configuration

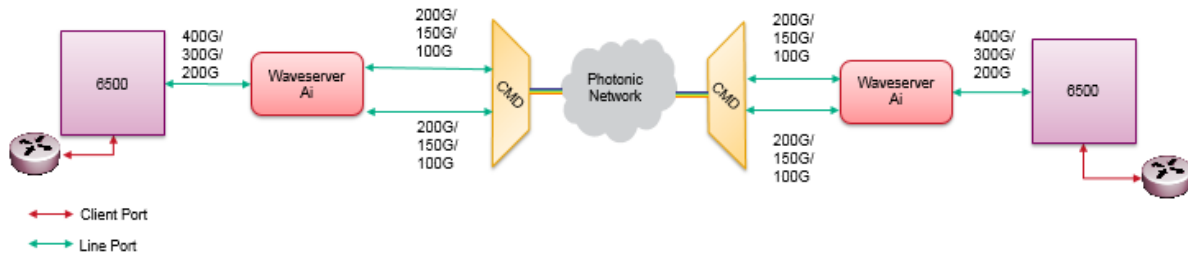
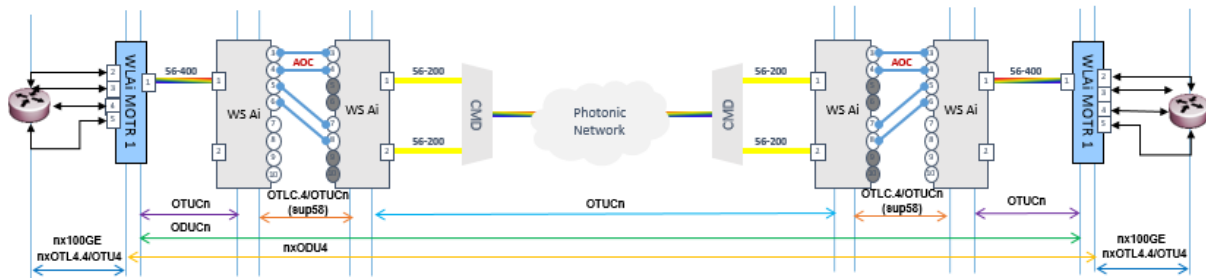


Figure 34 6500 WLAi MOTR interworking with Waveserver Ai 2xWLAi + 8xQSFP28 module circuit pack in a hybrid regen configuration for 1x400G <-> 2x200G



The following table provides the baud rates and line rates supported in [Figure 33](#).

Table 31 Supported baud rates and line rates

Baud rate (GBaud)	Line rate (Gbit/s)
56	1x200G <-> 2x100G (Waveserver Ai Release 1.2 and higher)
	1x300G <-> 2x150G (Waveserver Ai Release 1.2 and higher)
	1x400G <-> 2x200G (Waveserver Ai Release 1.2 and higher)

The following interworking considerations apply:

- Supports 56GBaud and 35/56GBaud interworking on line ports.
- Supported on Waveserver Ai modules 186-1601-900 and 186-1801-900. See [Note](#).

Note: For Waveserver Ai Release 1.4 (and higher), Waveserver Ai 2x400G 35/56GBaud EDFA C-Band Premium 8xQSFP28 (186-1601-901) is also supported for this configuration.

- Supports all client traffic types, alarms, conditioning, Performance Monitoring, and L0CP.

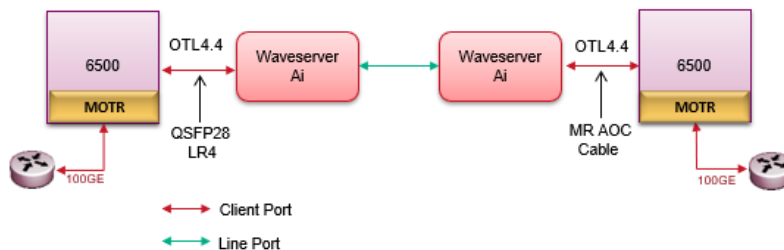
- AOC cables support at the regen site.
- OTUCn at the regen site.

For additional Waveserver Ai and 6500 WLAI interworking considerations, see [Waveserver Ai and 6500 WLAI MOTR interworking considerations](#).

Waveserver Ai 2xWLAi + 8xQSFP28 module interworking with 6500 WLAi MOTR circuit pack

This section outlines the Waveserver Ai 2xWLAi + 8xQSFP28 module interworking with the 6500 WLAi MOTR circuit pack. The Waveserver Ai OTU4 (OTL4.4) client port supports interworking with the 6500 WLAi MOTR (NTK538Dx) OTL4.4 clients. See the following figure:

Figure 35 Waveserver Ai OTU4 (OTL4.4) client port interworking with the 6500 WLAi MOTR (NTK538Dx) OTL4.4 clients



The following interworking considerations apply:

- Supported on Waveserver Ai Release 1.2 and higher.
- Supports 56GBaud and 35/56GBaud interworking on line ports.
- Supports OTL4.4 client pluggables: QSFP28 LR4 (160-9402-900), MR AOC.
- Supports alarms and Performance Monitoring.
- MCP 18.06 supports
 - Waveserver Ai and 6500 discovery
 - alarms and PMs reported on the Waveserver Ai and 6500

For information on Waveserver Ai Release 1.3, 6500 Release 12.3, and MCP 18.06 interworking considerations, see [Waveserver Ai Release 1.3, 6500 Release 12.3, and MCP 18.06 interworking considerations](#).

Waveserver Ai 2xWLAi + 8xQSFP28 module interworking with 6500 100G OCI circuit pack

This section outlines the Waveserver Ai 2xWLAi + 8xQSFP28 module interworking with the 6500 100G OCI circuit pack. The Waveserver Ai OTU4 (OTL4.4) client port supports interworking with the 6500 100G OCI (NTK529AC) OTU4 (OTL4.4) clients. See the following figures.

Figure 36 Waveserver Ai OTU4 (OTL4.4) client port interworking with the 6500 100G OCI (NTK529AC) OTU4 (OTL4.4) clients

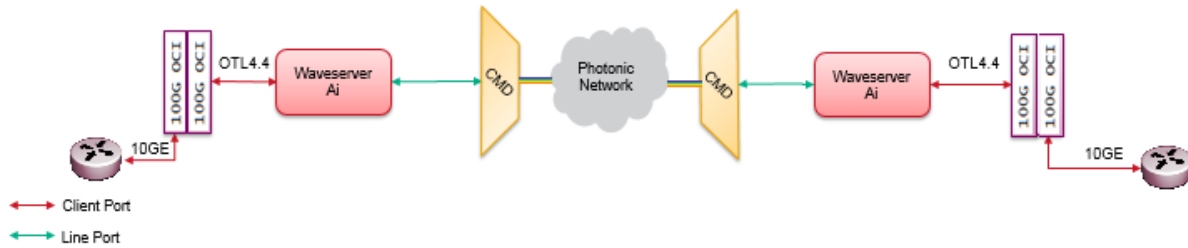
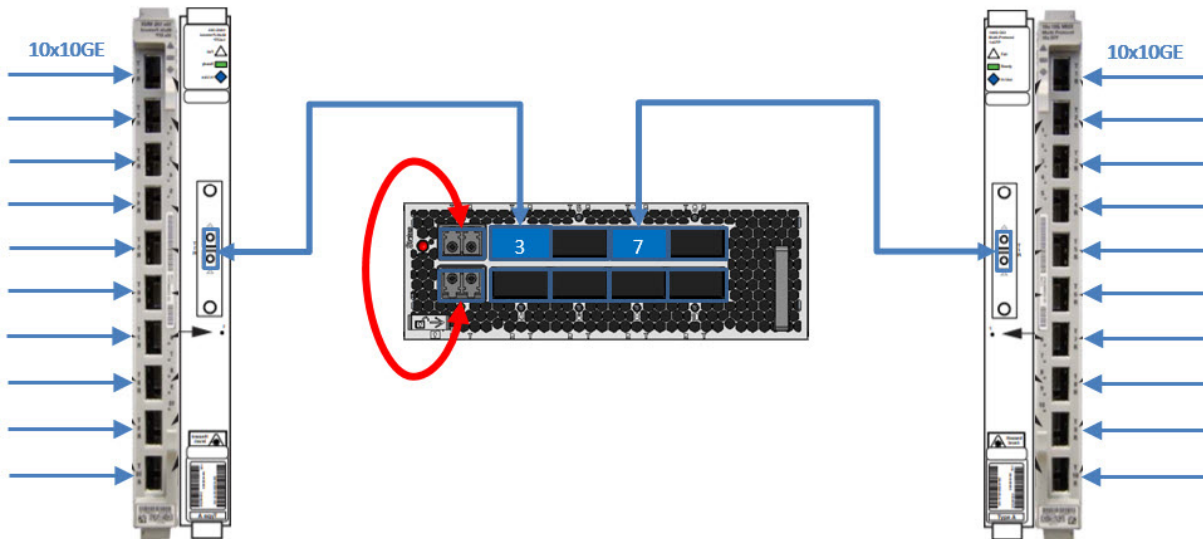


Figure 37 6500 100G OCI interworking with Waveserver Ai client side handoff



The following interworking considerations apply:

- Supported on Waveserver Ai Release 1.2 and higher
- Supports alarms and Performance Monitoring.
- Waveserver Ai client side handoffs are supported on all 8 client ports.
- Use the 103.1G-111.8G, 4x25G, WDM, SMF, 1310 nm, 10km CFP pluggable (160-9114-900) in the 100G OCI (NTK529AC) circuit pack.
- Only OTL4.4 client interoperability is supported.

- Ensure that the FEC is disabled on the Waveserver Ai client port, as the 100G OCI port does not support FEC.

Waveserver Ai 2xWLAi + 8xQSFP28 module interworking with 6500 200G OCI circuit pack

This section outlines the Waveserver Ai 2xWLAi + 8xQSFP28 module interworking with the 6500 200G OCI circuit pack. The Waveserver Ai OTU4 (OTL4.4) client port supports interworking with the 6500 200G OCI (NTK529HA) OTU4 (OTL4.4) clients. See the following figures.

Figure 38 Waveserver Ai OTU4 (OTL4.4) client port interworking with the 6500 200G OCI (NTK529HA) OTU4 (OTL4.4) clients

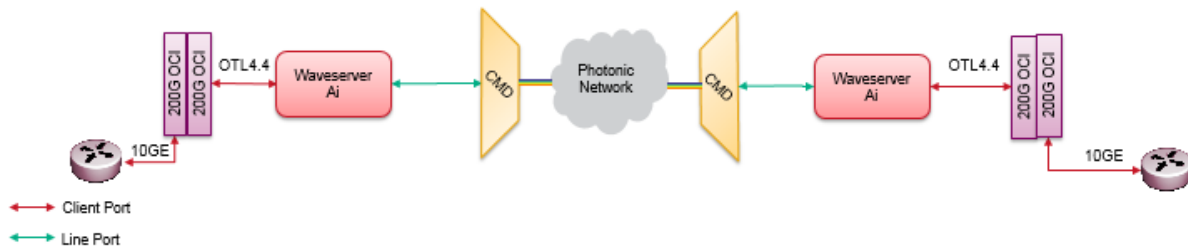
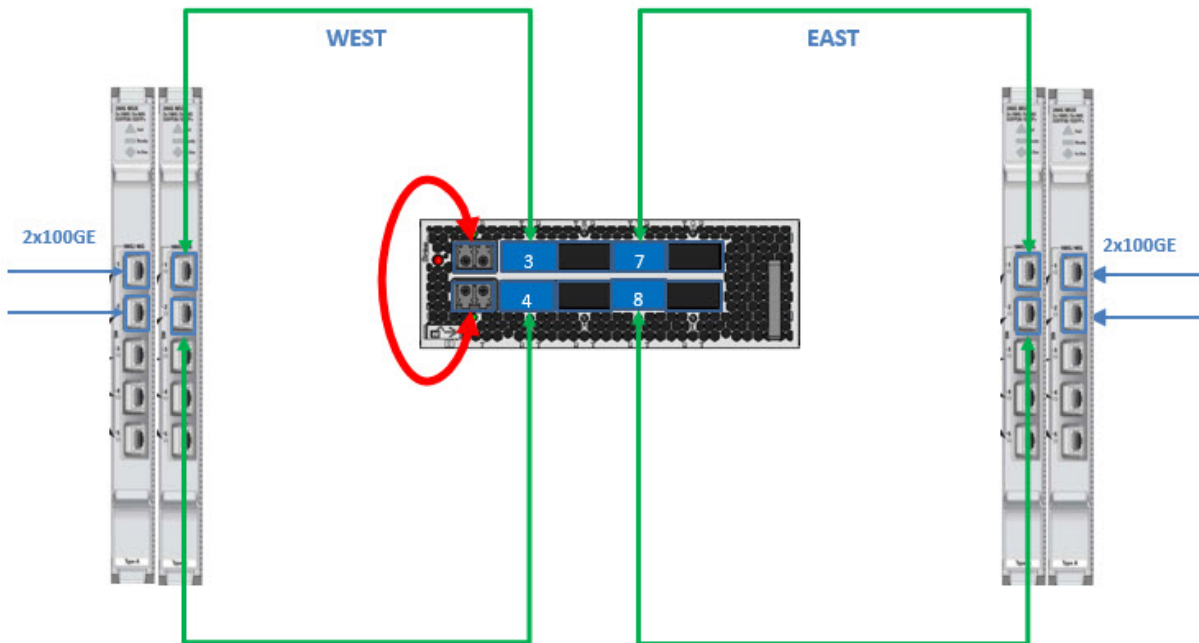


Figure 39 6500 200G OCI interworking with Waveserver Ai client handoff



The following interworking considerations apply:

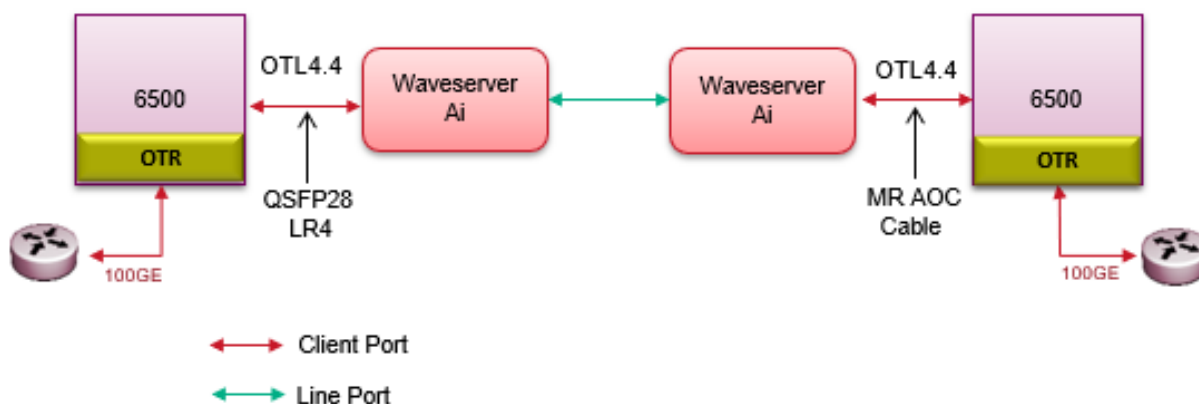
- Supported on Waveserver Ai Release 1.2 and higher.

- Supports 56GBaud and 35/56GBaud interworking on line ports.
- Only OTL4.4 client interoperability is supported.
- Supports OTL4.4 client pluggables: QSFP28 LR4, MR AOC.
- Supports 10GE client traffic.
- Supports alarms and Performance Monitoring.
- Waveserver Ai client side handoffs are supported on all 8 client ports.
- x00G Waveserver Ai regens are supported using pluggable transceiver 103.1G-111.8G, 4X25G, WDM, SMF, 1310nm, 10 km QSFP28 (160-9402-900) or Active Optical Cables.
- Ensure that FEC is enabled on the Waveserver Ai client port and the 200G OCl port.

Waveserver Ai 2xWLAi + 8xQSFP28 module interworking with the 6500 100G OTR circuit pack

This section outlines the Waveserver Ai 2xWLAi + 8xQSFP28 module interworking with the 6500 100G OTR circuit pack. The Waveserver Ai OTU4 (OTL4.4) client port supports interworking with the 6500 100G OTR (NTK538Ux) OTU4 (OTL4.4) clients. See the following figure:

Figure 40 Waveserver Ai OTU4 (OTL4.4) client port interworking with 6500 100G OTR (NTK538Ux) OTU4 (OTL4.4) clients



The following interworking considerations apply:

- Supported on Waveserver Ai Release 1.2 and higher.
- Supports 56GBaud and 35/56GBaud interworking on line ports.
- Supports OTL4.4 client pluggables: QSFP28 LR4 (160-9402-900), MR AOC.
- Supports alarms and Performance Monitoring.
- MCP 18.06 supports

- Waveserver Ai and 6500 discovery
- alarms and PMs reported on the Waveserver Ai and 6500

For information on Waveserver Ai Release 1.3, 6500 Release 12.3 and MCP 18.06 interworking considerations, see [Waveserver Ai Release 1.3](#), [6500 Release 12.3](#), and [MCP 18.06 interworking considerations](#).

Waveserver Ai Release 1.3, 6500 Release 12.3, and MCP 18.06 interworking considerations

For MCP to manage the Waveserver Ai and 6500, the following prerequisites apply:

- On the Waveserver Ai:
 - the member-name must be configured, use CLI command `system set member-name`.
 - SNMP must be provisioned and enabled on the Waveserver Ai for SNMP traps to be sent to the MCP server. For information on SNMP on the Waveserver Ai, see *Waveserver Ai User Guide*, 323-4002-100.
 - To launch 6500 Site Manager within MCP, you must increase the user login sessions on the 6500s.

To change the user login sessions in 6500 Site Manager:

 - Select **Configuration -> Comms Setting Management -> Services**.
 - Select **SSH/Telnet** in the Service Type drop-down list. Click **Edit**.
 - In the Edit SSH/Telnet Parameters dialog box, set the Maximum Number of Sessions for SSH to 12 and Telnet to 6.

The following limitations exist on MCP 18.06:

- You cannot create or add child entities on the OTNCn line port of the 6500 MOTR cards.
- MCP does not automatically discover or allow you to input the “number of tributary slots” required to create cross-connects on the 6500 MOTR cards.
- You cannot create cross-connects between the client PTP OTUCTP and the line PTP OTUCTP on the 6500 MOTR cards.
- You cannot discover links between the Waveserver Ai and 6500 over the client OTN connections in MCP.
- You cannot provision the client hold-off timer on the Waveserver Ai or 6500 using MCP.
- MCP allows you to provision transceiver type, ports, laser frequency, mode etc. on the 6500 MOTR cards, but you cannot edit these objects or attributes. To edit these objects or attributes, use Site Manager.

- MCP does not support provisioning TTI (Trail-Trace Identifier) on the Waveserver Ai.
- MCP does not support running ITS (Integrated Test Set) on the 6500 or Waveserver Ai.
- MCP supports warm/cold restarts of the Waveserver A chassis/Waveserver Ai modules. MCP does not support warm/cold restarts of the 6500 shelf/6500 cards.

Waveserver Ai Release 1.4 and 6500 Release 12.4 interworking

This section provides information on Waveserver Ai Release 1.4 and 6500 Release 12.4 interworking, including:

- [Waveserver Ai Release 1.4 fast tuning mode support](#)
- [Waveserver Ai 1xWLAi + 10xQSFP+/QSFP28 module interworking with 6500 WLAi MOTR circuit pack](#)
- [Waveserver Ai 2x400G 35/56GBaud EDFA C-Band w/OPS 8xQSFP28 module interworking with 6500 photonic line systems](#)

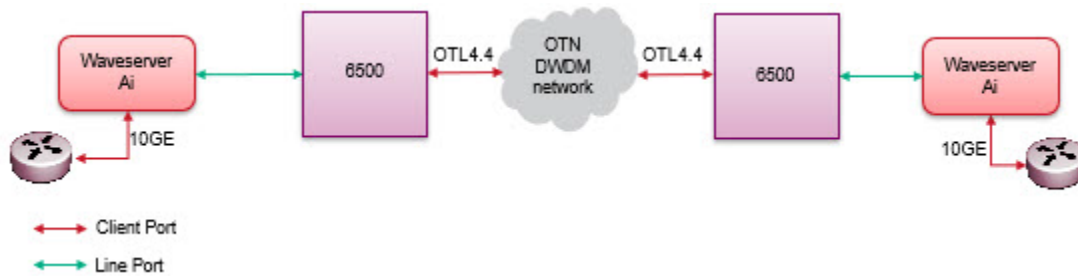
Note: For additional Waveserver Ai and 6500 WLAi interworking considerations, see [Waveserver Ai and 6500 WLAi MOTR interworking considerations](#).

Waveserver Ai Release 1.4 fast tuning mode support

Waveserver Ai Release 1.4 introduces support for fast tuning mode via SPLI. When a new channel is created on the 6500 (Release 12.4) with the channel capacity modes “Mode-2” or “Mode-3”, the ADJ-Tx Tuning mode is set to fast tuning and this mode is pushed to the Waveserver Ai via SPLI. The Tx Tuning mode attribute can be displayed on the Waveserver Ai using CLI command `spli adjacency show ptp <ptpList>` and it can also be retrieved on the 6500 as Discovered Tx Tuning Mode.

Waveserver Ai 1xWLAi + 10xQSFP+/QSFP28 module interworking with 6500 WLAi MOTR circuit pack

This section outlines the Waveserver Ai 1xWLAi + 10xQSFP+/QSFP28 module interworking with 6500 WLAi MOTR circuit pack. The 10GE traffic coming into the Waveserver Ai 1xWLAi + 10xQSFP+/QSFP28 module hands off to the 6500 MOTR card, which extends into an OTN DWDM network through the OTL4.4 client interfaces. See the following figure.

Figure 41 Waveserver Ai 1x WLAi + 10x QSFP+/QSFP28 module interworking with 6500 WLAi MOTR circuit pack

The following table provides the supported baud rates and line rates for this configuration.

Table 32 Supported baud rates and line rates

Baud rate (GBaud)	Line rate (GBit/s)
56	100G and 100G SOP (Fast-Tracking)
	200G and 200G SOP (Fast-Tracking)
	300G
	400G
35	100G and 100G SOP (Fast-Tracking)
	200G

The following interworking considerations apply:

- Supports traffic, alarms, Performance Monitoring, ITS, and L0CP.
- MCP 3.0.1 supports
 - launching Waveserver Ai CLI and 6500 Site Manager
 - Waveserver Ai and 6500 discovery
 - alarms and PMs reported on the Waveserver Ai and 6500

For information on Waveserver Ai Release 1.4, 6500 Release 12.4 and MCP 3.0.1 interworking considerations, see [Waveserver Ai Release 1.4](#), [6500 Release 12.4](#), and [MCP 3.0.1 interworking considerations](#).

Waveserver Ai Release 1.4, 6500 Release 12.4, and MCP 3.0.1 interworking considerations

For MCP to manage the Waveserver Ai and 6500, the following prerequisites apply:

- On the Waveserver Ai:

- the member-name must be configured, use CLI command `system set member-name`.
- SNMP must be provisioned and enabled on the Waveserver Ai for SNMP traps to be sent to the MCP server. For information on SNMP on the Waveserver Ai, see *Waveserver Ai User Guide*, 323-4002-100.
- To launch 6500 Site Manager within MCP, you must increase the user login sessions on the 6500s.
To change the user login sessions using 6500 Site Manager:
 - Select **Configuration -> Comms Setting Management -> Services**.
 - Select **SSH/Telnet** in the Service Type drop-down list. Click **Edit**.
 - In the Edit SSH/Telnet Parameters dialog box, set the Maximum Number of Sessions for SSH to 12 and Telnet to 6.

The following limitations exist on MCP 3.0.1:

- You cannot create or add child entities on the OTNCn line port of the 6500 MOTR cards.
- MCP does not automatically discover or allow users to input the “number of tributary slots” required to create cross-connects on the 6500 MOTR cards.
- You cannot create cross-connects between the client PTP OTUCTP and the line PTP OTUCTP on the 6500 MOTR cards.
- MCP allows you to provision transceiver type, ports, laser frequency, mode etc. on the 6500 MOTR cards, but you cannot edit these objects or attributes. To edit these objects or attributes, use Site Manager.
- MCP does not support provisioning TTI (Trail-Trace Identifier) on the Waveserver Ai.
- MCP does not support running ITS (Integrated Test Set) on the 6500 or Waveserver Ai.
- MCP supports warm/cold restarts of the Waveserver A chassis/Waveserver Ai modules. MCP does not support warm/cold restarts of the 6500 shelf/6500 cards.

Waveserver Ai 2x400G 35/56GBaud EDFA C-Band w/OPS 8xQSFP28 module interworking with 6500 photonic line systems

The Waveserver Ai 2x400G 35/56GBaud EDFA C-Band w/OPS 8xQSFP28 module supports 6500 photonic line systems listed in section [Waveserver Ai interworking with 6500 photonics configurations](#).

If an “Adjacency Mismatch” alarm is raised on the 6500 and the 6500 parameter settings are not being pushed to the Waveserver Ai PTP, verify that the required working and protection port parameters being received from the 6500 to the Waveserver Ai match. Below are the required parameters that must match on the working and protection ports:

- Line System Type
- Tx Tuning Unit
- Frequency
- Wavelength
- Transmission Mode
- Tx Blanking
- Laser Centering
- Laser Centering Range
- Link Optimization Mode
- Dispersion Mode
- Pre-compensation
- Tx Tuning Mode

To verify the working and protection port parameters being received on the Waveserver Ai from the 6500, on the Waveserver Ai, use CLI command `spli adjacency show ptp <slot>/<ptp>`. If there are differences in the working and protection port parameters that must match (based on the list above), modifications must be made on the 6500 ADJ-TXs to ensure those working and protection port parameters are matched.

Note: For SPLI considerations for this module, see “SPLI with 2x400G 35/56GBaud EDFA C-Band w/OPS 8xQSFP28 module (186-1607-900) considerations” in the *Waveserver Ai User Guide*, 323-4002-100.

Waveserver Ai and 6500 WLAi MOTR interworking considerations

For x00G transmission modes on Waveserver Ai, use the default settings for the OTSi Group Identifier (GID) and the Physical Identification (PID) on the 6500. The GID default value is 1 and the PID default value is 0. Only one OTSi entity is supported per OTSi group. Refer to “Retrieving equipment and facility details” in *6500 Configuration - Provisioning and Operating*, 323-1851-310 for details. Also, changing the transmission mode on Waveserver Ai automatically updates the OTUCn rate and the associated number of Tributary slots; this may raise a “Service Mismatch” alarm on the 6500 nodes, as the 6500 is expecting to receive the original OTUCn rate and associated number of Tributary Slots. To prevent this alarm and subsequent loss of traffic on the 6500, set the OTUCn rate to OTUC4 and the Number of Tributary Slots

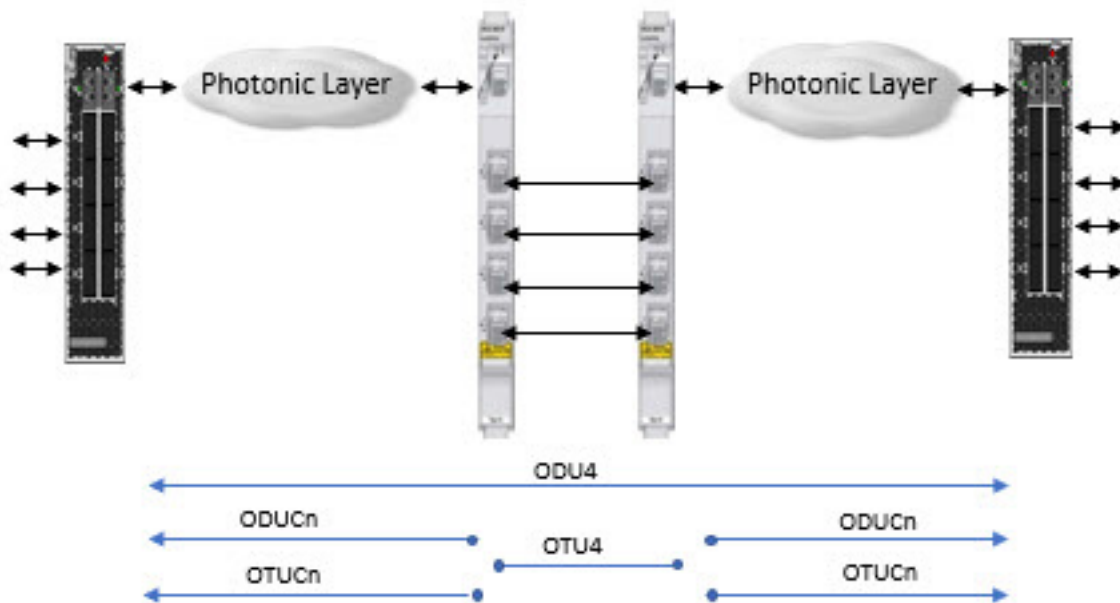
to 80. Refer to *6500 Configuration - Provisioning and Operating*, 323-1851-310.1 for information about configuring the OTUCn rate and Number of Tributary Slots.

For x50G transmission modes on Waveserver Ai, use the default setting for the OTSi Group Identifier (GID) and Physical Identifier (PID) for both the Primary and Mate MOTRs on the 6500. The GID default value is 1. The PID default value for the Primary is 0 and the PID default value for the Mate is 1. Also, on the 6500, set the OTUCn rate to OTUC8 and the Number of Tributary Slots to 160.

Waveserver Ai and 6500 WLAI MOTR interworking considerations for x00G regen configurations

6500 WLAI MOTR can be configured to support OTU4 regen and OTUC4 regen. The OTU4 method terminates, deconstructs and repackages the client payload into an OTU4 format, while the OTUC4 method is transparent to client data and transmits OTUC4 over 4 WLAI client ports. Both options maintain end-to-end authentication of the encrypted OTU4 link and client traffic, but the OTUC4 is the only method which provides true regen, as the OTU4 method terminates the Waveserver Ai line as shown in the figure below.

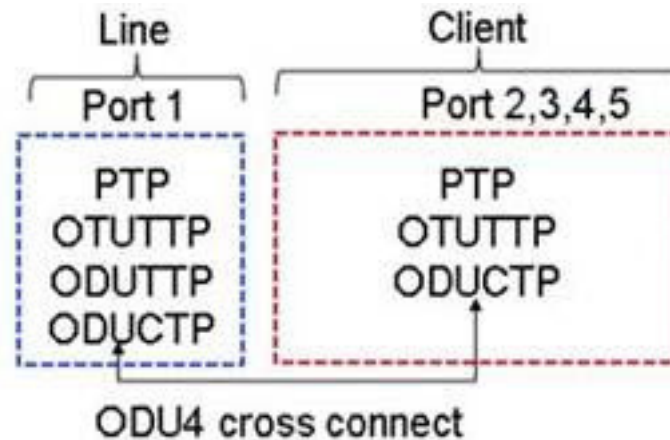
Figure 42 OTU4 method: ODU4 regen configuration showing termination of ODUcN



OTU4 method provisioning considerations

The client ports in the 6500 WLAi MOTR must be provisioned as OTU4 at the OTUTTP layer and the line port must be provisioned as an OTUC4 at the OTUTTP layer. For Waveserver Ai (Release 1.3 and lower), all 4 ODUCTPs must be created on the 6500 line port, even if all client connections are not required. See the following figure.

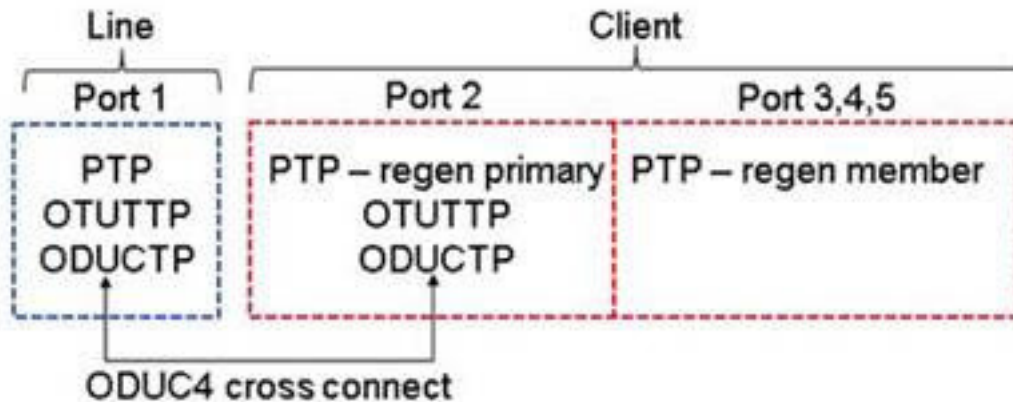
Figure 43 Site manager line and client facility model for ODU4 regen on a 6500 WLAi MOTR circuit pack



Note: For the line facility at the OTUTTP layer, OTUC4 must be selected.

OTUC4 method provisioning considerations

If you are using the 6500 WLAi MOTR cards in a x00G regen configuration, all four client ports must be provisioned as DAC or AOC cables, even if you are using less than 400G for the regen. Only when all client port PTP facility types are provisioned (one primary-port 2 and three members-ports 3, 4, 5) will the option for OTUTTP type OTUC4 be available on the primary client port. If any active client port fails in this scenario, the entire regen fails. The line port must not be provisioned as an ODUTTP for regen, it must be set to ODUCTP. For the OTUC4 regen provisioning method, there is one ODUCTP on the Primary line port 1 under the OTUTTP facility and one ODUCTP on the regen Primary client port 2 underneath the OTUTTP, these two ODUCTPs must have a cross-connection in order for traffic to flow. See the following figure.

Figure 44 Site manager line and client facility model for ODU4 regen

Note: For the 6500 WLAi regen to be functional, the regen primary and regen members must be provisioned.

Waveserver Ai (Release 1.3 and higher) and 6500 WLAi MOTR interworking considerations for x50G regen configurations

There are two methods of provisioning regens for the 6500 WLAi site for Waveserver Ai (Release 1.3 and higher) and 6500 WLAi interworking: the OTU4 method of provisioning and the OTUC8 method of provisioning. The OTUC8 method is the only method which provides true regen.

Waveserver Ai (Release 1.3 only) OTU4 method provisioning considerations

For the OTU4 regen provisioning method, all 8 ODUCTPs must be created on the 6500 line ports, even if all client connections are not required.

Waveserver Ai (Release 1.3 and higher) OTUC8 method provisioning considerations

For the OTUC8 regen provisioning method, there is one ODUCTP on the Primary line port 1 under the OTUTTP facility and one ODUCTP on the regen Primary client port 2 underneath the OTUTTP, these ODUCTPs must have a cross-connection in order for traffic to flow.

Testing traffic in x50G regen configuration for Waveserver Ai (Release 1.3 only) and 6500 interworking

To bring up traffic for Waveserver Ai and 6500 WLAi MOTR x50G regen configurations:

- 1 Connect all fibers for the Waveserver Ai and 6500 WLAi MOTR x50G regen configuration.
- 2 On the 6500, create 8 line ODUCTPs (for the OTU4 method provisioning) or create one line ODUCTP (for OTUC8 method provisioning). Ensure there are no connections to the client ODUCTP. If these connections exist, delete these connections.

- 3 Confirm line side traffic is running between the Waveserver Ai and 6500 WLAI MOTR cards. Clear any unexpected alarms.
- 4 Once traffic has been confirmed from the previous step, create connections between the line and client ODUCTPs on the 6500.

Waveserver Ai Release 1.4 and 6500 Release 12.4 considerations

When Waveserver Ai Release 1.4 and 6500 Release 12.4 are interworking, the discovered frequency on the 6500 may be automatically set to 0.

The discovered frequency on the 6500 is automatically set to 0 but the Waveserver Ai frequency remains unchanged (does not get set to 0), when all these conditions are met:

- You are using Waveserver Ai Release 1.4 and 6500 Release 12.4.
- On the Waveserver Ai, you are using the Waveserver Ai 2x400G 35/56GBaud EDFA C-Band w/OPS 8xQSFP28 module (186-1607-900) with 6500 SPLI in a protected configuration.
- On the 6500, the frequency is set to xxx.xxxxxyz, where either y and z are non-zero.
- The transmission mode on the 6500 ADJ-TX, which is connected to one of the Waveserver Ai iOPS ports is user modified. Following this, the transmission mode on the 6500 ADJ-TX, which is connected to the other Waveserver Ai iOPS protected port is also modified (to match the other protection port).

To recover from the 6500 discovered frequency being set to 0, modify a parameter on the 6500 ADJ-TX or warm restart the Waveserver Ai; these actions repopulate the 6500 discovered frequency.

Note: When the 6500 discovered frequency is set to 0, no adjacency mismatch alarms are raised on the 6500 or the Waveserver Ai.

List of procedures

- [Procedure 8, “Verifying 6500 and Waveserver Ai parameters for interworking configurations on the line port”](#)
- [Procedure 9, “Configuring 6500 and Waveserver Ai parameters for interworking configurations on the client port”](#)
- [Procedure 10, “Creating 6500 and Waveserver Ai fiber connections for Fixed-64 or Flex-12 configuration”](#)
- [Procedure 11, “Verifying 6500 threshold and power level settings for protected configurations”](#)
- [Procedure 12, “Setting Line B target gain for Extended Reach and line-amp configurations”](#)

- Procedure 13, “Setting 6500 Line B Target Peak Power/Gain Offset and Waveserver Ai Output Power for configurations using Waveserver Ai with EDFA”
- Procedure 14, “Resetting the TCA baselines”
- Procedure 15, “Calibrating the SRA circuit packs”
- Procedure 16, “Verifying traffic continuity for unprotected Fixed-64 and Flex-12 configurations”
- Procedure 17, “Verifying traffic continuity for Fixed-64 OPS protected configuration”
- Procedure 18, “Configuring 6500-based SLTE configurations using Waveserver Ai wayside channel”

Procedure 8 Verifying 6500 and Waveserver Ai parameters for interworking configurations on the line port

Use this procedure to verify that the 6500 configuration matches the Waveserver Ai for interoperability on the line. The transmission mode must match on the 6500 nodes and on the Waveserver Ai nodes. Also, the 6500 bay number must match the Waveserver Ai group identifier.

Requirements

To perform this procedure you must:

- have configured the 6500 nodes in the solution, according to the Commissioning section of this document
- have configured Waveserver Ai nodes in the solution
- be logged into the 6500 and Waveserver Ai CLI
- have the engineering documentation package (EDP) or installation documentation package (IDP) or equivalent site/network engineering plans

Steps

Verifying the transmission mode on the 6500 and Waveserver Ai

- 1 The transmission mode on the Waveserver Ai should match the transmission mode on the 6500.

If you are interworking Waveserver Ai with	Then go to
6500 WLAi	step 2
6500 photonics with SPLI on the CMD	step 3
- 2 To verify the transmission mode on the 6500 WLAi, enter

```
> port ptp show ptp-shelf-slot-port
```

Output example:

```
trans-mode | "56GBAUD_200G"
```

Go to [step 4](#).
- 3 To verify the transmission mode for 6500 photonics with SPLI on the CMD, enter

```
adjacencies tx show ADJ-shelf-slot-port
```
- 4 To verify the transmission mode on the near-end Waveserver Ai shelf, enter

```
> xcvr show xcvr <slot>/<xcvr>
```

Output example:

```
Mode | 56-200
```
- 5 Repeat [step 1](#) for the far-end 6500 and [step 4](#) for the far-end Waveserver Ai.

- 6 Verify the transmission modes on 6500 and Waveserver Ai match.
- | If | Then go to |
|---|-------------------------|
| the transmission modes on 6500 and Waveserver Ai match | step 13 |
| the 6500 transmission mode need to be changed | step 7 |
| the Waveserver Ai transmission mode needs to be changed | step 11 |
- 7 Consult with your network documentation to verify the correct 6500 transmission mode for the near-end 6500 shelf.
- | If you want to change the 6500 transmission mode on the | Then |
|---|------------------------|
| 6500 WLAi | step 8 |
| 6500 photonics with SPLI on the CMD | step 9 |
- 8 On the 6500 WLAi, enter
- ```
> port ptp set ptp-shelf-slot-port transmode
<56GBAUD_100G | 56GBAUD_150G | 56GBAUD_200G |
56GBAUD_250G | 56GBAUD_300G | 56GBAUD_400G | 35GBAUD_100G
| 35GBAUD_150G | 35GBAUD_200G>
```
- Go to [step 10](#).
- 9 On the 6500 CMD, enter
- ```
> adjacencies tx set ADJ-shelf-slot-port mod-class
<56GBAUD_100G | 56GBAUD_150G | 56GBAUD_200G |
56GBAUD_250G | 56GBAUD_300G | 56GBAUD_400G | 35GBAUD_100G
| 35GBAUD_150G | 35GBAUD_200G>
```
- 10 If required, repeat [step 7](#) for far-end 6500, otherwise go to [step 13](#).
- 11 Consult with your network documentation to verify the correct transmission mode for each Waveserver Ai. To change the near-end Waveserver Ai transmission mode, enter
- ```
> xcvr disable xcvr <slot>/<xcvr>
> xcvr set xcvr <slot>/<xcvr> mode <56-100 | 56-150 | 56-
200 | 56-250 | 56-300 | 56-400 | 35-100 | 35-150 | 35-200>
```
- Note 1:** After the mode is set, the XCVR will be automatically enabled.
- Note 2:** If SPLI is enabled, editing the transmission mode should always be performed on the 6500 CMD ADJ-TX using CLI command `adjacencies tx set ADJ-shelf-slot-port mod-class` `<56GBAUD_100G | 56GBAUD_150G | 56GBAUD_200G | 56GBAUD_250G | 56GBAUD_300G | 56GBAUD_400G | 35GBAUD_100G | 35GBAUD_150G | 35GBAUD_200G>`.
- 12 If required, repeat [step 11](#) for far-end Waveserver Ai.

#### Verifying 6500 bay number with Waveserver Ai group identifier

- 13 The group identifier on Waveserver Ai shelves should match the bay number on the 6500 nodes. To verify the bay number of the near-end 6500 shelf, enter
- ```
> shelves shelf show
```

Output example:

```
bay-num | 17
```

- 14** To verify the group identifier of the near-end Waveserver Ai shelf, enter
> system show configuration

Output example:

```
| Group | | Identifier | 17
```

- 15** Repeat [step 13](#) for the far-end 6500 and [step 14](#) for the far-end Waveserver Ai.
- 16** Verify the 6500 bay numbers and Waveserver Ai group identifiers match.

If	Then go to
the 6500 bay numbers and Waveserver Ai group identifiers match	step 21
the 6500 bay numbers need to be changed	step 17
the Waveserver Ai group identifiers need to be changed	step 19

- 17** Consult with your network documentation to verify the correct 6500 bay number for the near-end 6500 shelf. To change the 6500 bay-number, enter
> shelves shelf set shelf-1 bay-num [< 6500 bay number >]
- 18** If required, repeat [step 17](#) for the far-end 6500 shelf.
Go to [step 21](#)
- 19** Consult with your network documentation to verify the correct group-id for each Waveserver Ai. To change the near-end Waveserver Ai group-id, enter
> system set group-id [< Waveserver Ai Group-id >]
- 20** If required, repeat [step 19](#) for the far-end Waveserver Ai.
- 21** Go to procedure [Creating 6500 and Waveserver Ai fiber connections for Fixed-64 or Flex-12 configuration](#).

—end—

Procedure 9 Configuring 6500 and Waveserver Ai parameters for interworking configurations on the client port

Use this procedure to verify the 6500 configuration matches the Waveserver Ai for interoperability on the client.

Requirements

To perform this procedure you must:

- have configured the 6500 nodes in the solution, according to the Commissioning section of this document
- have configured Waveserver Ai nodes in the solution
- have the 6500 and Waveserver Ai fibered in the interworking configuration
- be logged into the 6500 and Waveserver Ai CLI
- have the engineering documentation package (EDP) or installation documentation package (IDP) or equivalent site/network engineering plans

Steps

Setting up client port on the 6500

- 1 To set the port service-type for the client port on the 6500.
`port ptp create ptp-shelf-slot-port service-type OTU4`
- 2 To verify the port service-type for the client port on the 6500.
`port ptp show ptp-shelf-slot-port`
- 3 Perform [step 1](#) to [step 2](#) on far-end 6500.

Setting up client port on the Waveserver Ai

- 4 To set the transceiver mode for the client port on the Waveserver Ai.
`xcvr set xcvr <slot>/<xcvr> mode OTL4.4`
- 5 To verify the transceiver mode for the client port on the Waveserver Ai.
`xcvr show xcvr <slot>/<xcvr>`
- 6 Perform [step 4](#) to [step 5](#) on far-end Waveserver Ai.
—end—

Procedure 10 Creating 6500 and Waveserver Ai fiber connections for Fixed-64 or Flex-12 configuration

Use this procedure to connect fibers between the 6500 and Waveserver Ai for a fixed channel configuration.

Requirements

To perform this procedure, you must:

- have installed all 6500 and Waveserver Ai nodes in the solution
- have completed procedure [Verifying 6500 and Waveserver Ai parameters for interworking configurations on the line port](#) and verified that all 6500 and Waveserver Ai nodes are configured correctly
- be logged into the 6500 and Waveserver Ai CLI

Steps

Verifying Waveserver Ai line-port frequencies and CMD ports

- 1 To verify the frequency for the near-end Waveserver Ai line-ports (1 and 2) along with the specified physical CMD port connection, enter:

```
> ptp show ptp <slot>/<ptp>
```

where <slot> = 1 to 3 and <ptp> = 1 or 2

Output example:

```
| Transmitter Wavelength (nm) | 1528.71 | 1528.71 |
```

```
> spli adjacency show
```

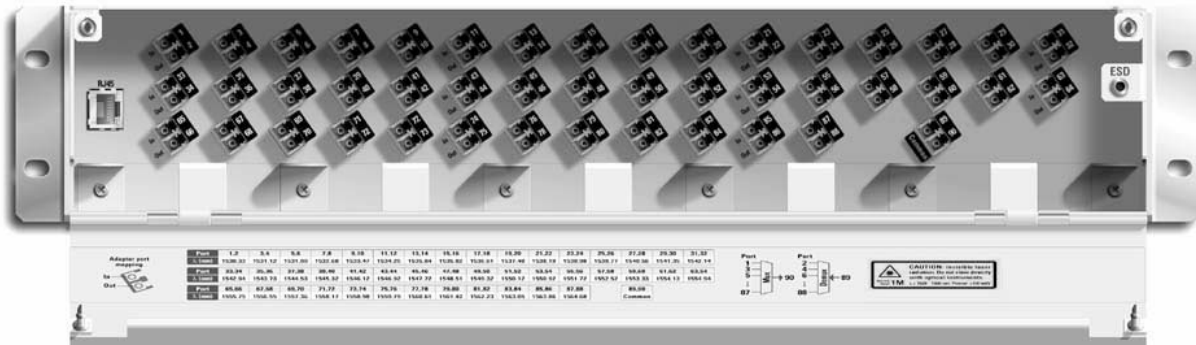
Output example:

SPLI PORT ADJACENCIES									
Port	Admin	Oper	Remote Information						
(*=dup)	State	State	Shelf ID	Slot	SubSlot	Port	SubPort	Shelf	TID
1/1	Enabled	Up	1	83	0	3	0	Site-B	
1/2	Enabled	Up	1	83	0	65	0	Site-B	

- 2 Repeat [step 1](#) for the far-end Waveserver Ai.

6500 CMD ports

- 3 Use the front-plate of the CMD you are connecting to in order to verify the ports specified by Waveserver Ai. The front-plate has a schematic, listing the wavelengths, and which set of Tx/Rx ports match up with that particular wavelength. See the following figure.

Figure 45 Wavelengths and Tx/Rx ports on front plate of CMD**Connecting fibers****DANGER****Risk of laser radiation exposure**

Do not look directly into the optical beam. Invisible light can severely damage your eyes.

**CAUTION****Risk of damage to modules**

Wear an antistatic wrist strap to protect the equipment from static damage. Connect the wrist strap to the ESD jack on the shelf or module.

- 4 The fiber connections between the 6500 CMD and the Waveserver Ai line ports require single-mode fiber with LC-LC connectors. After the port connections have been identified, refer to the “Inspecting and cleaning optical interface connectors” procedure, found in the *6500 Installation-General Information*, 323-1851-201 before cleaning the connectors.
Note: The following steps refer to the Waveserver Ai line ports 1 and 2.
- 5 Using a power meter, verify the Tx power being transmitted from Waveserver Ai Line port 1.
Note: The Tx power from a Waveserver Ai line port is set to -10dB so the value registered on the power meter should be very close to -10dB.
- 6 Connect the Tx from port 1 on the near-end Waveserver Ai to the odd-numbered port of the identified pair on the CMD and connect the Rx from port 1 to the even-numbered port on the CMD.
- 7 Using a power meter, verify the Tx power being transmitted from Waveserver Ai line port 2.
Note: The Tx power from a Waveserver Ai line port is set to -10dB so the value registered on the power meter should be very close to -10dB.
- 8 Connect the Tx from port 2 on the near-end Waveserver Ai to the odd-numbered port of the identified pair on the CMD and connect the Rx from port 2 to the even-numbered port on the CMD.

9 Repeat [step 3](#) to [step 8](#) on the far-end 6500 and Waveserver Ai. If the far end fiber connections are added, continue with next step.

10	If you have	Then go to
	a protected configuration	Verifying 6500 threshold and power level settings for protected configurations
	an Extended Reach or line-amp configuration	Setting Line B target gain for Extended Reach and line-amp configurations
	an unprotected Standard Reach configuration	Resetting the TCA baselines

—end—

Procedure 11 Verifying 6500 threshold and power level settings for protected configurations

Use this procedure to complete the configuration for 6500 terminal nodes that are in protected configurations.

Requirements

To complete this procedure, make sure you have:

- performed the commissioning procedures for terminal nodes as described in the Commissioning chapter of this technical publication
- connected the Waveserver Ai wavelengths to the 6500 photonic system and have the minimum number of channels as required in table [LOS Thresholds for OPS SW1 In and OPS SW2 In ports for protected configurations](#). If this procedure is attempted with fewer than the number of required channels, there will not be enough optical power in the photonic system to clear the OPS LOS alarm.
- logged into the 6500 and Waveserver Ai CLI

Note: To set the OPTMON LOS thresholds in ROADM networks where the OPS is located between the Waveserver Tx/Rx and the CMD, and where the channel count is higher than the minimum channel count required, the OPS LOS threshold for SW1/SW2 IN OPTMON facility is provisioned 6 dB below the SW1/SW2 IN OPTMON facility OPR-OTS PM value.

Steps

Configuring OPS power levels and LOS Thresholds

- 1 For protected configurations, the Input power level to OPS SW1 In and SW2 In ports must be within 2 dB of each other. To view the OPS SW1 In and OPS SW2 port power levels, enter

```
> pm optmon counts show optmon-1-3-13 montype opr-ots
and
```

```
> pm optmon counts show optmon-1-3-15 montype opr-ots
```

If the power differs by more than 2 dB, insert a pad on the SW with higher input power, so that its input power is within ± 2 dB of the other SW.

Note: This operational consideration exists to ensure that the modem receiver does not experience a dramatic change in power level when the OPS switches. The modem receiver may take a longer time to re-acquire a signal if the new signal has a different power level leading to higher than expected protection switch times.

- 2 For all protected configurations, the OPS LOS Thresholds that were set by the “install create” command for the OPS SW1 In and OPS SW2 In ports must be manually configured. The new OPS LOS Threshold will depend on the type of amplifier used, the span count of the photonic system, and the power loss for each span. Refer to the following table to calculate the value that applies to your configuration.

Note: If your network has one or more spans with losses greater than those listed in the table, it is considered to be an Extended Reach configuration. For Extended Reach configurations, the OPS LOS thresholds must be calculated by Ciena. Contact your Ciena support team.

Table 33 LOS Thresholds for OPS SW1 In and OPS SW2 In ports for protected configurations

Amp type	Span count	Maximum span loss (per span)	Minimum Waveserver Ai channels required	OPS LOS Threshold (dBm)
SLA	1	< 10.2 dB	Fixed-64: N/A	-1.0 – Pad value at SW input
MLA3	1	< 24.7 dB	Fixed-64: 3	0 – Pad value at SW input
MLA3	2	< 22 dB	Fixed-64: 4	+4.0 – Pad value at SW input
MLA2/MLA3	3	< 22.0 dB	Fixed-64: 5	+3.0 – Pad value at SW input

- 3 Use the following CLI commands to set the LOS Threshold from [step 2](#).
- Set the LOS Threshold for the OPS SW1 In port:

```
> optmons optmon set optmon-1-3-13 los-thres <los-thres Value>
```
 - Set the LOS Threshold for the OPS SW2 In port:

```
> optmons optmon set optmon-1-3-15 los-thres <value>
```

- 4
- | If you have | Then go to |
|--|---|
| an Extended Reach configuration or 6500 line-amp | Setting Line B target gain for Extended Reach and line-amp configurations |
| a Standard Reach configuration | Resetting the TCA baselines |
- end—

Procedure 12 Setting Line B target gain for Extended Reach and line-amp configurations

Use this procedure to set the Line B target gain for Extended Reach 6500 terminal nodes and 6500 line-amp nodes.

Requirements

To complete this procedure, make sure you have:

- the engineering data plan for the 6500 Extended Reach and line-amp shelves
- connected the Waveserver Ai wavelengths to the 6500 photonic system

For more information on Standard and Extended Reach configurations, see [Standard reach and Extended reach configurations](#).

Steps

Setting Line B target gain for Extended Reach terminal configurations (protected or unprotected)

- 1 For Extended Reach configurations (protected or unprotected), you must set the Line B (port 6) Target Gain on the 6500 terminal nodes. Adjust the value so that Line B output power matches table [Target Power at Line B \(dBm\)](#) for all amplifier cards in the shelf (the Target Gain value will be ~23 dB).

To set the Line B Target Gain, enter

```
> amps amp set amp-1-<1|2>-6 targ-gain <targ-gain Value>
```

To view the Line B output power, enter

```
> pm amp counts show amp-1-<1|2>-6 montype opout-ots
```

- 2 Repeat [step 1](#) for the far-end 6500 node.

Setting Line B target gain for line-amps

- 3 For configurations with line-amps, you must set the Line B (port 6) Target Gain for each amplifier card. Adjust the value so that Line B output power matches table [Target Power at Line B \(dBm\)](#) for all amplifier cards in the shelf (the Target Gain value will be ~23 dB).

To set the Line B Target Gain, enter

```
> amps amp set amp-1-<1|2>-6 targ-gain <targ-gain Value>
```

To view the Line B output power, enter

```
> pm amp counts show amp-1-<1|2>-6 montype opout-ots
```

- 4 Repeat [step 3](#) for each line-amp shelf in the photonic system.
- 5 Go to procedure [Resetting the TCA baselines](#).

—end—

Table 34 Target Power at Line B (dBm)

Channel count	Band plan and amplifier type		
	fixed64	Channel count	fixed64
	MLA3		MLA3
	1, 2, or 3 spans		1, 2, or 3 spans
1	4.9	49	21.8
2	7.9	50	21.9
3	9.7	51	22.0
4	10.9	52	22.0
5	11.9	53	22.1
6	12.7	54	22.2
7	13.4	55	22.3
8	13.9	56	22.4
9	14.4	57	22.5
10	14.9	58	22.5
11	15.3	59	22.6
12	15.7	60	22.7
13	16.0	61	22.7
14	16.4	62	22.8
15	16.7	63	22.9
16	16.9	64	23.0
17	17.2	65	-
18	17.5	66	-
19	17.7	67	-
20	17.9	68	-
21	18.0	69	-
22	18.2	70	-
23	18.4	71	-
24	18.6	72	-
25	18.8	73	-

Band plan and amplifier type			
Channel count	fixed64	Channel count	fixed64
	MLA3		MLA3
	1, 2, or 3 spans		1, 2, or 3 spans
26	19.0	74	-
27	19.2	75	-
28	19.3	76	-
29	19.5	77	-
30	19.6	78	-
31	19.8	79	-
32	19.9	80	-
33	20.0	81	-
34	20.2	82	-
35	20.3	83	-
36	20.4	84	-
37	20.6	85	-
38	20.7	86	-
39	20.8	87	-
40	20.9	88	-
41	21.0	89	-
42	21.1	90	-
43	21.2	91	-
44	21.3	92	-
45	21.4	93	-
46	21.5	94	-
47	21.6	95	-
48	21.7	96	-

Procedure 13 Setting 6500 Line B Target Peak Power/Gain Offset and Waveserver Ai Output Power for configurations using Waveserver Ai with EDFA

Use this procedure to set the 6500 Line B Target Peak Power/Gain Offset and Waveserver Ai Output Power for configurations using Waveserver Ai with EDFA.

Requirements

To perform this procedure you must:

- have configured the 6500 nodes in the solution, according to the Commissioning chapter of this document
- have configured both Waveserver Ai nodes in the solution
- be logged into the 6500 and Waveserver Ai CLI
- have the engineering documentation package (EDP) or installation documentation package (IDP) or equivalent site/network engineering plans

Overview

For Greenfield systems using Waveserver Ai with EDFA, use this procedure to:

- Provision the Line B Target Peak Power and Gain Offset to the values indicated in the following tables:
 - [Waveserver Ai - greenfield Line B target peak powers/gain offsets and output powers - Single ch Budget](#)
 - [Waveserver Ai - greenfield Line B target peak powers/gain offsets and output powers - 2 ch Budget](#)
 - [Waveserver Ai - greenfield Line B target peak powers/gain offsets and output powers - 12 ch Budget](#)
 - [Waveserver Ai - greenfield Line B target peak powers/gain offsets and output powers - 64 ch Budget](#)
- If the system uses a CMD64 with either MLA2 or MLA3:
 - Provision the ADJ-TX between the CMD port and the Waveserver Ai port. OCLD Tx Power will be set by SPLI.
The Waveserver Ai IP address and node name will have already been set in the 6500 SPLI window (**Node Information** -> **SPLI**).
or
 - Provision the Waveserver Ai Tx Output to the value indicated in the following tables:

- Waveserver Ai - greenfield Line B target peak powers/gain offsets and output powers - Single ch Budget
- Waveserver Ai - greenfield Line B target peak powers/gain offsets and output powers - 2 ch Budget
- Waveserver Ai - greenfield Line B target peak powers/gain offsets and output powers - 12 ch Budget
- Waveserver Ai - greenfield Line B target peak powers/gain offsets and output powers - 64 ch Budget

Steps

Setting 6500 Line B target peak power and gain offset for Waveserver Ai with EDFA configurations

- 1 For Waveserver Ai with EDFA configurations, you must set the Line B (port 6) Target Peak Power and Gain Offset on the 6500 terminal nodes. Adjust the value so that Line B Target Peak Power and Gain Offset matches [Table 35](#) to [Table 38](#) for all amplifier cards in the shelf.
To set the Line B Target Peak Power and Gain Offset, enter

```
> amps amp set amp-1-<1|2>-6 targ-peak-pow <targ-peak-pow Value> gain-offset <gain offset value>
```
- 2 Repeat [step 1](#) for the far-end 6500 node.

Setting the Waveserver Ai Tx output power for Waveserver Ai with EDFA configurations

- | | | | | | | | |
|---|---|---|-------------------|---------------------|------------------------|-------------------------|------------------------|
| 3 | <table border="0" style="width: 100%;"> <tr> <td style="width: 60%;">If the Waveserver Ai Tx output power</td> <td style="width: 40%;">Then go to</td> </tr> <tr> <td>will be set by SPLI</td> <td>step 4</td> </tr> <tr> <td>will not be set by SPLI</td> <td>step 6</td> </tr> </table> | If the Waveserver Ai Tx output power | Then go to | will be set by SPLI | step 4 | will not be set by SPLI | step 6 |
| If the Waveserver Ai Tx output power | Then go to | | | | | | |
| will be set by SPLI | step 4 | | | | | | |
| will not be set by SPLI | step 6 | | | | | | |

Provisioning 6500 ADJ-TX facilities

- 4 If the system uses aCMD64 with either MLA2 or MLA3, provision the ADJ-TX between the CMD port and the Waveserver Ai port.
To create an adjacency (ADJ), enter

```
> adjacencies adjacency create <AID>
[adj-type <value>]
[clfi <STRING>]
[layer <L1OTN|L2MON|L2PHYS|LINE|OTU|PHYS|SECT|SUBPORT>]
[prov-addr-form <format>]
[prov-fe-addr <STRING>]
```

Note: The provisioned far end address should be in a pair of double quotes.

To edit and ADJ-TX, enter

```
> adjacencies tx set <AID>
[active <TRUE | FALSE>]
[auto-disc <AUTO | MANUAL>]
```

```
[bias <DECIMAL>]
[center-frequency <DECIMAL>]
[ckt-id <string value>]
[ctrl-freq-offset<DECIMAL>]
[cur-pow <DECIMAL>]
[express-delete <TRUE | FALSE>]
[exp-res-time <DECIMAL>]
[freqresolution <DECIMAL>]
[laser-centering <DECIMAL>]
[laser-centering-range <TRUE | FALSE>]
[link-opt-mode <LNKSTDMODE | LNKTRACKINGMODE>]
[loopback-status <status>]
[max-freq-guard-band <DECIMAL>]
[max-pow <DECIMAL>]
[min-pow <DECIMAL>]
[min-spectral-width <DECIMAL>]
[mod-class <mod-class>]
[och-tx-blanking <OFF | ON>]
[paired-rx <NO | YES>]
[path-disp <DECIMAL>]
[prov-fe-pec <STRING>]
[rate <rate>]
[sig-bw-10dB <DECIMAL>]
[sig-bw-3dB <DECIMAL>]
[syncprov <FALSE | TRUE>]
[trans-mode <STRING>]
[tx-ctrl <NO | YES>]
[tx-disp-mode <AUTOMATIC | FIXEDTX | EXPERT]
[tx-pre-comp <STRING>]
[type <type>]
```

The OCLD Tx Power will be set by SPLI.

The Waveserver Ai IP address and node name will have already been set in the 6500 SPLI window (**Node Information-> SPLI**).

- 5 Repeat [step 4](#) for the far-end node.

The procedure is complete.

- 6 Provision the Waveserver Tx Output to the value indicated in [Table 35](#) to [Table 38](#).

To set the Waveserver Ai Tx output power on the Waveserver Ai line port, enter:

```
> ptp set ptp <slot>/<port> tx-power <Tx power (dBm)
Value>
```

For more information, see the *Waveserver Ai User Guide*, 323-4002-100.

7 Repeat [step 6](#) for the far-end node.

The procedure is complete.

—end—

Table 35 Waveserver Ai - greenfield Line B target peak powers/gain offsets and output powers - Single ch Budget

Configuration						Provisioning					
						Terminal Line B				Waveserver Ai Tx	
Data rate (GB/s)	Channels	OPS	Mux/Demux	Amps	Span Count	Default Gain (dB)	Gain Offset (dB)	Total Gain (dB)	Target Peak Power (dBm)	Output Power (dBm)	SPLI Setpoint (dBm)
400	1	no	none	none	1	—	—	—	—	4.0	—
400	1	yes	none	none	1	—	—	—	—	4.0	—
300	1	no	none	none	1	—	—	—	—	4.0	—
300	1	yes	none	none	1	—	—	—	—	4.0	—

Table 36 Waveserver Ai - greenfield Line B target peak powers/gain offsets and output powers - 2 ch Budget

Configuration						Provisioning					
						Terminal Line B				Waveserver Ai Tx	
Data rate (GB/s)	Channels	OPS	Mux/Demux	Amps	Span Count	Default Gain (dB)	Gain Offset (dB)	Total Gain (dB)	Target Peak Power (dBm)	Output Power (dBm)	SPLI Setpoint (dBm)
400	2	no	BMD2	none	1	—	—	—	—	4.0	—
400	2	yes	BMD2	none	1	—	—	—	—	4.0	—
300	2	no	BMD2	none	1	—	—	—	—	4.0	—
300	2	yes	BMD2	none	1	—	—	—	—	4.0	—

Table 37 Waveserver Ai - greenfield Line B target peak powers/gain offsets and output powers - 12 ch Budget

Configuration						Provisioning					
						Terminal Line B				Waveserver Ai Tx	
Data rate (GB/s)	Channels	OPS	Mux/Demux	Amps	Span Count	Default Gain (dB)	Gain Offset (dB)	Total Gain (dB)	Target Peak Power (dBm)	Output Power (dBm)	SPLI Setpoint (dBm)
400	12	no	CCMD12	none	1	—	—	—	—	-2.0	—
300	12	no	CCMD12	none	1	—	—	—	—	-2.0	—

Table 38 Waveserver Ai - greenfield Line B target peak powers/gain offsets and output powers - 64 ch Budget

Configuration						Provisioning					
						Terminal Line B				Waveserver Ai Tx	
Data rate (GB/s)	Channels	OPS	Mux/Demux	Amps	Span Count	Default Gain (dB)	Gain Offset (dB)	Total Gain (dB)	Target Peak Power (dBm)	Output Power (dBm)	SPLI Setpoint (dBm)
400	64	no	CMD64	none	1	—	—	—	—	4.0	—
400	64	no	CMD64	MLA3	1	13.0	4.5	17.5	4.9	set by SPLI to	-6.6
400	64	no	CMD64	MLA3	2	13.0	3.5	16.5	4.9	set by SPLI to	-5.6
400	64	no	CMD64	MLA3	3	13.0	2.5	15.5	4.9	set by SPLI to	-4.6
400	64	yes	CMD64	none	1	—	—	—	—	4.0	—
400	64	yes	CMD64	MLA3	1	16.2	1.3	17.5	4.9	-6.6	—
400	64	yes	CMD64	MLA3	2	16.2	0.3	16.5	4.9	-5.6	—
400	64	yes	CMD64	MLA3	3	16.2	-0.7	15.5	4.9	-4.6	—
300	64	no	CMD64	none	1	—	—	—	—	4.0	—
300	64	no	CMD64	MLA3	1,2,3	13.0	4.5	17.5	4.9	set by SPLI to	-6.6
300	64	yes	CMD64	none	1	—	—	—	—	4.0	—

Configuration						Provisioning					
						Terminal Line B				Waveserver Ai Tx	
Data rate (GB/s)	Channels	OPS	Mux/Demux	Amps	Span Count	Default Gain (dB)	Gain Offset (dB)	Total Gain (dB)	Target Peak Power (dBm)	Output Power (dBm)	SPLI Setpoint (dBm)
300	64	yes	CMD64	MLA3	1,2,3	16.2	1.3	17.5	4.9	-6.6	—

Procedure 14 Resetting the TCA baselines

Use this procedure to set the Threshold Crossing Alert (TCA) baseline power levels on the 6500 amplifier and OPS cards. Completing this procedure allows the optical power baseline to be set correctly. In the event of a change in the power profile, Gauge TCA alarms will be raised, allowing for better fault isolation. This procedure must be completed on all 6500 nodes in the photonic system, once all of the Waveserver Ai units have been connected to the 6500 terminal nodes. If any additional Waveserver Ai channels are added to the photonic system, this procedure should be repeated.

Requirements

To perform this procedure you must:

- have connected all Waveserver Ai channels to the 6500 CMDs
- be logged into the 6500 and Waveserver Ai CLI

Steps

Resetting the TCA baselines

- 1 To view the current TCA baseline settings for each 6500 circuit pack, enter

```
< pm amp counts show shelf-1-all tmper basln
```


and

```
< pm optmon counts show shelf-1-all tmper basln
```
- 2 The mon value indicates the current TCA baseline for a specific port.
- 3 To reset the min/max values and the TCA baselines on the near-end 6500 shelf, enter the following four commands:

```
< pm optmon reset shelf-1-all
```



```
< pm optmon reset optmon-1-all tmper basln
```



```
< pm amp reset shelf-1-all
```



```
< pm amp reset amp-1-all tmper basln
```
- 4 Repeat [step 1](#) to [step 3](#) for:
 - the far-end 6500 terminal shelf
 - any line-amp 6500 shelves located between the terminal shelves
 - if you have a protected configuration, any line-amp 6500 shelves on both the working and protected path.
- 5 Once the TCA baselines have been reset, you can verify the new baseline settings by entering

```
< pm optmon counts show shelf-1-all tmper basln
```


and

```
< pm amp counts show shelf-1-all tmper basln
```

6	If you have	Then
	an unprotected Fixed-64 or a Flex-12 configuration	Verifying traffic continuity for unprotected Fixed-64 and Flex-12 configurations
	a protected Fixed-64 configuration	Verifying traffic continuity for Fixed-64 OPS protected configuration
	—end—	

Procedure 15 Calibrating the SRA circuit packs

If a SRA is used in the thin terminal configuration, use this procedure to manually calibrate the SRA circuit packs at each end before adding channels.

Requirements

To perform this procedure, you must:

- have the engineering documentation package (EDP) or installation documentation package (IDP) or equivalent site/network engineering plans
- be logged into the 6500
- have set the SRA “FORCEDSHUTOFF” parameter value to “FALSE” in order to bring up the RAMAN pump lasers to ON.



CAUTION

Possible damage to equipment

During this procedure, there must not be any input power to the XLA circuit pack otherwise the equipment may be damaged.

The following table shows the gain range per fiber type for the SRA calibration.

Table 39 SRA calibration - Gain range per fiber type

Fiber type	Min Gain (dB)	Max Gain (dB)	Min Span Low Gain (dB)	Max Gain Low Gain (dB)	Min Span High Gain (dB)	Max Span (dB)
NDSF	6	15	14	23	21	43
eLEAF (LEAF)	8	20	16	28	23	
TWRS	10	24	18	32	25	
TWC	10	23	18	31	25	
TWP	10	23	18	31	25	
LS	10	24	18	32	25	
DSF	10	24	18	32	25	
TWreach	10	24	18	32	25	
Teralight	8	20	16	28	23	
NDSF Low Water Peak	7	19	15	27	22	
Unknown	6	15	14	23	21	

Note: When commissioning the SRA, the “FORCEDSHUTOFF” parameter value must be set to “FALSE” in order to bring up the RAMAN pump lasers to ON.

Steps

- 1 Verify there is no input power to the XLA circuit pack



CAUTION

Possible damage to equipment

Not performing this action could result in damaging the equipment.

- 2 Set the XLA post-amplifier (facility port 6) AMP Mode to the Power:
 - a. Set the AMP facility to out-of-service (OOS).
 - b. Set the ALSO Disabled parameter to True.
 - c. Set the AMP Mode to Power.
 - d. Record the current power and set the Target Power to 20 dBm.
 - e. Record the current switch mode and set the Gain Mode to High.
 - f. Set the AMP facility to in-service (IS).

Note: The “Automatic Shutoff Disabled” alarm will be raised. The alarm will clear after [step 7](#) is completed.

- 3 At the downstream site, set the XLA pre-amplifier (facility port 8) to OOS.
- 4 Before sending the calibration command, check the following conditions are met:
 - No Optical Line Fail alarm
 - No OTDR alarm
 - No transmitter sending power into the CMD (that is, there must no input power to the XLA circuit pack)
- 5 Send the calibration request to SRA using the following TL1 command:
`OPR-RAMAN-CALIBRATION:[tid]:RAMAN-shelf-slot-8:CTAG`
Note: There is no Site Manager support for this command.
If calibration is successful, the “Calibration Required” alarm is cleared and the SRA RAMAN facility “Calibrated Flag” is set.
If calibration fails, the “Calibration Required” alarm does not clear, the SRA RAMAN facility “Calibrated Flag” is not set.
If calibration fails on multiple attempts, contact your next level of support.
- 6 After the calibration request completes, record the recommended gain on the SRA RAMAN facility:
 - a. Set the Raman gain mode to “Provisioned”.
 - b. Set the provisioned Gain to the recommended gain + 1 dB.
 - c. Re-run the calibration process at [step 5](#).
 - d. Once set, the gain should not be reverted back to recommended.**Note:** If calibration fails after performing [step 6](#) multiple times, contact your next level of support.
- 7 Set the XLA post-amplifier (facility port 6) AMP Mode to Gain:
 - a. Set the AMP facility to out-of-service (OOS).
 - b. Set the ALSO Disabled parameter to False.
 - c. Set the AMP Mode to Gain.
 - d. Set the Target Power to the value recorded in [step 2, substep d](#).
 - e. Record the Gain Mode to the value recorded in [step 2, substep e](#).
 - f. Set the AMP facility to in-service (IS).
- 8 At downstream site, set XLA pre-amplifier (facility port 8) to IS.
—end—

Procedure 16 Verifying traffic continuity for unprotected Fixed-64 and Flex-12 configurations

Use this procedure to verify that the 6500 and Waveserver Ai nodes are carrying traffic correctly once the shelves have been fibered together.

Requirements

To complete this procedure, you must:

- be logged into all applicable 6500 and Waveserver Ai CLI sessions
- have connected all fibers between Waveserver Ai and the 6500 CMD
- have verified the configuration of the 6500 and Waveserver Ai nodes

Steps

LED status

- 1 On the Waveserver Ai nodes, verify that line-port LEDs (ports 1 and 2) are green.
- 2

If you are	Then go to
verifying traffic on a Fixed-64 configuration	step 3
verifying traffic on a Flex-12 configuration	step 5
- 3 On the 6500 amplifiers in slot 1, verify that the following LEDs are dark:
 - OSC 3/4
 - Ln B 5/6
 - Ln A 7/8
- 4 Continue to [step 6](#).
- 5 On the 6500 CCMD12 cards, verify that the Common In Port 25 LEDs are dark.
- 6 On the 6500 SPAP-2s, verify that the OSC1 LEDs are dark.
Note: If an LED is yellow, it indicates that an alarm is raised against that port.

Verifying power levels and alarms

- 7 To verify that the Rx power on the Waveserver Ai line ports 1 and 2 are within the specified range, enter

```
> ptp show ptp <slot>/<1 | 2> status
```

 Output example:

```
| Rx
| Actual Power (dBm) | -8.60
```

Note: The Rx power range for Waveserver Ai line-ports is -22.5 dBm to +5.5 dBm.

- 8 To verify the presence of any threshold crossing, power level, or other unexpected alarms on the 6500 nodes, enter
`> alarms alarm show`
- 9 To clear any unexpected alarms, see *6500 Fault Management - Alarm Clearing*, 323-1851-543.
- 10 To verify the presence of any unexpected alarms on the Waveserver Ai nodes, enter
`> alarm show`
- 11 Use the procedures in the *Waveserver Ai User Guide*, 323-4002-100 to clear any unexpected alarms.

Seating the client plugs**CAUTION****Risk of damage to modules**

Wear an antistatic wrist strap to protect the equipment from static damage. Connect the wrist strap to the ESD jack on the shelf or module.

- 12 On the Waveserver Ai nodes, seat the client plugs.
- 13 Verify that the Waveserver Ai client-port LEDs are green.
- 14 To verify the presence of any unexpected alarms on the Waveserver Ai nodes, enter
`> alarm show`
- 15 Use the procedures in the *Waveserver Ai User Guide*, 323-4002-100 to clear any unexpected alarms.

Verifying Waveserver Ai port information

Use this section to verify the status and configuration of the Waveserver Ai ports.

- 16 To display the status of all of the Physical Termination Points (PTPs) on the Waveserver Ai nodes, enter
`> ptp show`
- 17 To display the summary information about all provisioned ports on the Waveserver Ai nodes, enter
`> port show`
- 18 To display the summary information about all transceiver ports on the Waveserver Ai nodes, enter
`> xcvr show`

Clearing the Waveserver Ai statistics and testing traffic

Verify the traffic on the Waveserver Ai client ports are working as expected using a test set or the client traffic source.

- 19 To clear the client port pm statistics on the Waveserver Ai nodes, enter
`> pm clear port <slot>/<port>`

- 20** Clear the traffic counters on the test set or client traffic source for the port you will be testing.
- 21** To show the current port pm statistics (verify received Ethernet Bytes/Packets (Rx Bytes, Rx Packets)) on the Waveserver Ai node, enter
- ```
> pm show port <slot>/<port>
```
- 22** Compare the number of Ethernet Bytes/Packets received on the Waveserver Ai client ports with the number of Ethernet Bytes/Packets received on the test source. The number should be close to the same, although some differences can arise due to delays in clearing the ports on all of the equipment.
- end—**

## Procedure 17 Verifying traffic continuity for Fixed-64 OPS protected configuration

Use this procedure to verify that the 6500 and Waveserver Ai nodes are carrying traffic correctly and that the protection switching on 6500 is working as expected.

### Requirements

To complete this procedure, you must:

- be logged into all applicable 6500 and Waveserver Ai CLI sessions
- have connected all fibers between Waveserver Ai and the 6500 CMD
- have verified the configuration of the 6500 and Waveserver Ai nodes
- have the minimum number of Waveserver Ai channels as listed in table [Minimum channel count](#).

### Steps

#### LED status

- 1 On the Waveserver Ai nodes, verify that line-port LEDs (ports 1 and 2) are green.
- 2 On the 6500 amplifiers in slot 1 and slot 2, verify that the following LEDs are dark:
  - OSC 3/4
  - Ln B 5/6
  - Ln A 7/8
- 3 On the OPS circuit pack, verify that the following LEDs are dark:
  - Sw1 13
  - Sw2 15
- 4 On the 6500 SPAP-2s, verify that the OSC1 LEDs are dark.
 

**Note:** A yellow LED on the 6500 or Waveserver Ai indicates that an alarm is raised against that port.

#### Verifying power levels and alarms

- 5 To verify that the Rx power on the Waveserver Ai line ports 1 and 2 are within the specified range, enter

```
> ptp show ptp <slot>/<1 | 2> status
```

Output example:

```
| Rx |
| Actual Power (dBm) | -8.60
```

**Note:** The Rx power range for Waveserver Ai line-ports is -22.5 dBm to +5.5 dBm.

- 6 To verify the presence of any threshold crossing, power level, or other unexpected alarms on the 6500 nodes, enter  
`> alarms alarm show`
- 7 To clear any unexpected alarms, see *6500 Fault Management - Alarm Clearing*, 323-1851-543.
- 8 To verify the presence of any unexpected alarms on the Waveserver Ai nodes, enter  
`> alarm show`
- 9 Use the procedures in the *Waveserver Ai User Guide*, 323-4002-100 to clear any unexpected alarms.

#### **Seating the client plugs**



#### **CAUTION**

##### **Risk of damage to modules**

Wear an antistatic wrist strap to protect the equipment from static damage. Connect the wrist strap to the ESD jack on the shelf or module.

- 10 On the Waveserver Ai nodes, seat the client plugs.
- 11 Verify that the Waveserver Ai client-port LEDs are green.
- 12 To verify the presence of any unexpected alarms on the Waveserver Ai nodes, enter  
`> alarm show`
- 13 Use the procedures in the *Waveserver Ai User Guide*, 323-4002-100 to clear any unexpected alarms.

#### **Verifying Waveserver Ai port information**

- 14 To display the status of all of the Physical Termination Points (PTPs) on the Waveserver Ai nodes, enter  
`> ptp show`
- 15 To display the summary information about all provisioned ports on the Waveserver Ai nodes, enter  
`> port show`
- 16 To display the summary information about all transceiver ports on the Waveserver Ai nodes, enter  
`> xcvr show`

#### **Clearing the Waveserver Ai statistics and testing traffic on working path**

Verify the traffic on the Waveserver Ai client ports are working as expected using a test set or the client traffic source.

- 17 To clear the client port pm statistics on the Waveserver Ai nodes, enter  
`> pm clear port <slot>/<port>`
- 18 Clear the traffic counters on the test set or client traffic source for the port you will be testing.

- 19 To show the current port pm statistics (verify received Ethernet Bytes/Packets (Rx Bytes, Rx Packets)) on the Waveserver Ai node, enter  
`pm show port <slot>/<port>`
- 20 Compare the number of Ethernet Bytes/Packets received on the Waveserver Ai client ports with the number of Ethernet Bytes/Packets received on the test source. The number should be close to the same, although some differences can arise due to delays in clearing the ports on all of the equipment.

#### **Verifying 6500 and Waveserver Ai protection status**

- 21 To verify the protection configuration of the 6500 nodes, enter  
`> optmons prot-group show all`  
 and verify that the prot-scheme is set to 1plus1.
- 22 To verify the protection configuration of the Waveserver Ai nodes, enter  
`> system show configuration`  
 and verify that the Line Protection is set to trunkOPS.
- 23 Use the following table to verify that the protection status between 6500 and Waveserver Ai nodes are configured correctly.

**Table 40**

| 6500 protection (> optmons prot-group show all output) | Waveserver Ai protection (> system show configuration output) |
|--------------------------------------------------------|---------------------------------------------------------------|
| 1plus1                                                 | trunkOPS                                                      |
| blank (no output)                                      | unprotected                                                   |

- 24 If the 6500 protection configuration is not set up correctly, use your network documentation to confirm the correct protection configuration and follow the commissioning section of this document.
- 25 If the Waveserver Ai protection configuration is not set up as Trunk-OPS protection, enter  
`> system set line-protection trunk-ops`  
**Note:** If the system line protection is set to trunk-ops, the default conditioning hold-off value is 100 ms.

#### **Clearing the Waveserver Ai statistics, traffic counters on the test set or client traffic source prior to testing manual protection switching**

- 26 To clear the client port pm statistics on the Waveserver Ai nodes, enter  
`> pm clear port <slot>/<port>`
- 27 Clear the traffic counters on the test set or client traffic source for the port you will be testing.

#### **Testing manual protection switching**

- 28 To verify the working and protection path of the 6500 nodes, enter

```
> optmons optmon show optmon-1-3-all
Ciena_Toronto-17# optmons optmon show optmon-1-3-all
```

```

aid | OPTMON-1-3-13
los-thres | -20.00
port-label | "OPS3 SW1 In"
also-disabled| FALSE
pst | IS
sst | WRKRX

```

```
aid | OPTMON-1-3-15
los-thres | -20.00
port-label | "OPS3 SW2 In"
also-disabled| FALSE
pst | IS
sst | STBYH

```

In the above example, the working path is on OPTMON-1-3-13 (indicated by `sst | WRKRX`) and the protection path is on OPTMON-1-3-15 (indicated by `sst | STBYH`).

**Note:** The default configuration is the working path on OPTMON-1-3-13 and the protection path on OPTMON-1-3-15.

**29** To operate a protection switch to the protect path, enter

```
> optmons prot-switch operate optmons-1-3-13 sc man
```

on the near-end and far-end 6500 nodes.

**30** To confirm that the traffic has switched to the protect path, enter

```
> optmons optmon show optmon-1-3-all
Ciena_Toronto-17# optmons optmon show optmon-1-3-all
```

```

aid | OPTMON-1-3-13
los-thres | -20.00
port-label | "OPS3 SW1 In"
also-disabled| FALSE
pst | OOS-AU
sst | FAF, STBYH

```

```
aid | OPTMON-1-3-15
los-thres | -20.00
port-label | "OPS3 SW2 In"
also-disabled| FALSE
pst | IS
sst | WRKRX

```

**31** Verify that the traffic is on the alternate path on both 6500 nodes. In the above example, the working path is now on OPTMON-1-3-15 (indicated by `sst | WRKRX`) and the protection path is on OPTMON-1-3-13 (indicated by `sst | FAF, STBYH`).

### **Verifying the Waveserver Ai port information on the 6500 protect path**

**32** To display the status of all of the Physical Termination Points (PTPs) on the Waveserver Ai nodes, enter

```
> ptp show
```

- 33** To display the summary information about all provisioned ports on the Waveserver Ai nodes, enter

```
> port show
```

- 34** To display the summary information about all transceiver ports on the Waveserver Ai nodes, enter

```
> xcvr show
```

***Verifying traffic on the 6500 protection path***

- 35** To show the current port pm statistics (received Ethernet Bytes/Packets (Rx Bytes, Rx Packets)) on the Waveserver Ai node, enter

```
> pm show port <slot>/<port>
```

- 36** Compare the number of Ethernet Bytes/Packets received on the Waveserver Ai client ports with the number of Ethernet frames received on the test source. The number should be close to the same, although some differences can arise due to delays in clearing the ports on all of the equipment.

***Clearing the Waveserver Ai statistics, traffic counters on the test set or client traffic source prior to moving traffic back to the working path***

- 37** To clear the client port pm statistics on the Waveserver Ai nodes, enter

```
> pm clear port <slot>/<port>
```

- 38** Clear the traffic counters on the test set or client traffic source for the port you will be testing.

***Moving traffic back to the working path***

- 39** To move traffic back to the working path, enter,

```
> optmons prot-switch operate optmons-1-3-15 sc man
```

on the near-end and far-end 6500 nodes.

- 40** To verify the working and protection path of the 6500 nodes, enter

```
> optmons optmon show optmon-1-3-all
```

```
Ciena_Toronto-17# optmons optmon show optmon-1-3-all
```

---

|               |               |
|---------------|---------------|
| aid           | OPTMON-1-3-13 |
| los-thres     | -20.00        |
| port-label    | "OPS3 SW1 In" |
| also-disabled | FALSE         |
| pst           | IS            |
| sst           | WRKRX         |

---

|               |               |
|---------------|---------------|
| aid           | OPTMON-1-3-15 |
| los-thres     | -20.00        |
| port-label    | "OPS3 SW2 In" |
| also-disabled | FALSE         |
| pst           | IS            |
| sst           | STBYH         |

---

In the above example, the working path is back on OPTMON-1-3-13 (indicated by `sst | WRKRX`) and the protection path is back on OPTMON-1-3-15 (indicated by `sst | STBYH`).

- 41 To verify the alarms on the Waveserver Ai nodes, enter  
`> alarm show`
- 42 To verify the PTP facility state on the Waveserver Ai nodes, enter  
`> ptp show`

#### **Verifying traffic on the working path**

- 43 To show the current port pm statistics (received Ethernet Bytes/Packets (Rx Bytes, Rx Packets)) on the Waveserver Ai node, enter  
`> pm show port <slot>/<port>`
- 44 Compare the number of Ethernet Bytes/Packets received on the Waveserver Ai client ports with the number of Ethernet frames received on the test source. The number should be close to the same, although some differences can arise due to delays in clearing the ports on all of the equipment.

#### **Clearing the Waveserver Ai statistics, traffic counters on the test set or client traffic source prior to testing automatic protection switching**

- 45 To clear the client port pm statistics on the Waveserver Ai nodes, enter  
`> pm clear port <slot>/<port>`
- 46 Clear the traffic counters on the test set or client traffic source for the port you will be testing.

#### **Testing automatic protection switching**



#### **DANGER**

##### **Risk of laser radiation exposure**

Do not look directly into the optical beam. Invisible light can severely damage your eyes.



#### **CAUTION**

##### **Risk of damage to modules**

Wear an antistatic wrist strap to protect the equipment from static damage. Connect the wrist strap to the ESD jack on the shelf or module.

- 47 On the working path, pull the Rx fiber (port 8) of the amplifier circuit pack on one of the 6500 nodes. This will cause a switch to the protection path for the near-end and far-end 6500 nodes.
- 48 To confirm that the traffic has switched to the protect path, enter

```
> optmons optmon show optmon-1-3-all
Ciena_Toronto-17# optmons optmon show optmon-1-3-all
aid | OPTMON-1-3-13
los-thres | -20.00
port-label | "OPS3 SW1 In"
also-disabled| FALSE
pst | OOS-AU
sst | FAF, STBYH

aid | OPTMON-1-3-15
```

|               |               |
|---------------|---------------|
| los-thres     | -20.00        |
| port-label    | "OPS3 SW2 In" |
| also-disabled | FALSE         |
| pst           | IS            |
| sst           | WRKRX         |

and verify that the traffic is on the alternate path on both 6500 nodes.

- 49** To verify the alarms on the Waveserver Ai nodes, enter  
> alarm show

***Verifying Waveserver Ai port information on the 6500 protection path***

- 50** To display the status of all of the Physical Termination Points (PTPs) on the Waveserver Ai nodes, enter  
> ptp show
- 51** To display the summary information about all provisioned ports on the Waveserver Ai nodes, enter  
> port show
- 52** To display the summary information about all transceiver ports on the Waveserver Ai nodes, enter  
> xcvr show

***Verifying traffic on the 6500 protection path***

Verify the traffic on the Waveserver Ai client ports are working as expected using a test set or the client traffic source.

- 53** To clear the client port pm statistics on the Waveserver Ai nodes, enter  
> pm clear port <slot>/<port>
- 54** Clear the traffic counters on the test set or client traffic source for the port you will be testing.
- 55** To show the current port pm statistics (verify received Ethernet Bytes/Packets (Rx Bytes, Rx Packets)) on the Waveserver Ai node, enter  
> pm show port <slot>/<port>
- 56** Compare the number of Ethernet Bytes/Packets received on the Waveserver Ai client ports with the number of Ethernet frames received on the test source. The number should be close to the same, although some differences can arise due to delays in clearing the ports on all of the equipment.
- 57** To verify the alarms on the Waveserver Ai nodes, enter  
> alarm show
- 58** To verify the state of the PTP facilities on the Waveserver Ai nodes, enter  
> ptp show

***Moving traffic back to the working path***

- 59** Reconnect the fiber into the Rx port of the 6500 amplifier.
- 60** To move traffic back to the working path, enter  
> optmons prot-switch operate optmons-1-3-15 sc man  
on the near-end and far-end 6500 nodes

**61** To verify the working and protection path of the 6500 nodes, enter

```
> optmons optmon show optmon-1-3-all
Ciena_Toronto-17# optmons optmon show optmon-1-3-all
```

```

aid | OPTMON-1-3-13
los-thres | -20.00
port-label | "OPS3 SW1 In"
also-disabled| FALSE
pst | IS
sst | WRKRX

```

```
aid | OPTMON-1-3-15
los-thres | -20.00
port-label | "OPS3 SW2 In"
also-disabled| FALSE
pst | IS
sst | STBYH
```

In the above example, the working path is back on OPTMON-1-3-13 (indicated by `sst | WRKRX`) and the protection path is back on OPTMON-1-3-15 (indicated by `sst | STBYH`).

**62** To verify the alarms on the Waveserver Ai nodes, enter

```
> alarm show
```

**63** To verify the state of the PTP facilities on the Waveserver Ai nodes, enter

```
> ptp show
```

**—end—**

## Procedure 18 Configuring 6500-based SLTE configurations using Waveserver Ai wayside channel

---

Use this procedure to set up Waveserver Ai and 6500 for 6500-based SLTE configurations using the Waveserver Ai wayside channel.

### Requirements

To complete this procedure, you must:

- be logged into all applicable Waveserver Ai CLI sessions
- be logged into all applicable 6500 Site Manager sessions
- have a LAN cable available

### Overview

The Waveserver Ai wayside channel allows 6500 SLTE shelves on either side of the submarine link to communicate for the purposes of managing that link. For more information, see [Comms channel for 6500-based SLTE configurations](#).

### Steps

#### *Configuring Waveserver Ai DCN ports*

- 1 Using CLI, provision the Waveserver Ai DCN ports on both the near-end and the far-end to use the wayside channel:  

```
chassis port set port dcn mode wayside-channel
```
- 2 Verify that the Waveserver Ai DCN ports are configured as wayside-channel:  

```
chassis port show
```

#### *Configuring the 6500 ILAN/COLAN ports*

- 3 In Site Manager select the required network element in the navigation tree and select **Configuration -> Comms Setting Management**.
- 4 Select the **Interfaces** tab and select the 6500 shelf number corresponding to the shelf that will be connected to the Waveserver Ai, then select **LAN** as the Interface Type.
- 5 Click the **Add** button. Select the appropriate 6500 LAN port (ILAN/COLAN) in the Name field. Ensure that the Configuration is set to **Automatic** and click **OK**. For more information, see *6500 Data Communications Planning and User Guide*, 323-1851-101.
- 6 Select **IP** from the Interface Type drop-down list, then click **Add** and select the appropriate 6500 LAN port (ILAN/COLAN) from the list. Ensure that the IP address is routable to the DCN. Ciena recommends that the IP address have a /30 subnet. The IP address must be unique and must not overlap with existing COLAN or Shelf IP addresses.
- 7 Add the 6500 LAN port to the OSPF area, by selecting the **Routers** tab and selecting **OSPF Circuit** from the Router Type drop-down list.

- 8 Click **Add** and select the appropriate 6500 LAN port.
- 9 In the Add OSPF Circuit Parameters dialog box, set the required Network Area. Set other OSPF parameters, such as Cost and Authentication Type, appropriately. Ensure that Opaque Link State Advertisement is set to **On**. Click **OK**. See *6500 Data Communications Planning and User Guide*, 323-1851-101 and *6500 Configuration - Control Plane*, 323-1851-330 for details.
- 10 Complete [step 3](#) to [step 9](#) on both the near-end and far-end 6500 nodes. Once completed, go to next step.

**Connecting the LAN cables**

- 11 At both the near-end and far-end sites, connect a LAN cable between the designated 6500 ILAN or COLAN port and the Waveserver Ai DCN port to be used for wayside access. Verify that the link negotiates to 100BT Full-duplex.

**Verifying traffic**

- 12 In Site Manager, open the IP Routing table and verify that the 6500 NE is receiving communications through the specified LAN port.

**Configuring OSRP**

- 13 Set the OSRP to use out-of-band communications. See *6500 Data Communications Planning and User Guide*, 323-1851-101 and *6500 Configuration - Control Plane*, 323-1851-330 for details.

—end—



---

## CHAPTER 6

# Waveserver Ai and 5170 interworking

---

This section includes the common configurations between Waveserver Ai and 5170.

Topics include:

- [Waveserver Ai and 5170 interworking summary](#)
- [Waveserver Ai Release 1.2 \(and higher\) and 5170 Release 8.6.1 interworking](#)

**Note:** For information on configuring Waveserver Ai, refer to *Waveserver Ai User Guide*, 323-4002-100 and for information on 5170 refer to the 5170 Service Aggregation Switch (5170) documentation library.

### Waveserver Ai and 5170 interworking summary

The Waveserver Ai connects to the 5170 through the 100GE client-side interfaces.

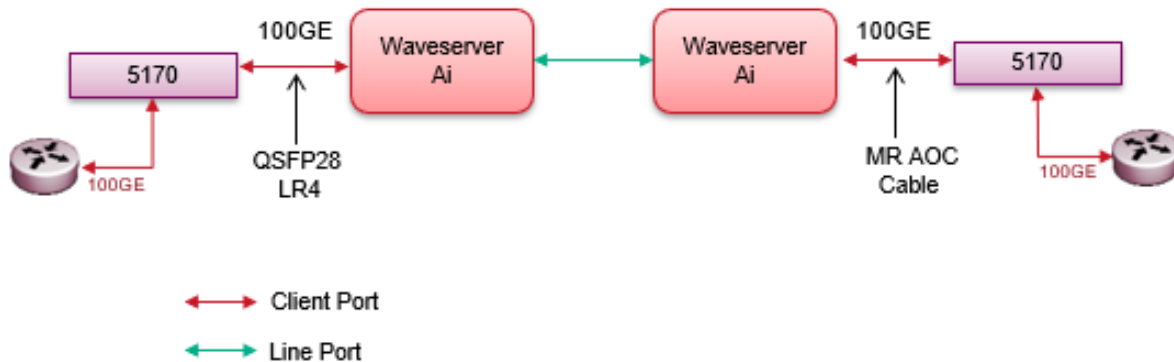
### Waveserver Ai Release 1.2 (and higher) and 5170 Release 8.6.1 interworking

The following Waveserver Ai and 5170 interworking configurations are supported:

- [Waveserver Ai 2xWLAi + 8xQSFP28 module interworking with 5170 using an AOC Ciena-branded cable](#)
- [Waveserver Ai 2xWLAi + 8xQSFP28 module interworking with 5170 using an SR4 Ciena-branded cable](#)

#### Waveserver Ai 2xWLAi + 8xQSFP28 module interworking with 5170 using an AOC Ciena-branded cable

This section outlines the Waveserver Ai 2xWLAi + 8xQSFP28 module interworking with 5170 using an AOC Ciena-branded cable. The Waveserver Ai 100GE clients interoperate with the 5170 100GE clients over the AOC Ciena-branded cable. See the following figure.

**Figure 46** Waveserver Ai 2xWLAi + 8xQSFP28 module interworking with 5170 using AOC Ciena-branded cable

The following interworking considerations apply:

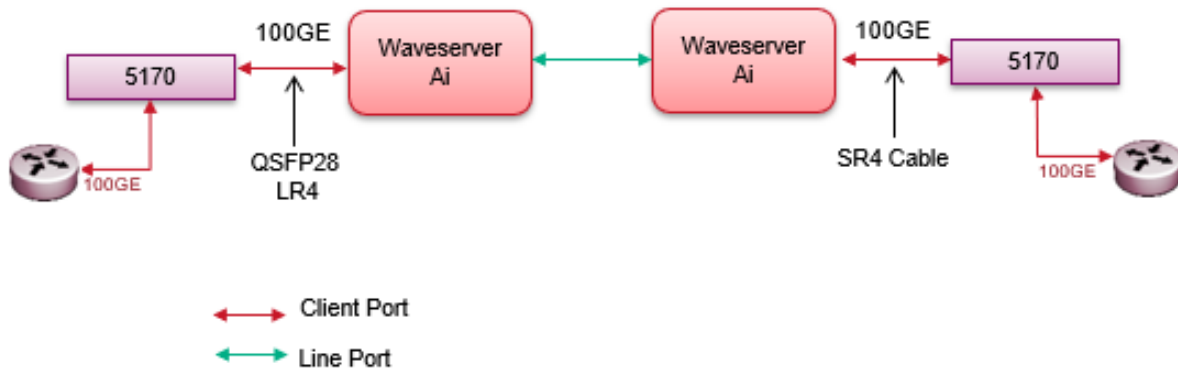
- Supports Waveserver Ai Release 1.2 and higher.
- Supports 5170 Release 8.6.1 and higher.
- Supports 56GBaud and 35/56GBaud interworking on line ports.
- This configuration supports alarms and Performance Monitoring.
- Supports 100GE client pluggables: QSFP28 LR4 (160-9402-900), MR AOC cable.
- MCP 18.06 supports
  - Waveserver Ai and 5170 discovery
  - alarms and PMs reported on the Waveserver Ai and 5170

For information on Waveserver Ai Release 1.2, 5170 Release 8.6.1 and MCP 18.06 interworking considerations, see [Waveserver Ai Release 1.2](#), [5170 Release 8.6.1](#), and [MCP 18.06 interworking considerations](#).

### **Waveserver Ai 2xWLAi + 8xQSFP28 module interworking with 5170 using an SR4 Ciena-branded cable**

This section outlines the Waveserver Ai 2xWLAi + 8xQSFP28 module interworking with 5170 using an SR4 Ciena-branded cable. Waveserver Ai 100GE clients interoperate with 5170 CES 100GE clients over an SR4 Ciena-branded cable. See the following figure.

**Figure 47** Waveserver Ai 2xWLAi + 8xQSFP28 module interworking with 5170 using an SR4 Ciena-branded cable



The following interworking considerations apply:

- Supported on Waveserver Ai Release 1.2 and higher
- Supports 5170 Release 8.6.1 and higher.
- Supports 56GBaud and 35/56GBaud interworking on line ports.
- Supports alarms and Performance Monitoring.
- Supports 100GE client pluggables: QSFP28 LR4 (160-9402-900), SR4 cable.
- MCP 18.06 supports
  - Waveserver Ai and 5170 discovery
  - alarms and PMs reported on the Waveserver Ai and 5170

For information on Waveserver Ai Release 1.2, 5170 Release 8.6.1 and MCP 18.06 interworking considerations, see [Waveserver Ai Release 1.2](#), [5170 Release 8.6.1](#), and [MCP 18.06 interworking considerations](#).

### Waveserver Ai Release 1.2, 5170 Release 8.6.1, and MCP 18.06 interworking considerations

For MCP to be able to manage the Waveserver Ai and 5170, the following prerequisites apply:

- On the Waveserver Ai, the member-name must be configured, use CLI command `system set member-name`.
- On the 5170:
  - SSH must be enabled and an SSH key must be generated, use CLI commands `enable ssh`, `ssh client enable`, and `ssh server key generate`.
  - Increase the super user accounts to 10, use CLI command `user set max-super-users 10`.

- SNMP must be provisioned and enabled on the Waveserver Ai/5170 for SNMP traps to be sent to the MCP server. For information on SNMP on the Waveserver Ai, see *Waveserver Ai User Guide*, 323-4002-100. For information on SNMP on the 5170, refer to the 5170 Service Aggregation Switch (5170) documentation library.



# Waveserver Ai

## Waveserver Ai and 6500/5170 Interworking

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